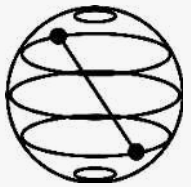
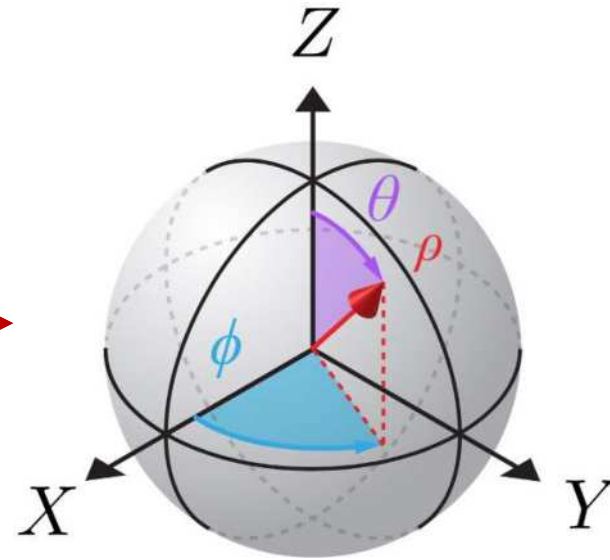
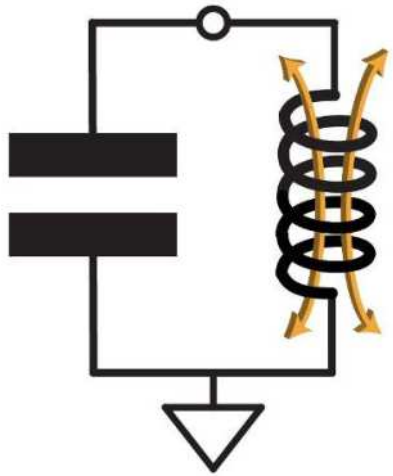


Superconducting Qubits 101

Making Your First Qubit From an Oscillator

Introduction to Circuit
Quantum Electrodynamics (cQED)

Zlatko K. Minev



IBM Quantum
IBM T.J. Watson Research Center, Yorktown Heights, NY



@zlatko_minev



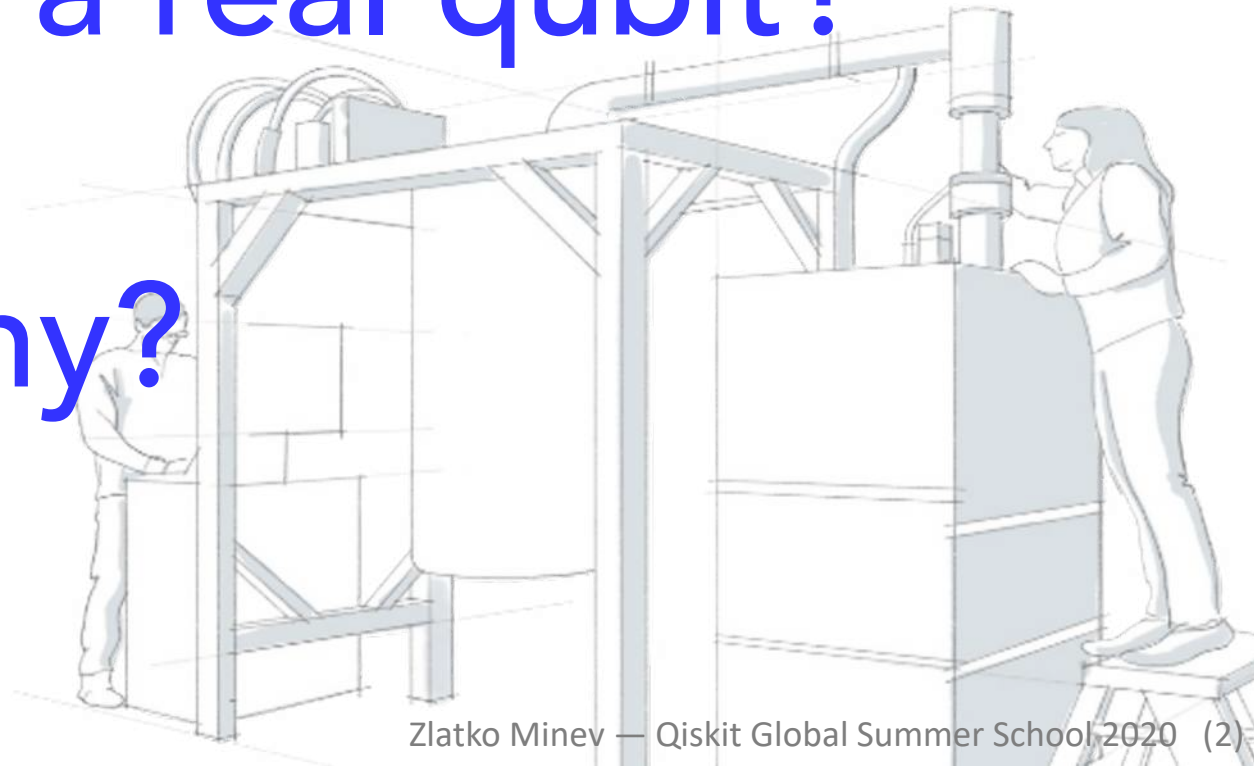
zlatko-minev.com

*Image copyright:
ZKM unless otherwise noted*

What is a real qubit?

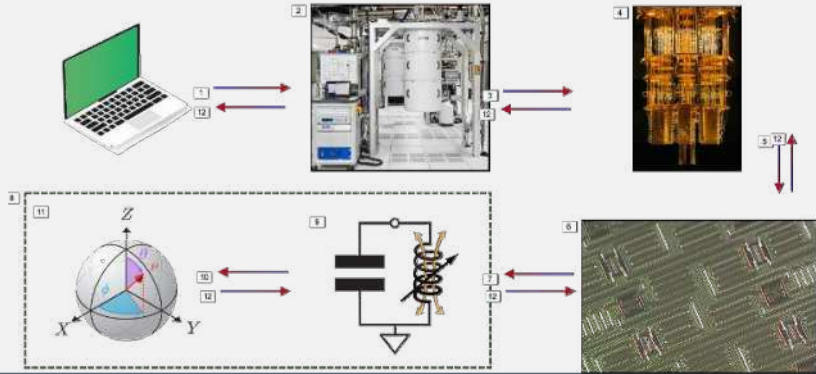
How can you design, control, and measure a real qubit?

Why?

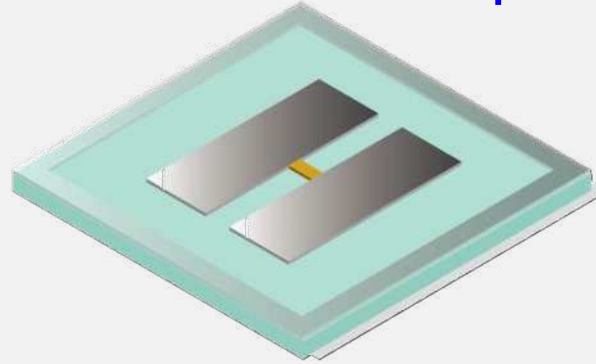


On the road ahead

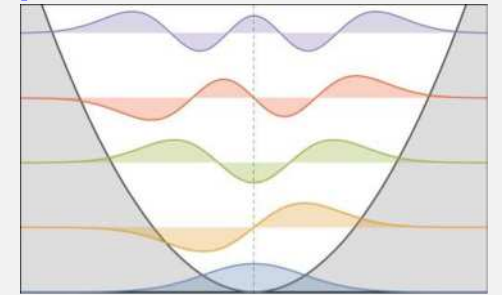
Qubit in the cloud



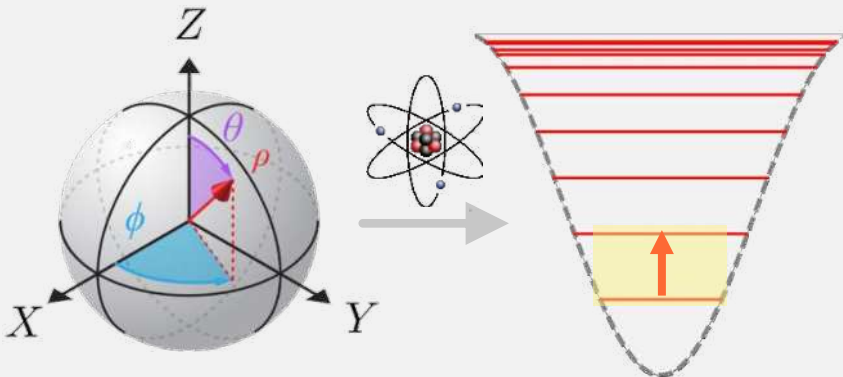
cQED: Transmon qubit



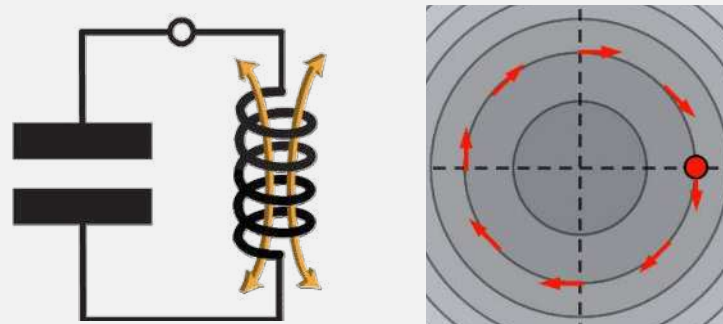
Unveiling the quantum oscillator



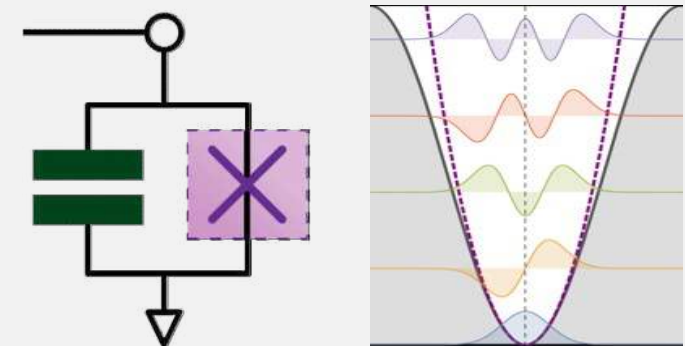
Qubit from atom / oscillator



Classical circuits & the LC



Transmon qubit



This Lecture

Introductory and skill reaffirming

Don't need to know much going in, but we will go far

Examples: simplest, most practical examples

Step by step

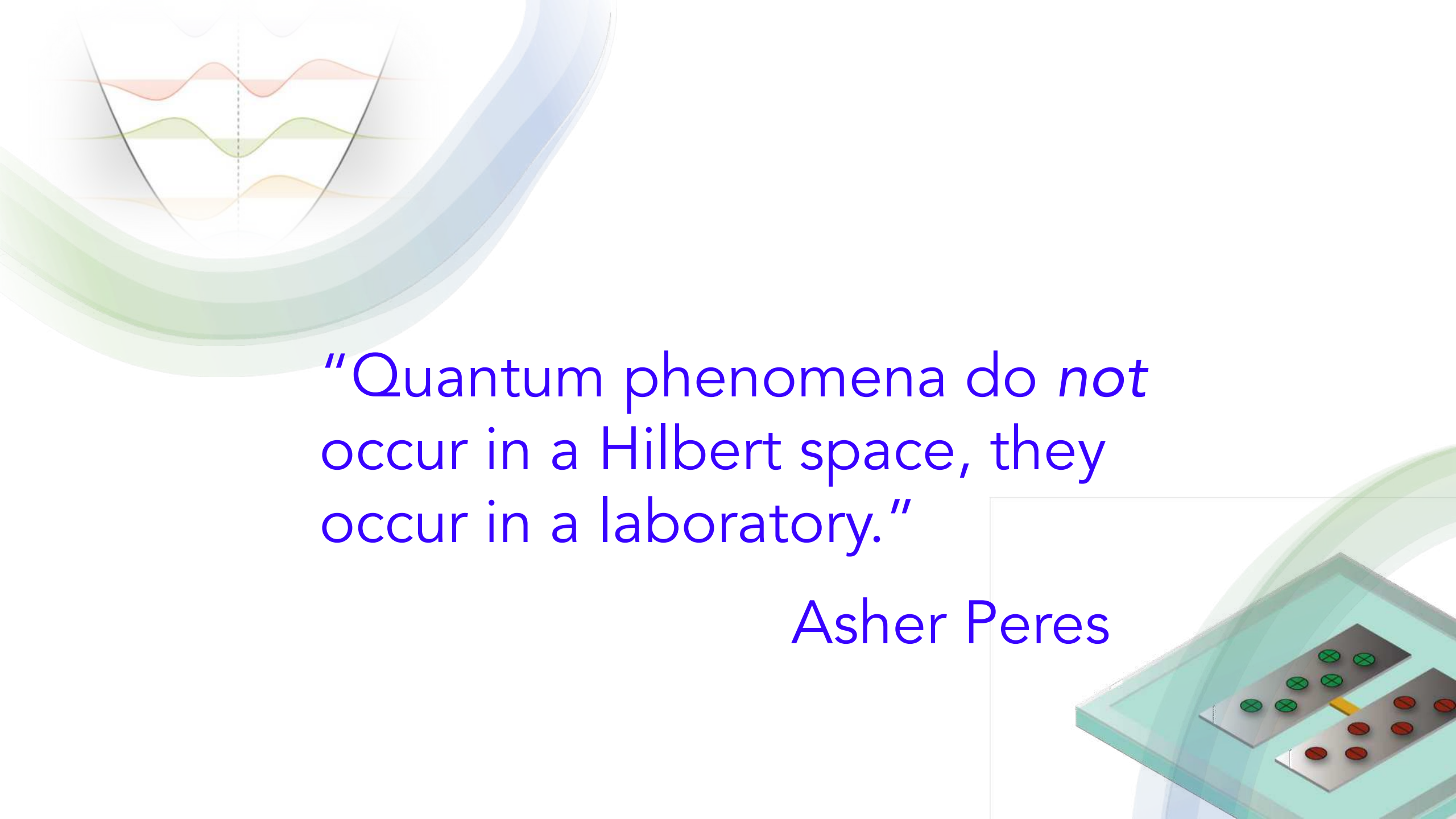
Ask questions!

Tightly integrated work by Dr. Thomas McConkey!



Advanced
material



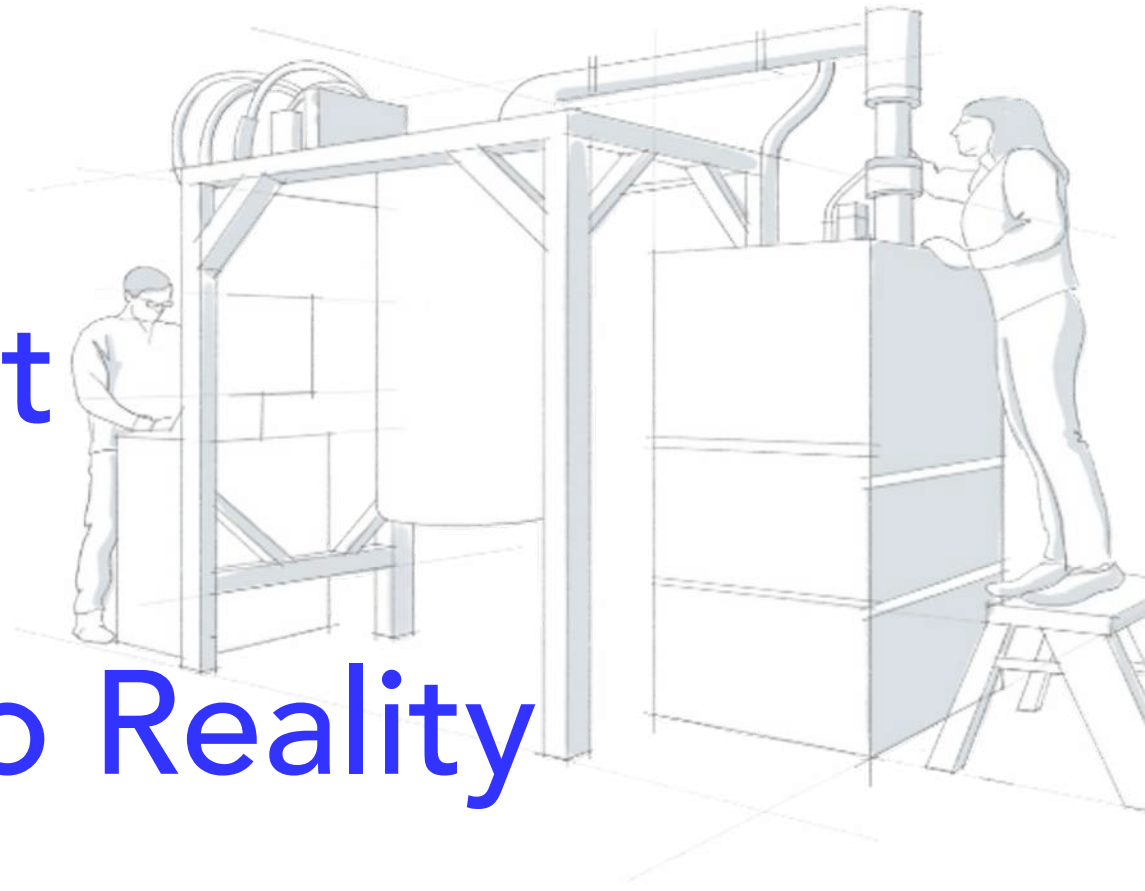


"Quantum phenomena do *not* occur in a Hilbert space, they occur in a laboratory."

Asher Peres

Qubit

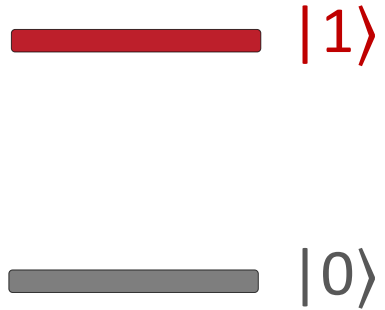
From Idea to Reality



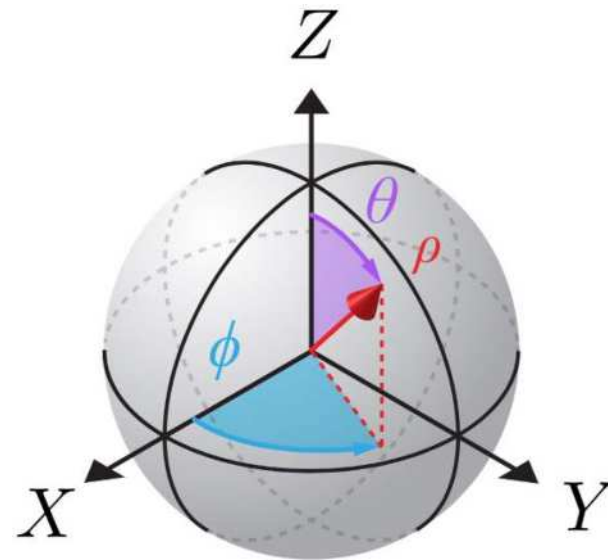
THE BIG PICTURE
before calculations

Qubit: idea

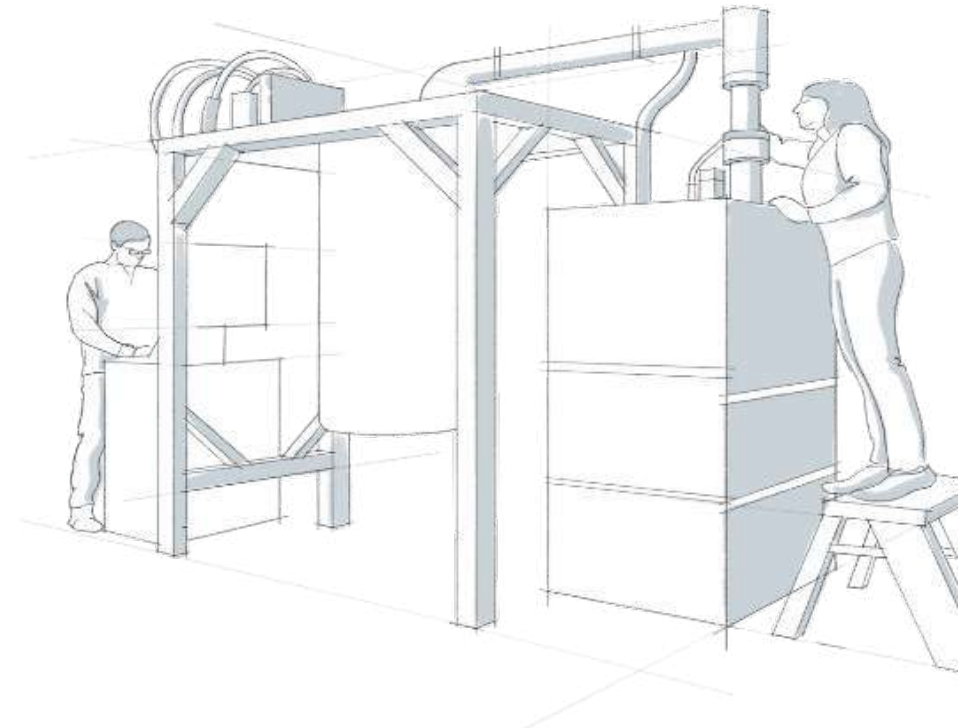
Energy levels



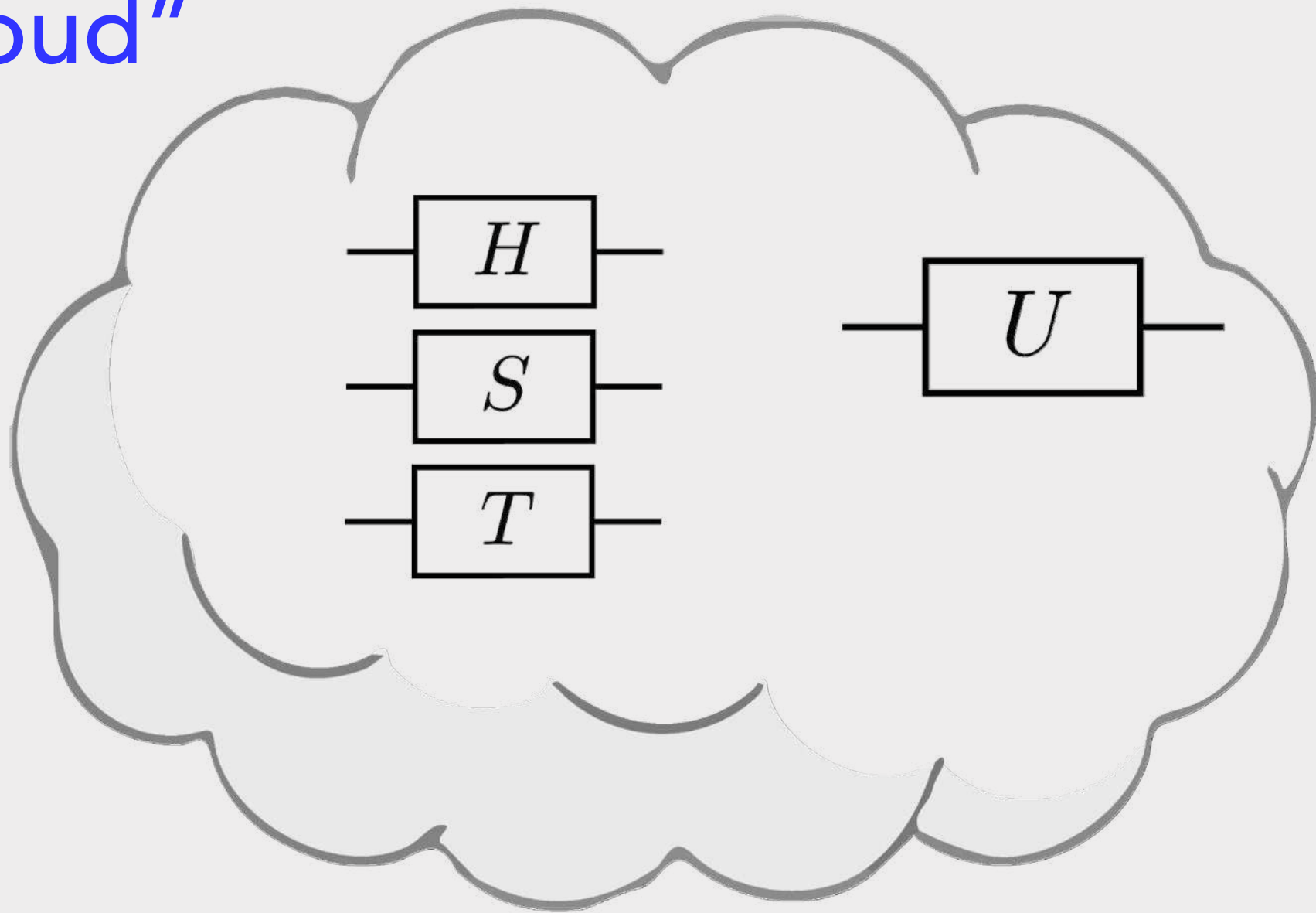
Hilbert space



Quantum cloud?



"Quantum cloud"



Quantum in the cloud or lab

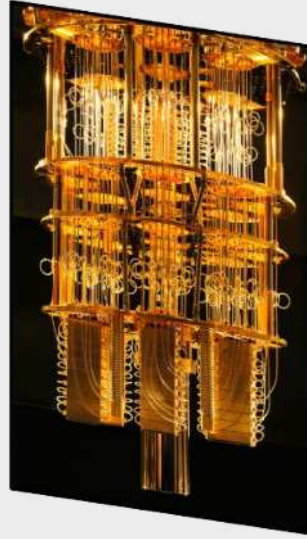
Superconducting qubits



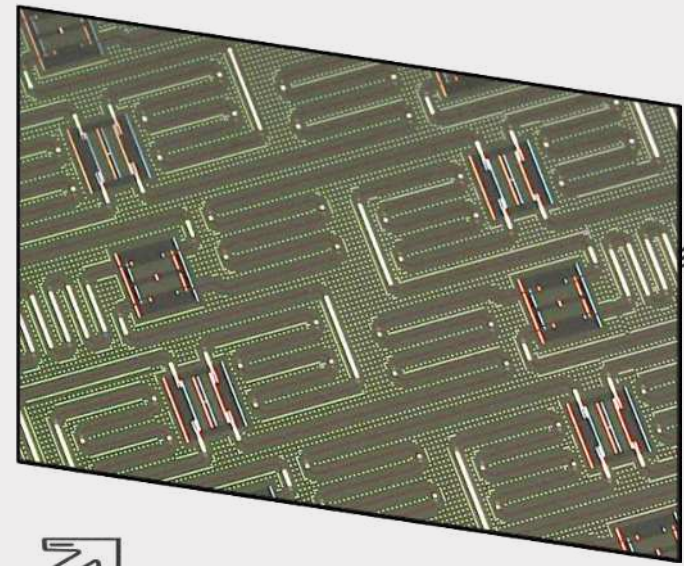
Quantum community



Lab / cloud facility

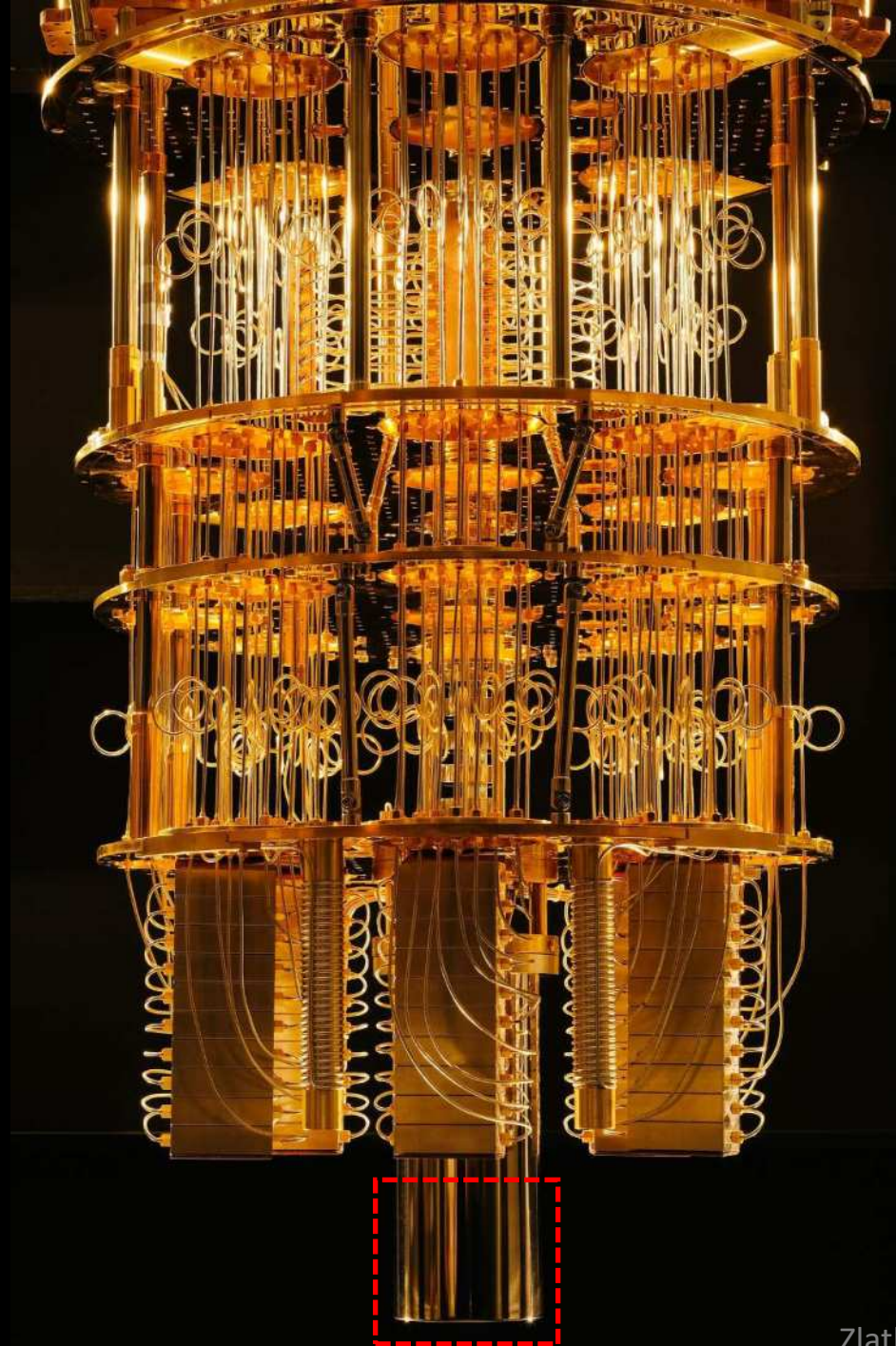


Cryogenic environment

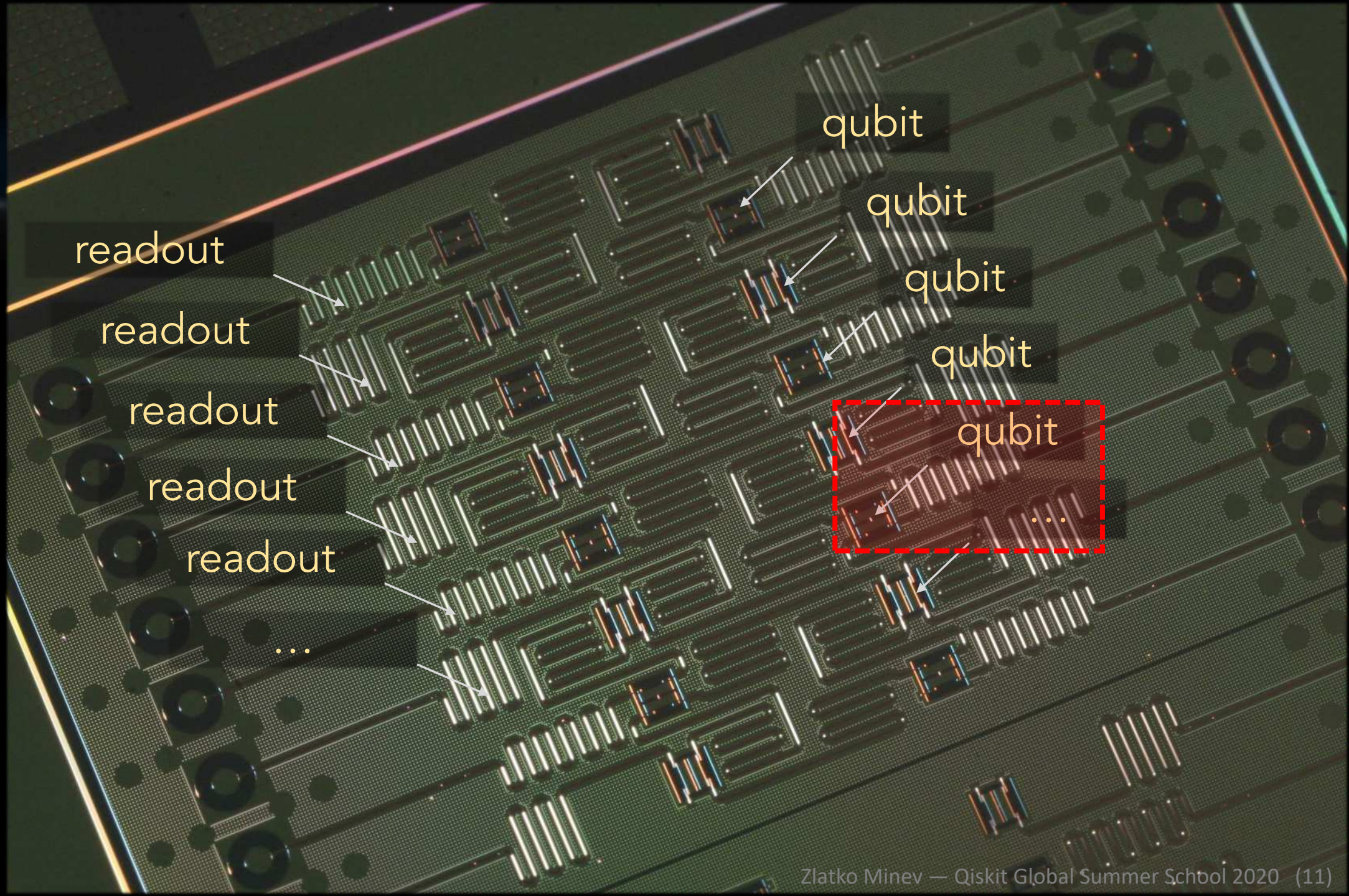
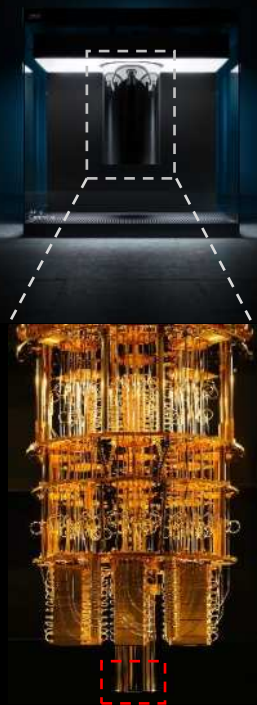


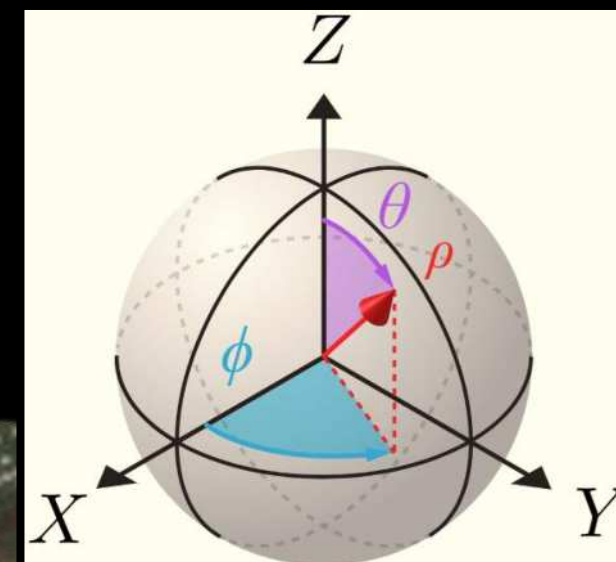
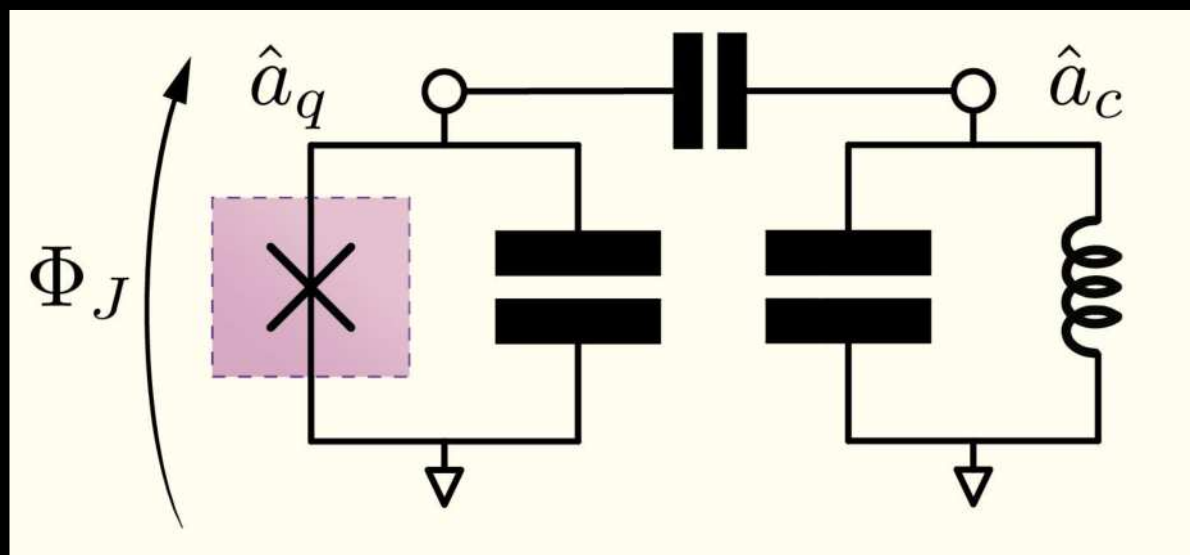
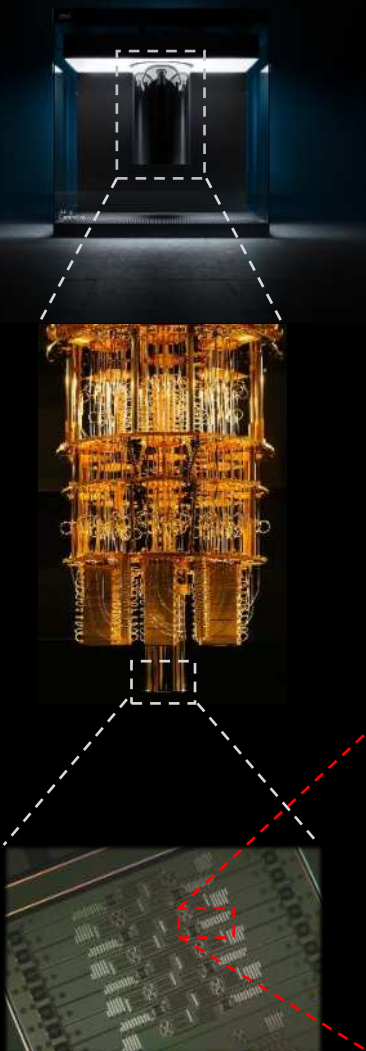
Quantum-device chip



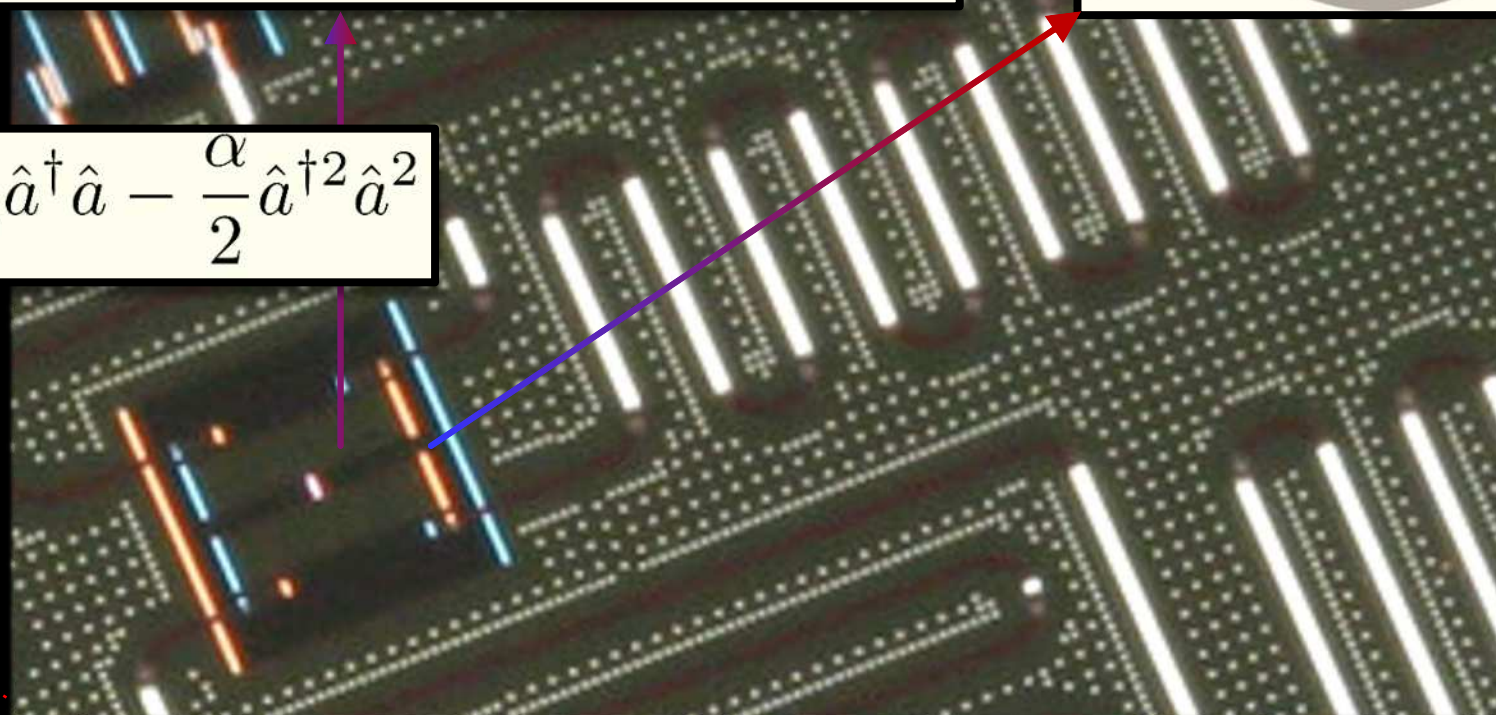


Operation at
15 mK (-273.13 °C)



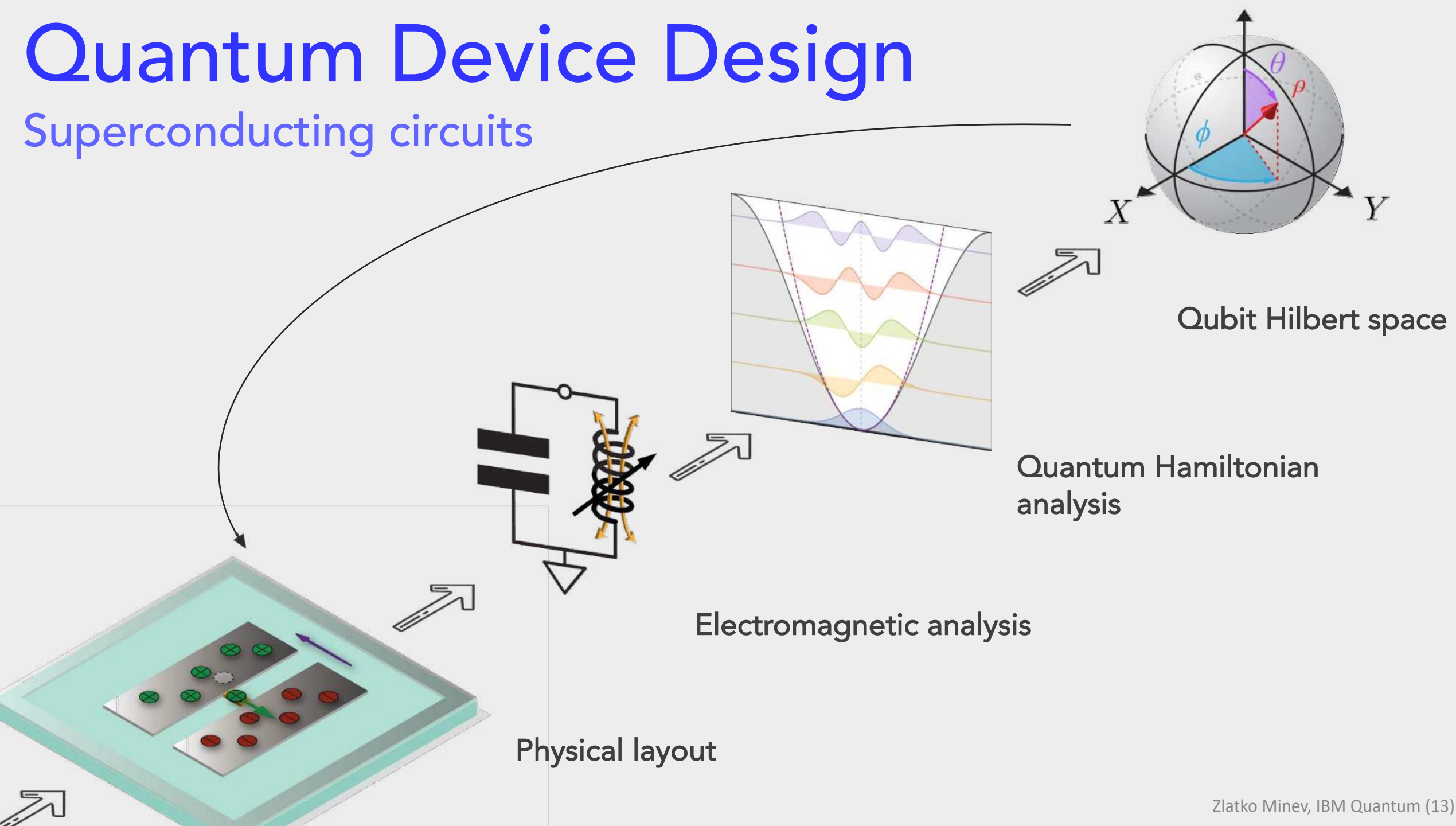


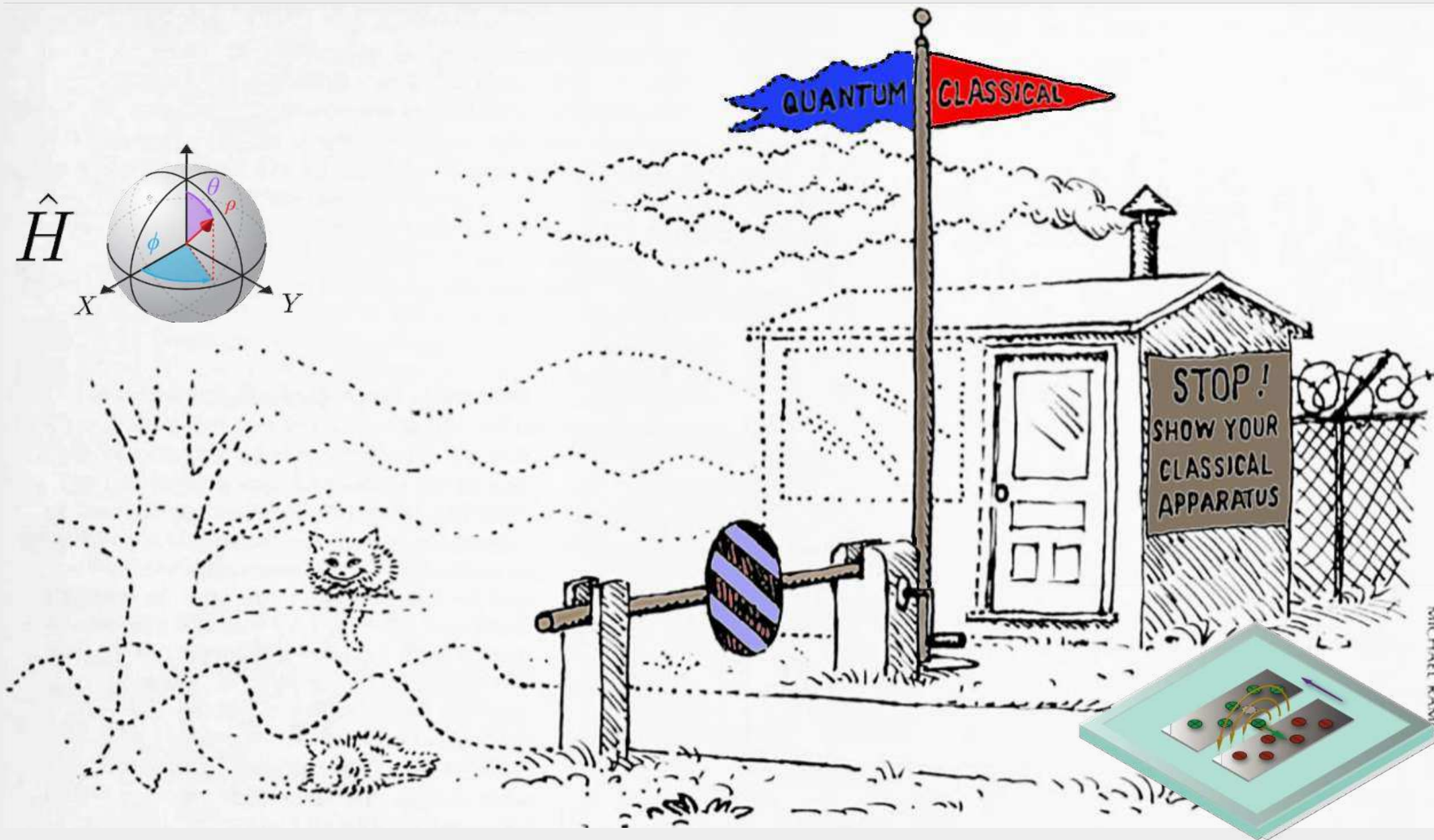
$$\hat{H} \approx \omega_0 \hat{a}^\dagger \hat{a} - \frac{\alpha}{2} \hat{a}^{\dagger 2} \hat{a}^2$$



Quantum Device Design

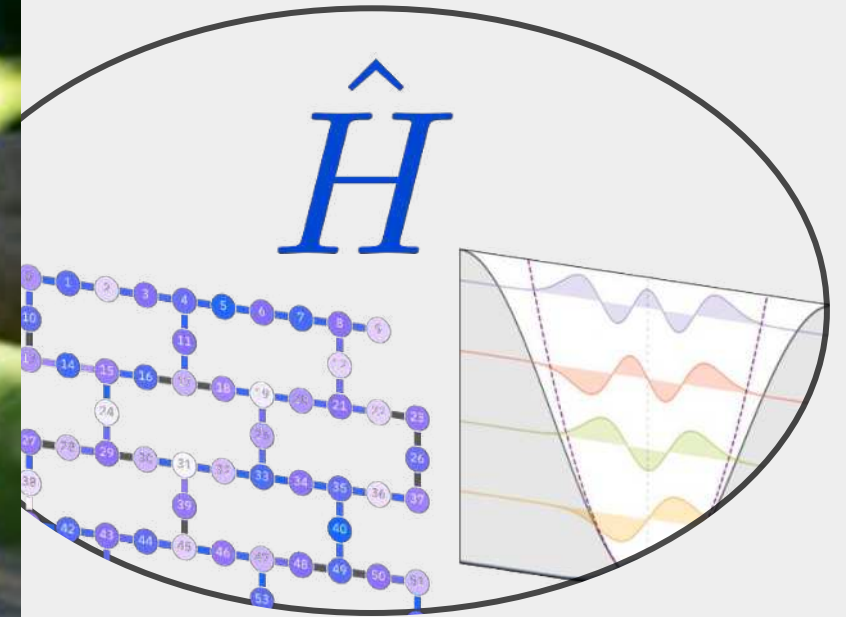
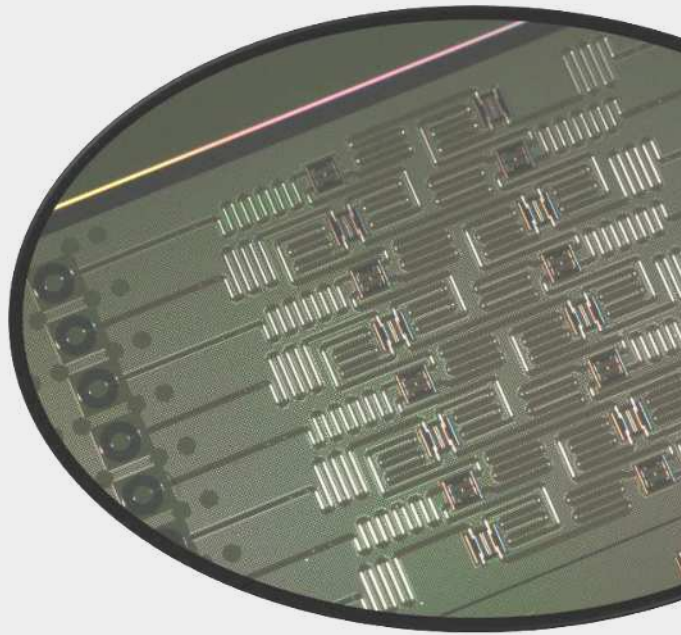
Superconducting circuits



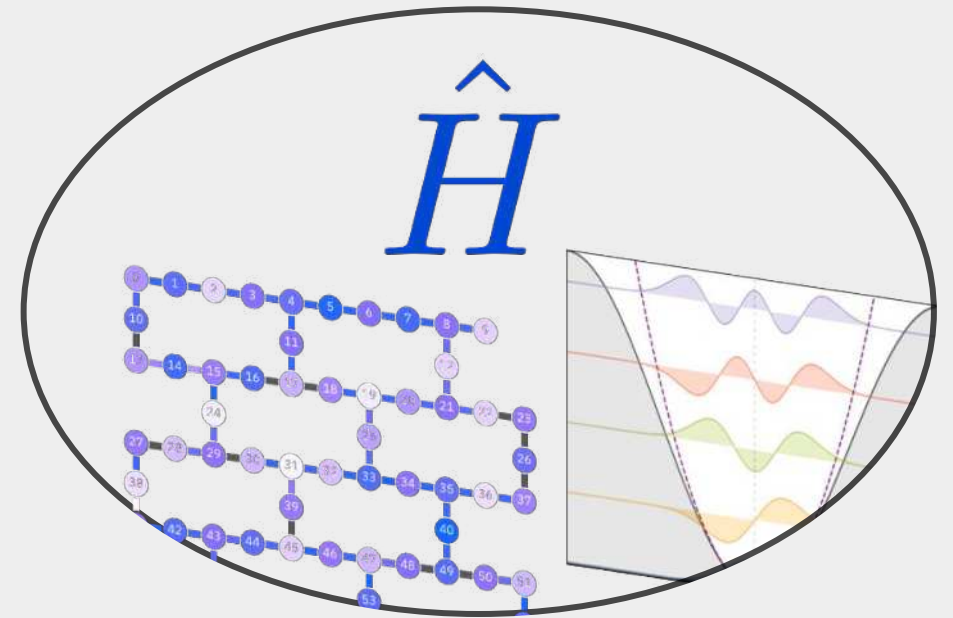
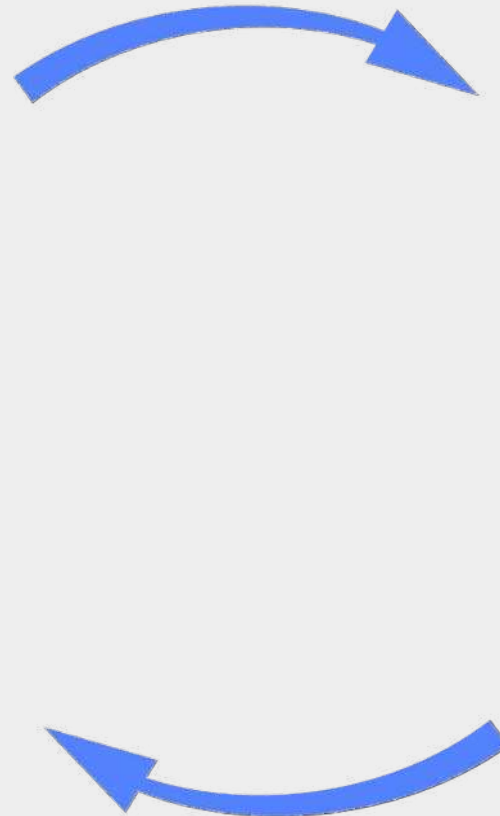
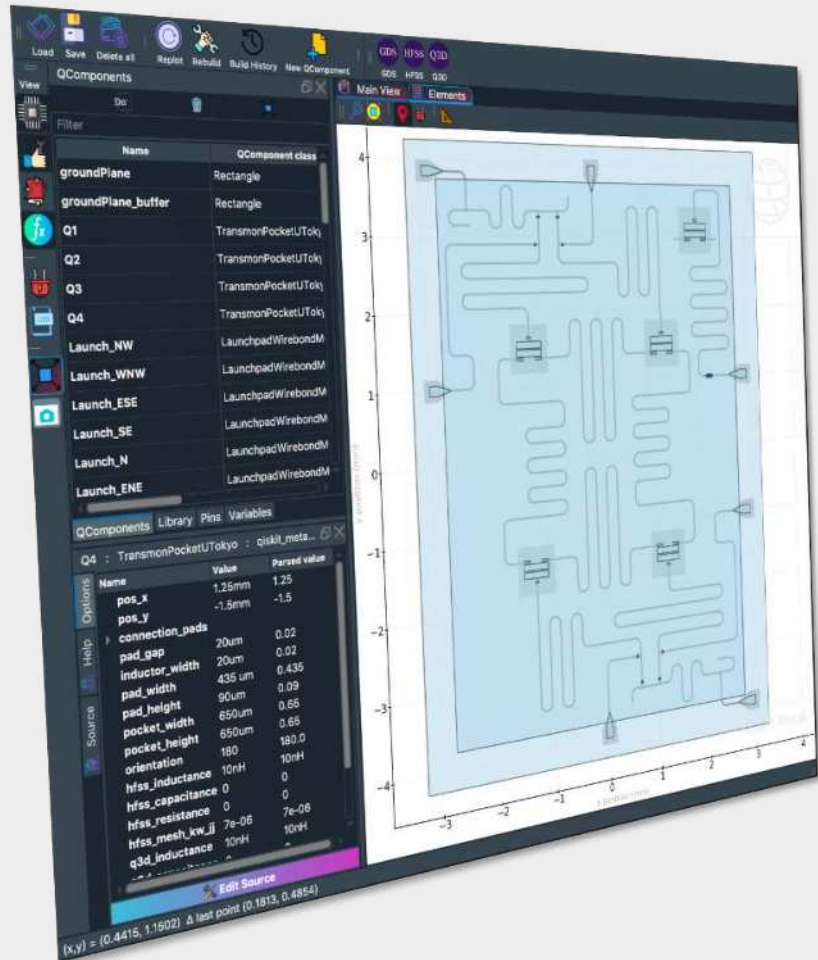


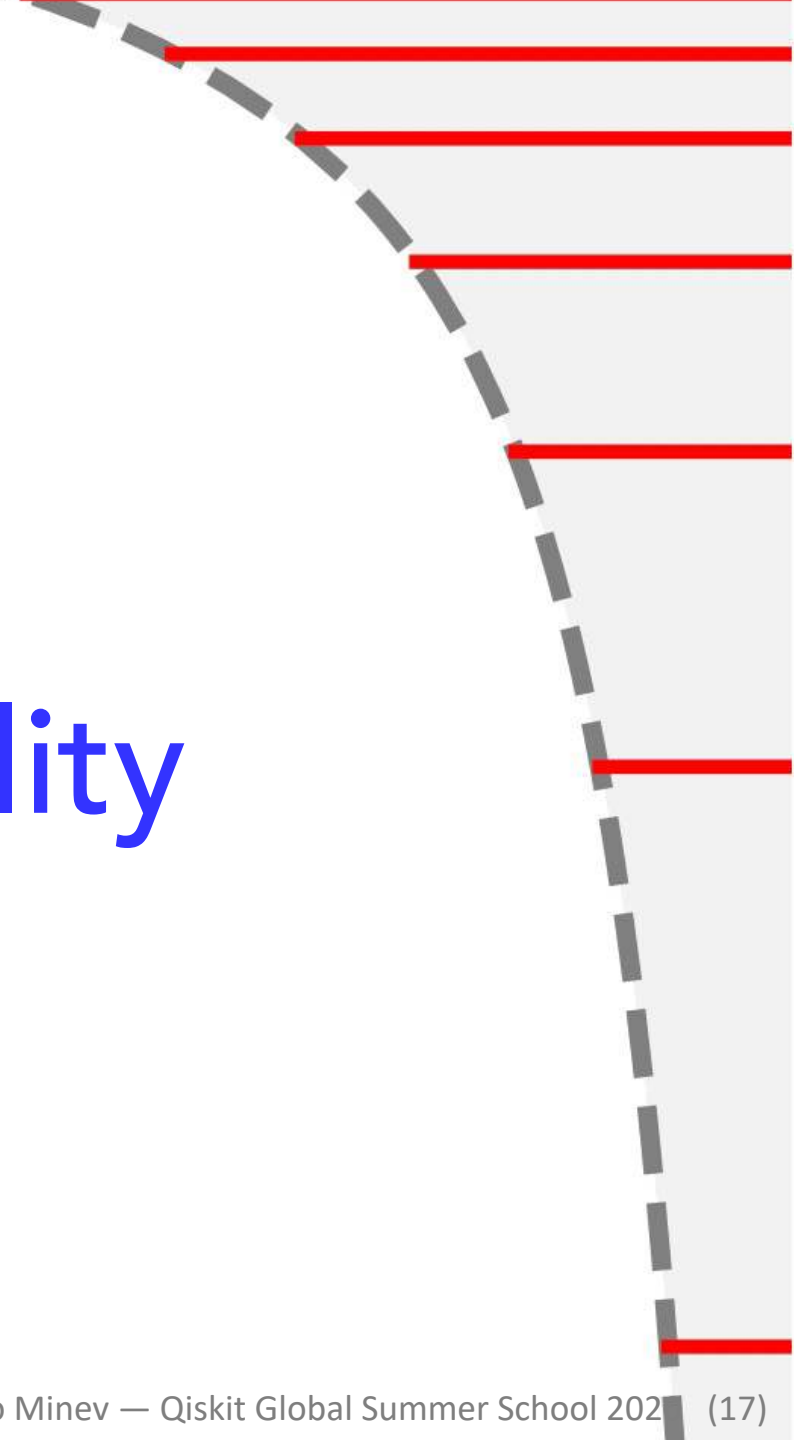
Drawing: Zurek, Physics Today (1991)

Physical Devices $\leftrightarrow$ Quantum Hamiltonian



Make easy?





Qubit

From Idea to Reality

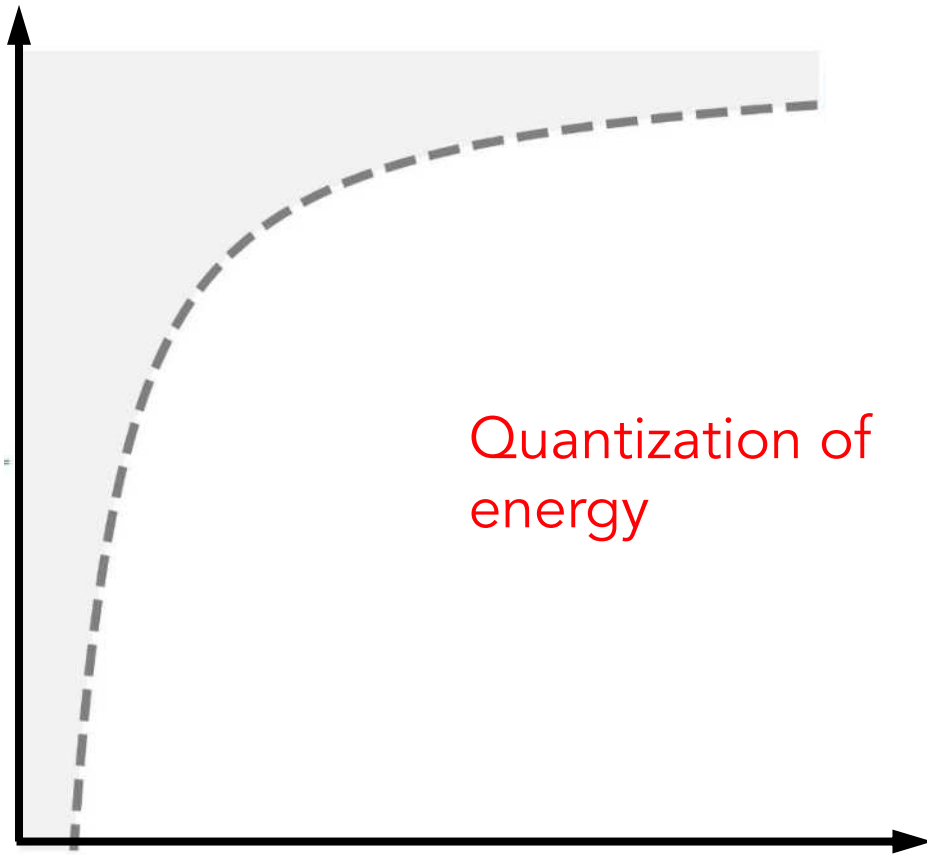
Concepts

The light of atoms



Atomic energy levels and transitions

Electron potential-energy landscape



Radial distance from nucleus

Qubit from Atom

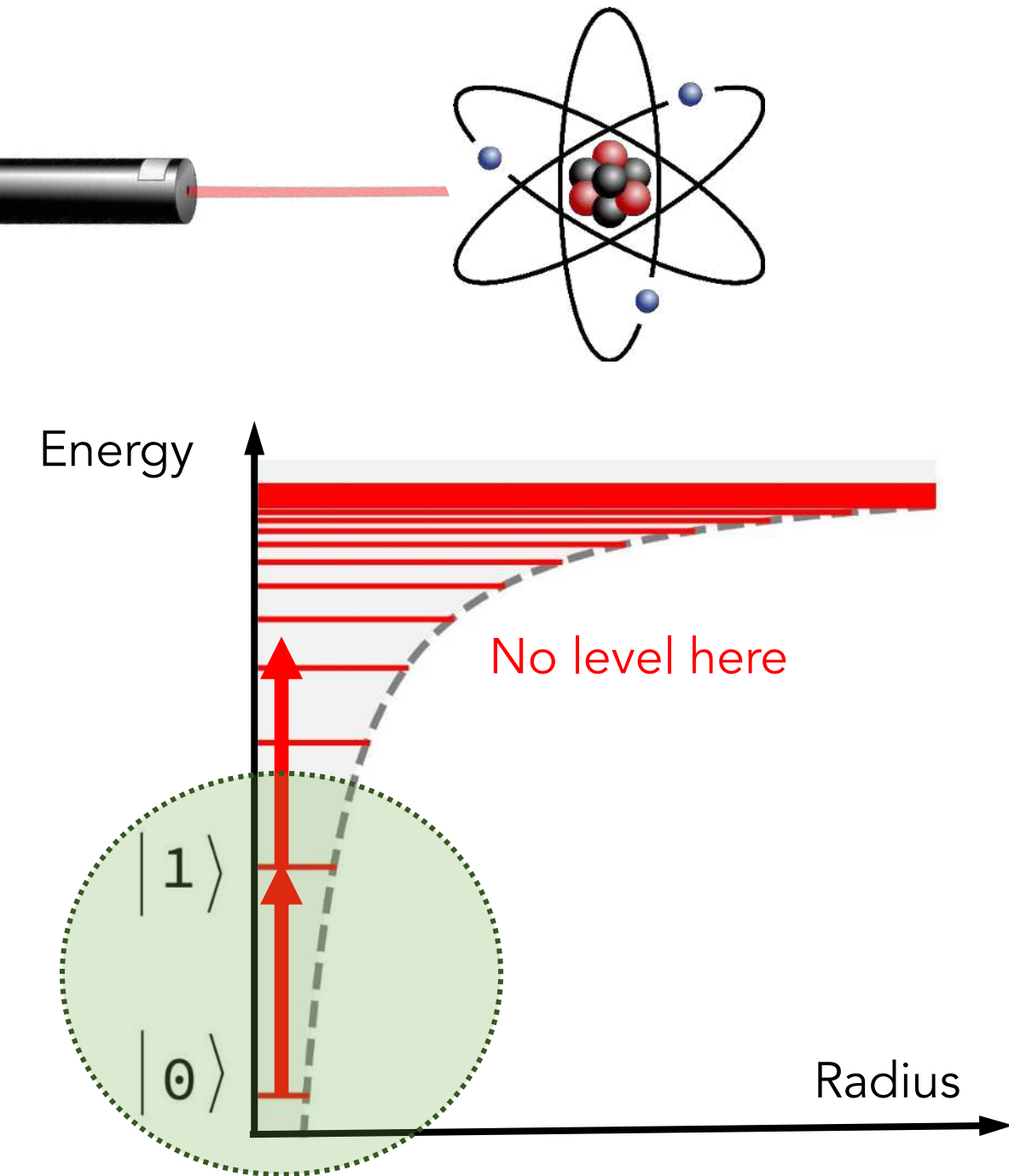
Anharmonic degree of freedom and spin

Isolated from environment and thermal bath

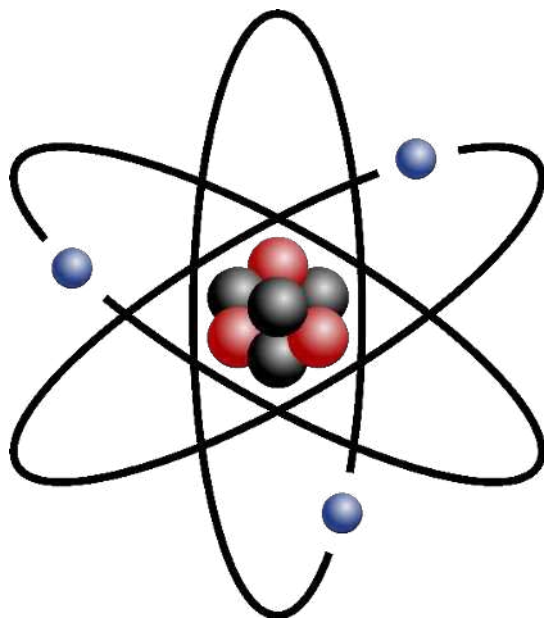
Low-loss

Level diagram allows for qubit-specific control and readout

There are always more than two levels

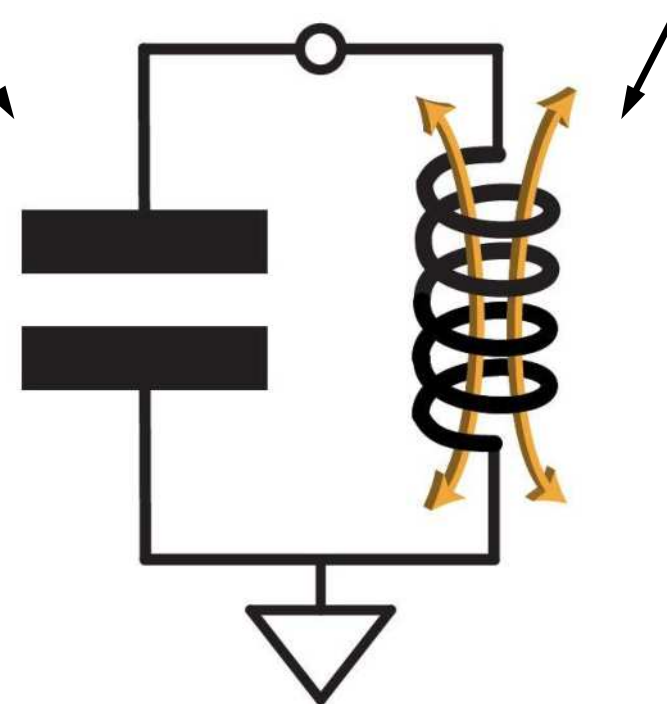


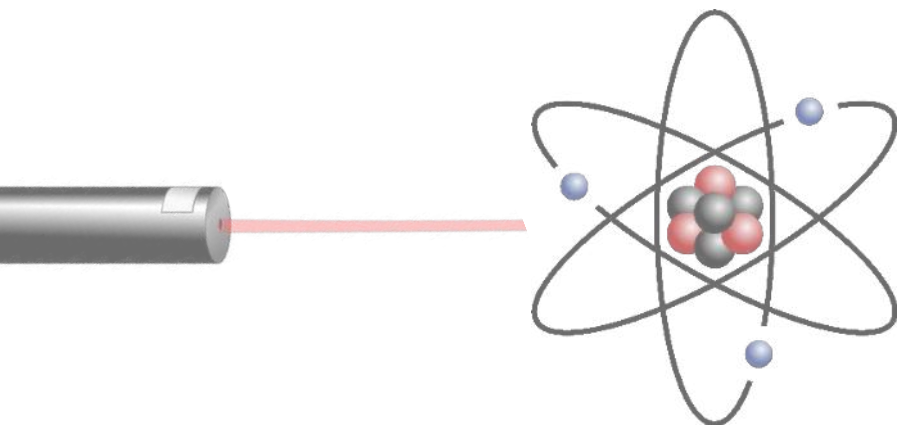
Artificial atoms



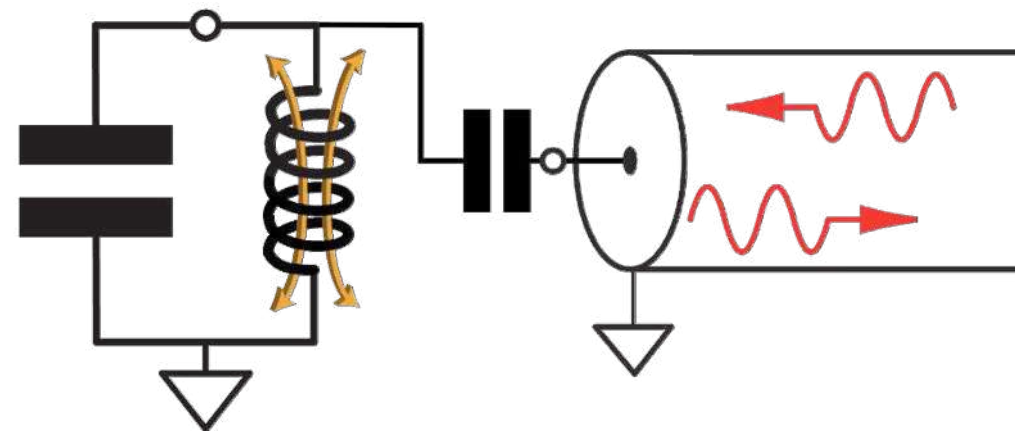
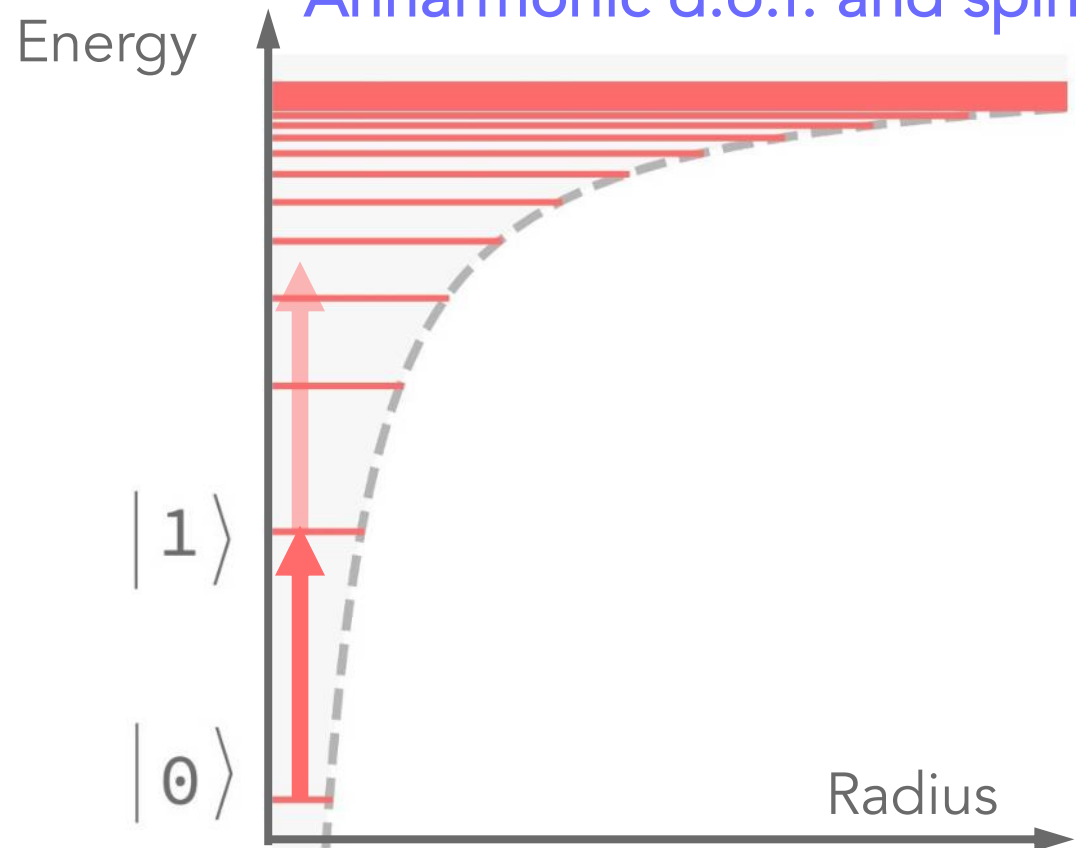
capacitor

inductor

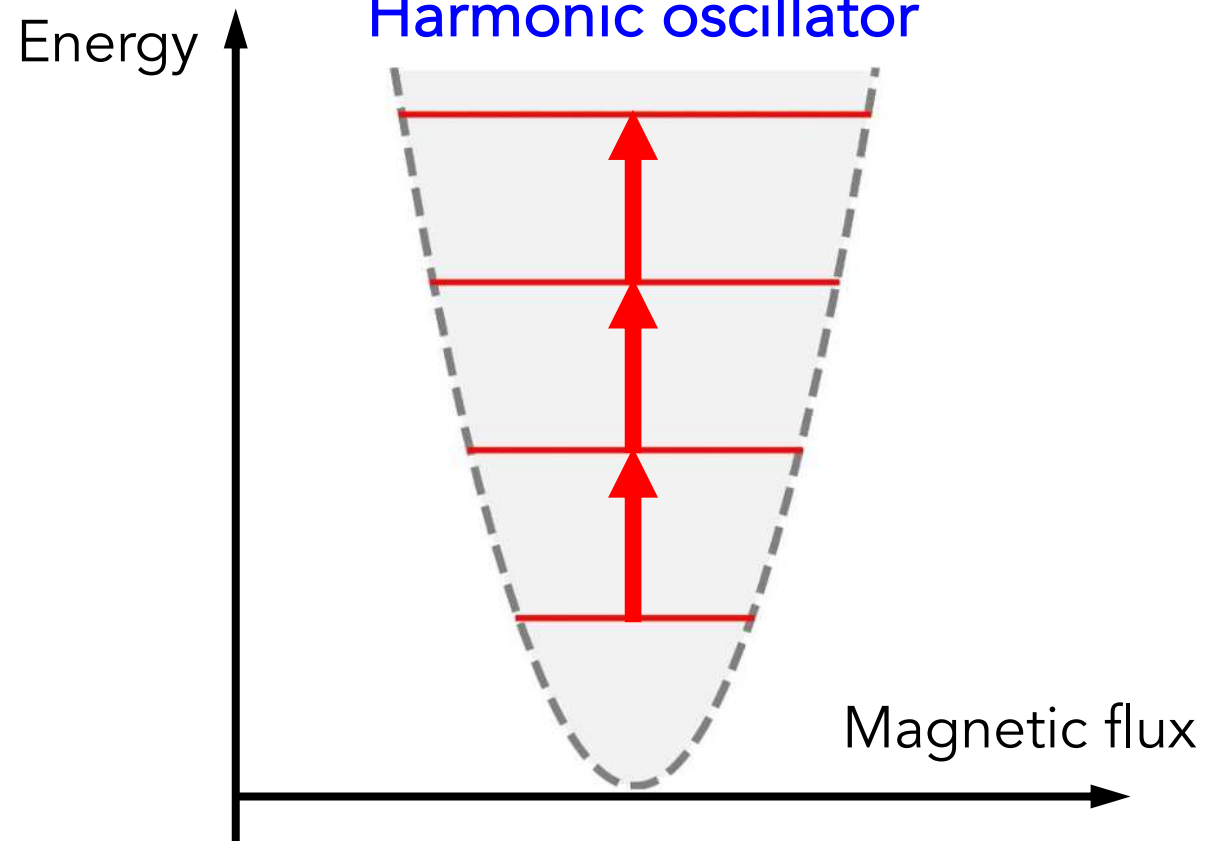


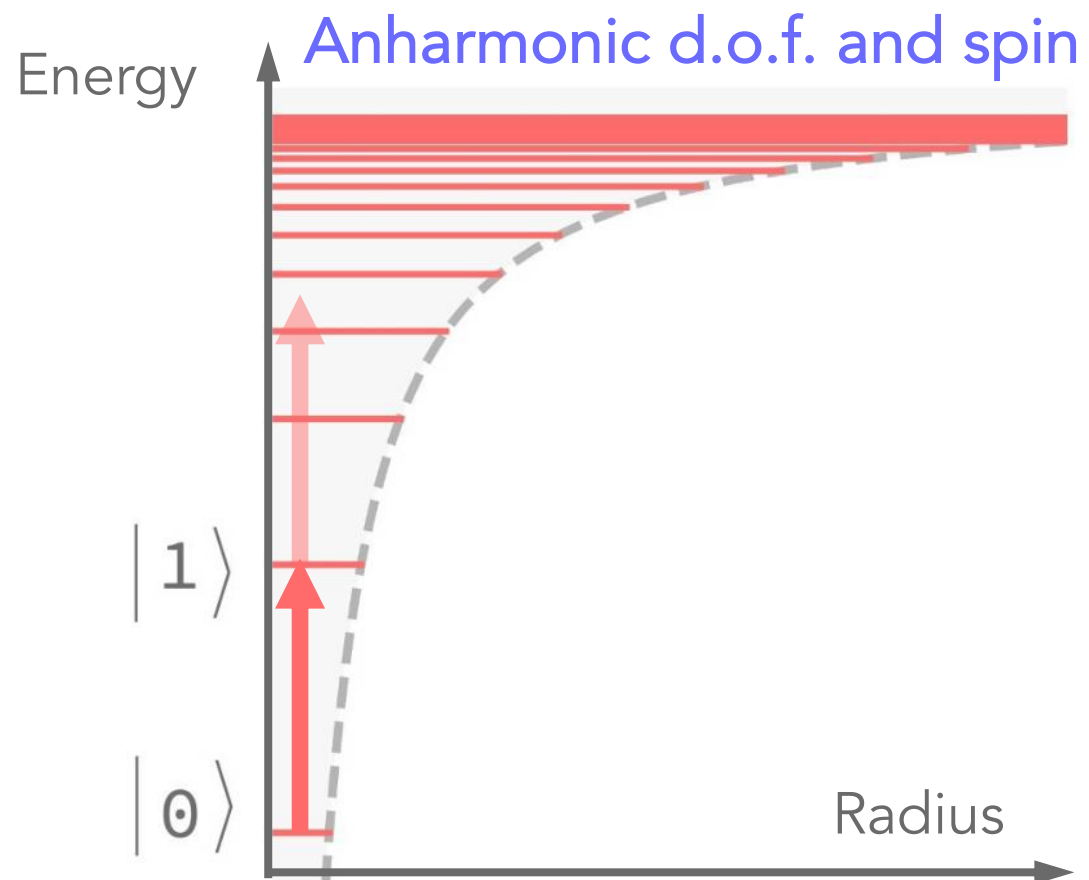
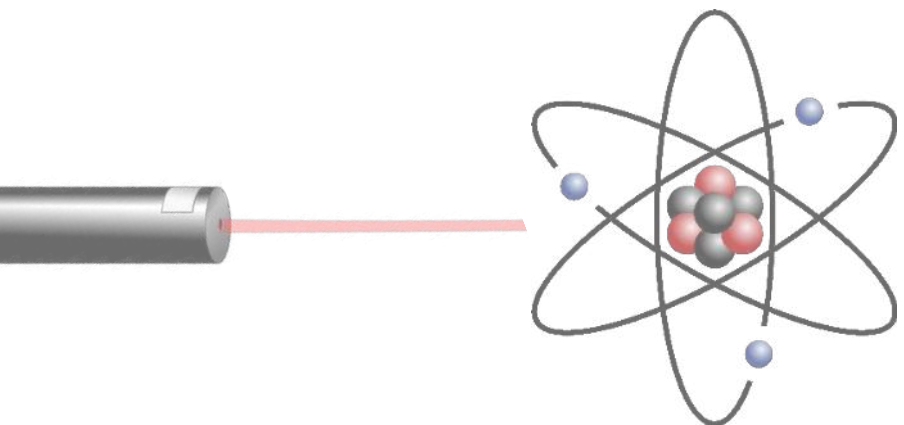


Anharmonic d.o.f. and spin

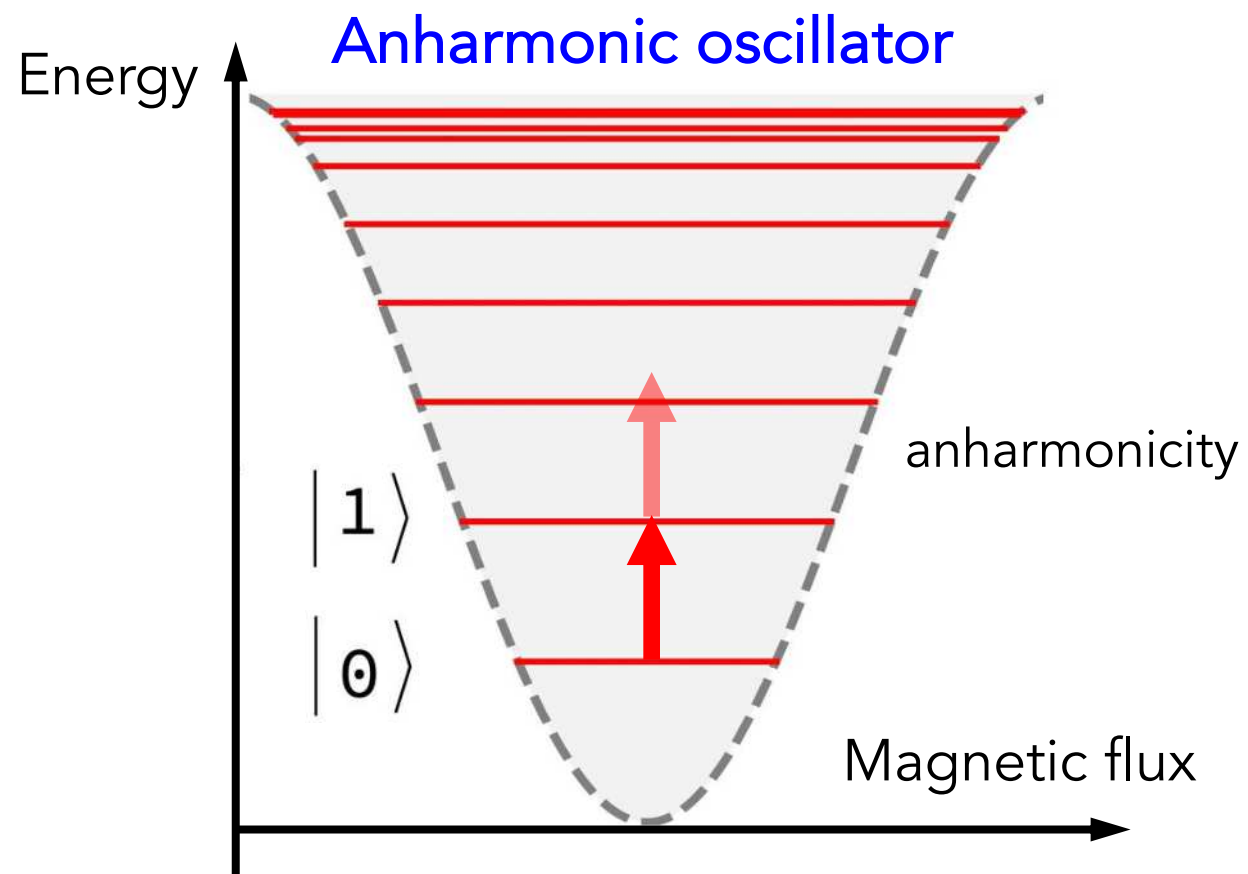
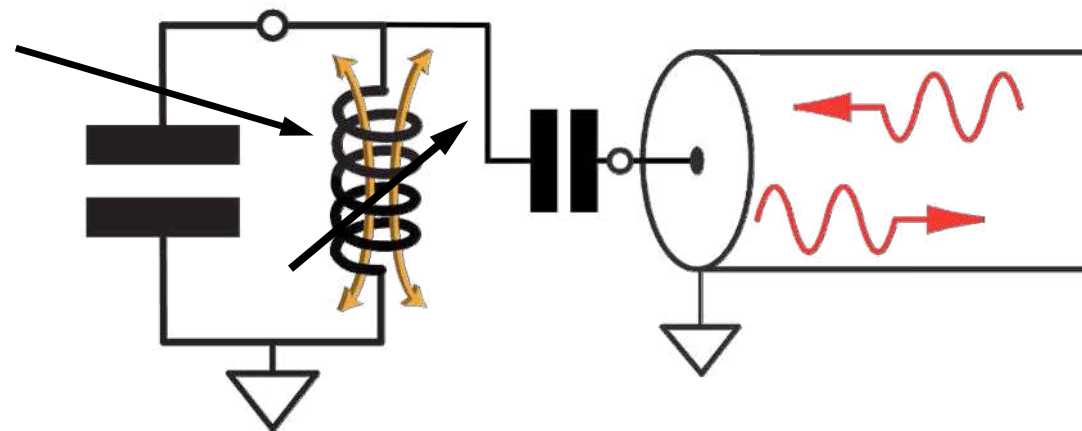


Harmonic oscillator



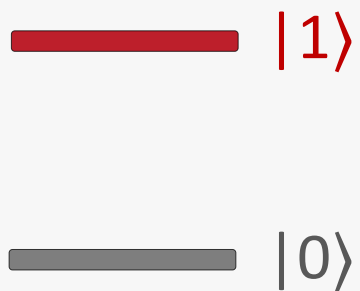


non-linear
inductor

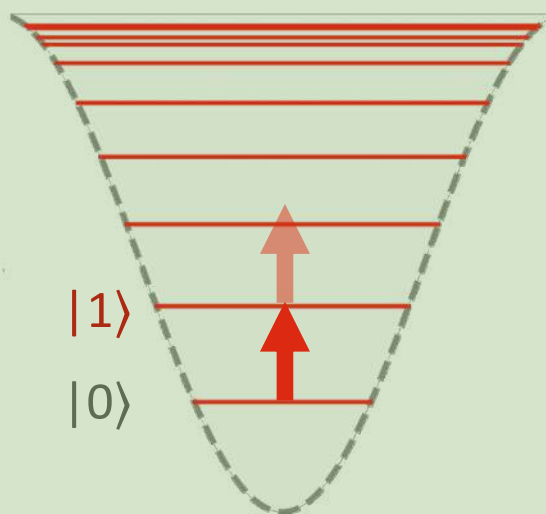


Big-picture connections

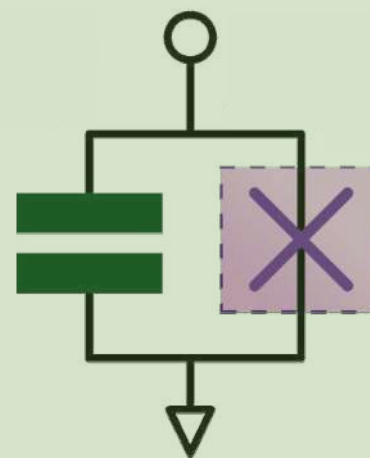
Idealization of
qubit



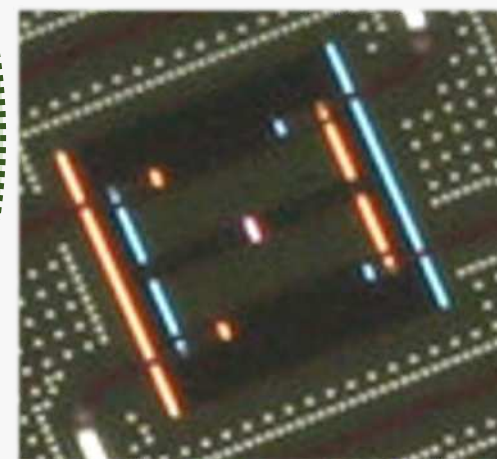
Anharmonic
oscillator



Physical circuit
model



Physical layout



Idealization



Physical reality

Circuit Quantum Electrodynamics (cQED)

Macro
overview

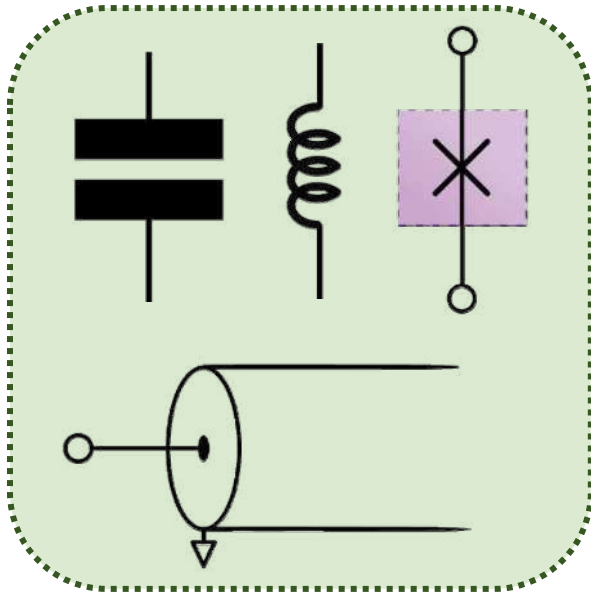
There are two kinds of physicists:

Those who believe all of physics is *spins*.

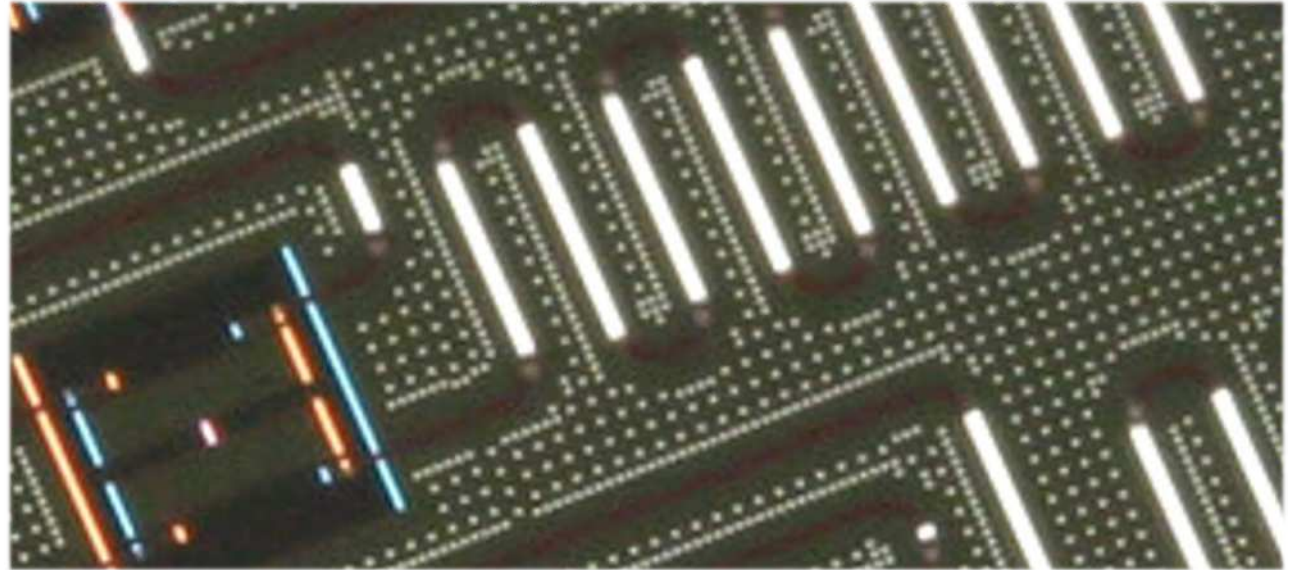
Those who believe all of physics is *oscillators*.

cQED Ingredients

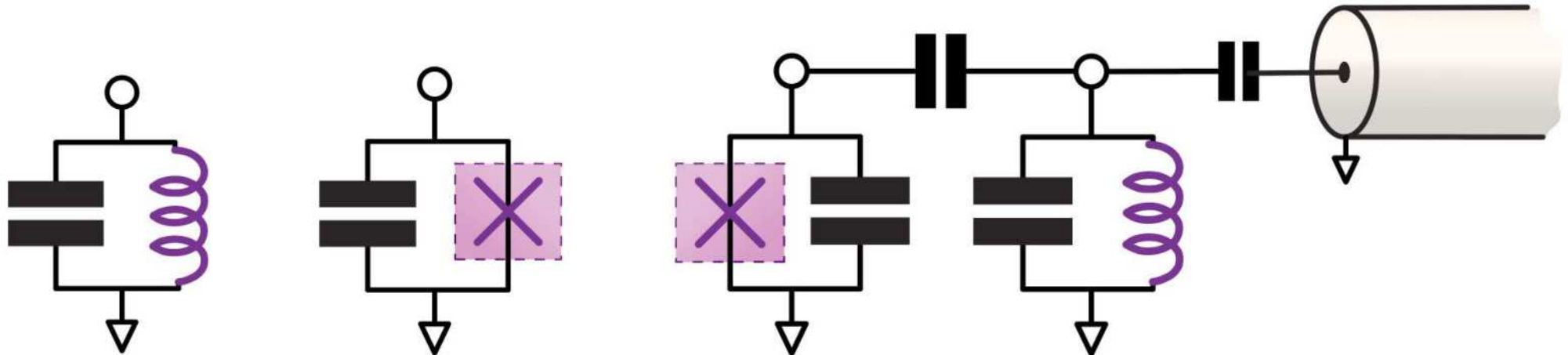
Circuit elements



Microwave oscillators

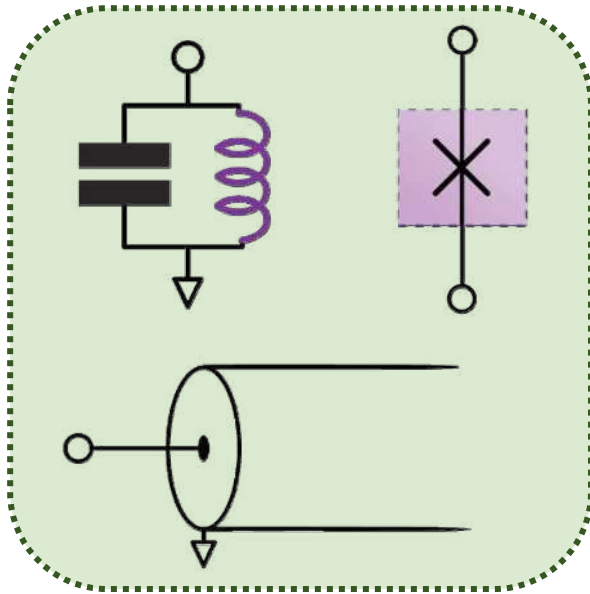


Combine



...

cQED Ingredients



Small dissipation

Isolation from environment

Low temperature

Nonlinearity

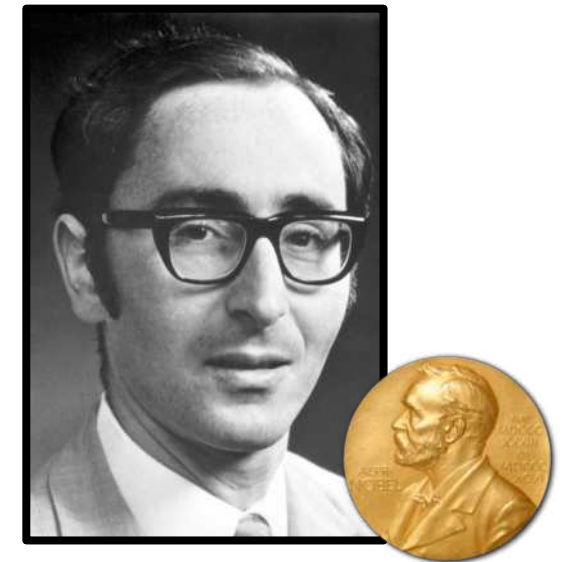
Large vacuum fluctuations

Superconductivity

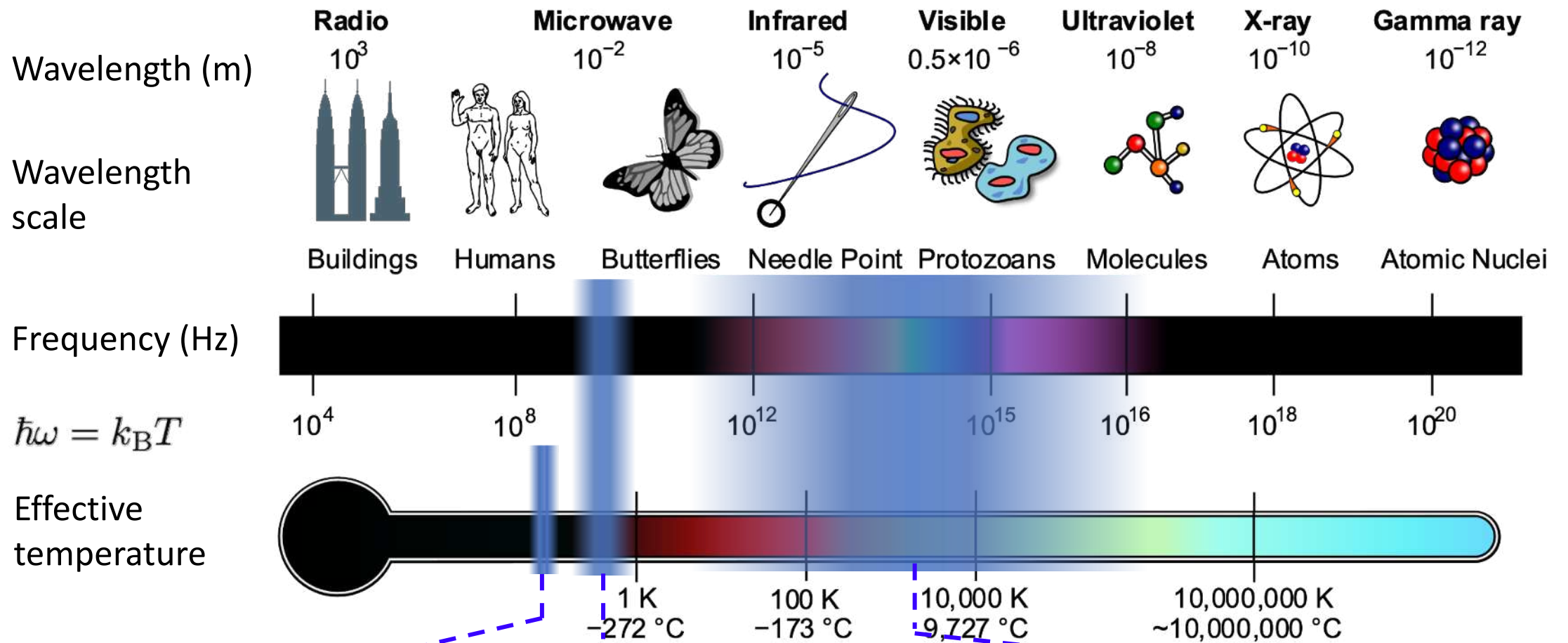
Nominally zero intrinsic dissipation and heat

Nominal temperature far below energy level splitting

Non-linear, robust Josephson tunnel junction effect



* Low energy dynamics. Microwave image: vectorpocket; Nobel photograph: Jonathunder; Josephson photograph: Nobel Foundation archive.

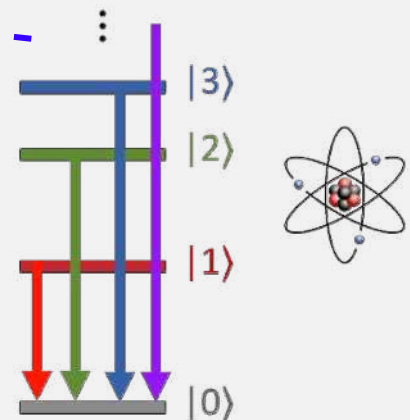
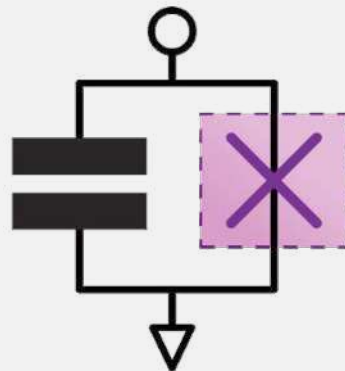


Order of magnitude



10^{-2} K

0.5 K, 10 GHz, 3 cm



Spectrum image: Inductiveload, NASA

A few introductory reviews

And many more... check online or ask us for specific topic

Qiskit Textbook (2020; more chapters coming)

Blais, A., Grimsmo, A. L., Girvin, S. M., & Wallraff, A. (2020)
Circuit Quantum Electrodynamics (arXiv:2005.12667)

Kjaergaard, M., Schwartz, ... Oliver, W. D. (2020)
Superconducting Qubits: Current State of Play
Annual Reviews of Condensed Matter Physics 11, 369-395

Krantz, P., Kjaergaard, M., Yan, F., ... & Oliver, W. D. (2019)
A quantum engineer's guide to superconducting qubits
Applied Physics Reviews, 6(2), 021318

Corcoles, A. D., Kandala, A., ... Gambetta, J. M. (2019)
Challenges and Opportunities of Near-Term Quantum Computing Systems. *Proceedings of the IEEE*, 1–15.

Wendin, G. (2017)
Quantum information processing with superconducting circuits. *Reports on Progress in Physics*, 80(10), 106001

Gambetta, J. M., Chow, J. M., & Steffen, M. (2017)
Building logical qubits in a superconducting quantum computing system. *Npj Quantum Information*, 3(1), 2

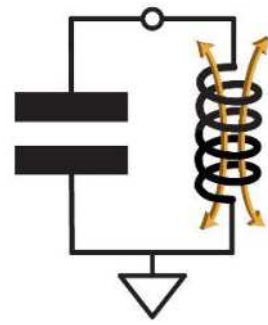
Girvin, S. M. (2011) Circuit QED: superconducting qubits coupled to microwave photons. *Quantum machines: measurement and control of engineered quantum systems*, 113, 2.

Clerk, A. A., Girvin, S. M., Marquardt, F., & Schoelkopf, R. J. (2010)
Introduction to quantum noise, measurement, and amplification
Reviews of Modern Physics, 82(2), 1155–1208

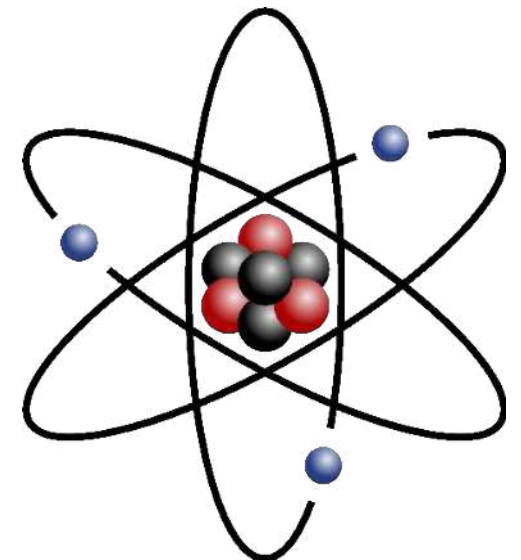
Clarke, J., & Wilhelm, F. K. (2008)
Superconducting quantum bits. *Nature*, 453(7198), 1031–1042

Devoret, M. H. (1997)
Quantum Fluctuations in Electrical Circuits.
In *Fluctuations Quantiques/Quantum Fluctuations* (p. 351)

...



Circuit Quantum Electrodynamics

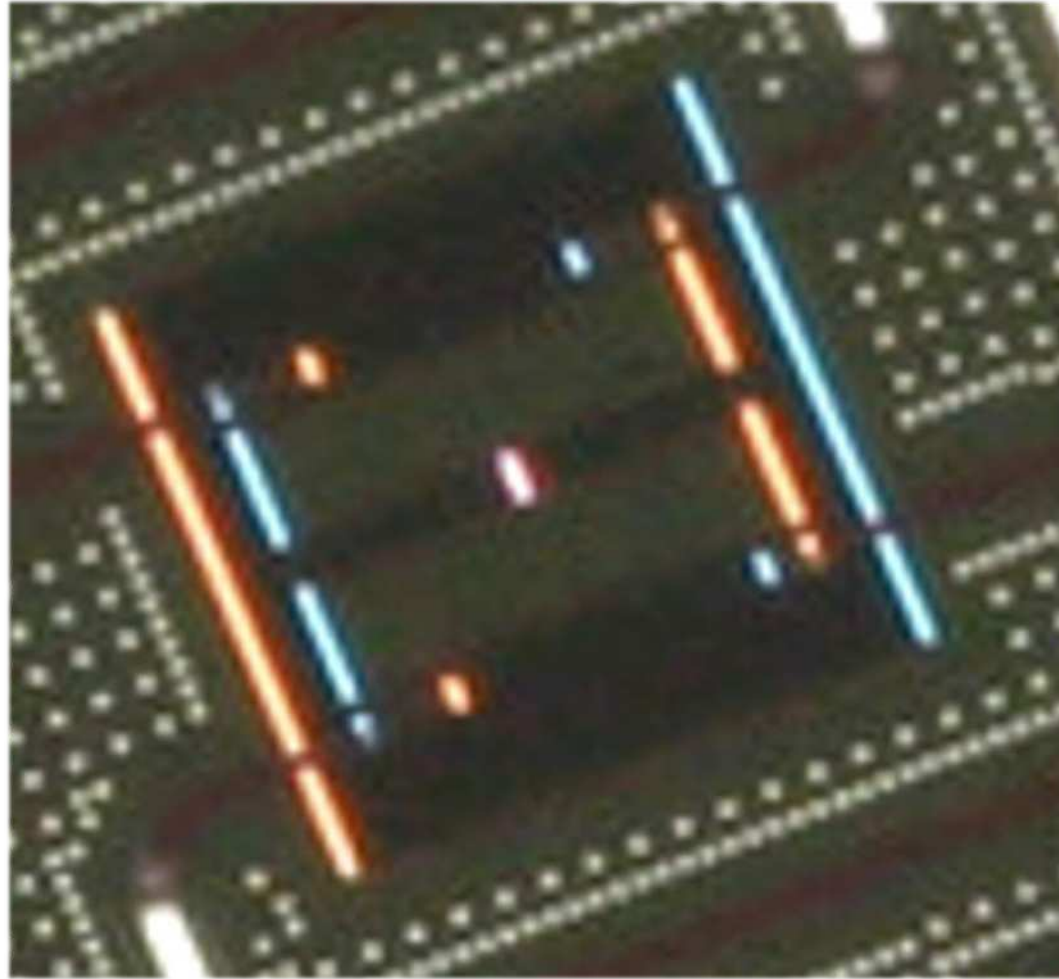




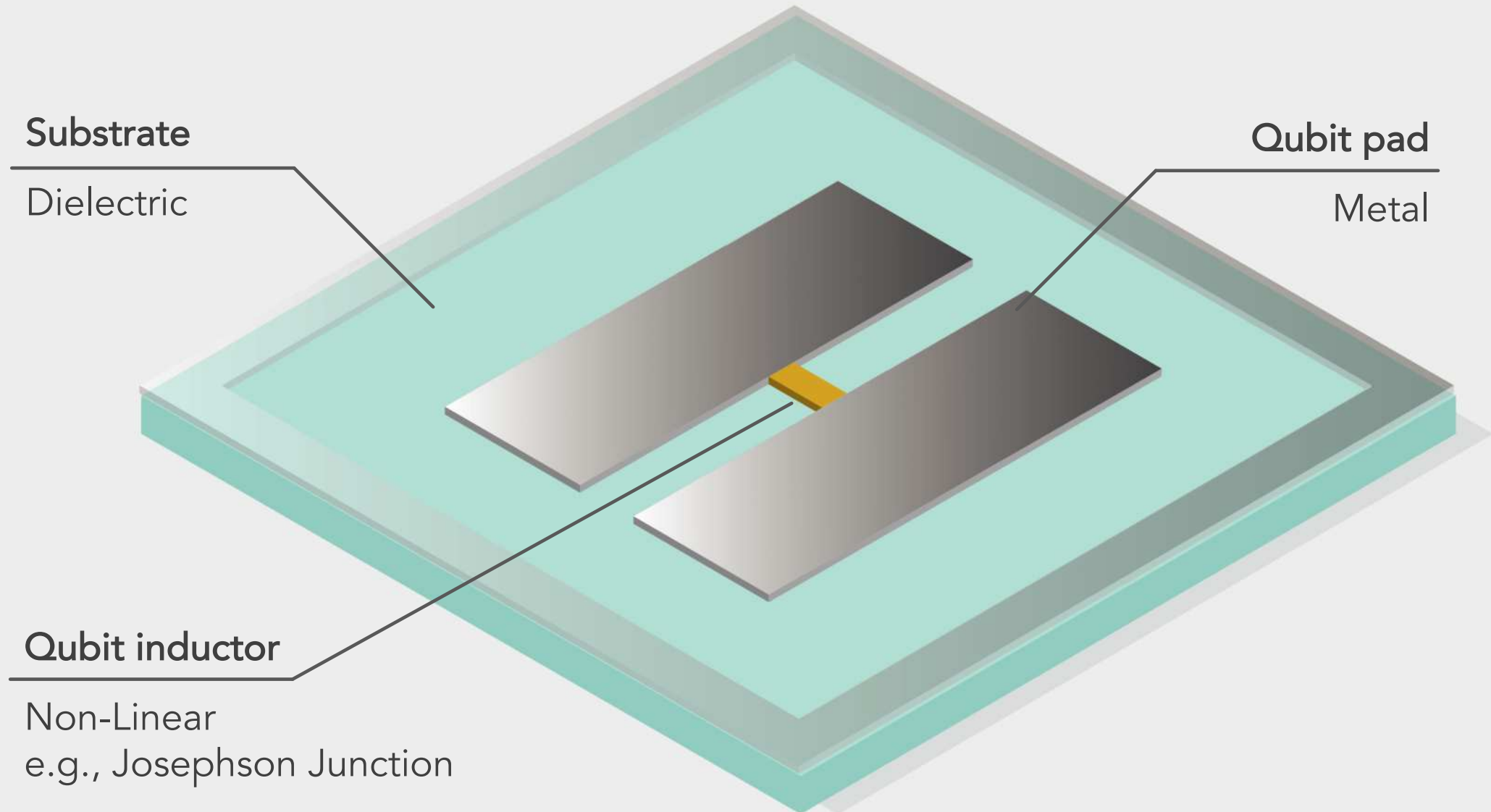
To do quantum,
start with classical

Classical oscillator

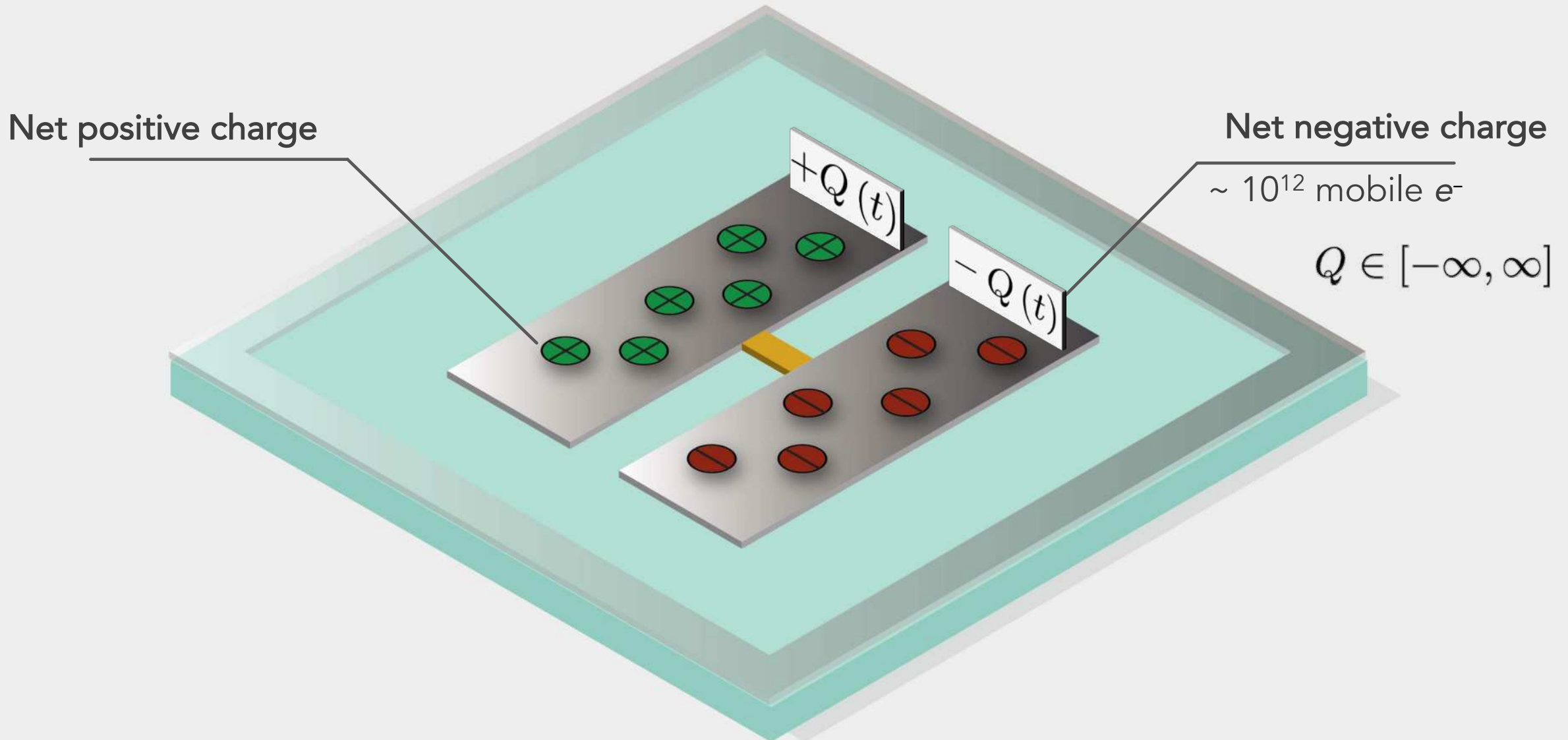
Transmon qubit



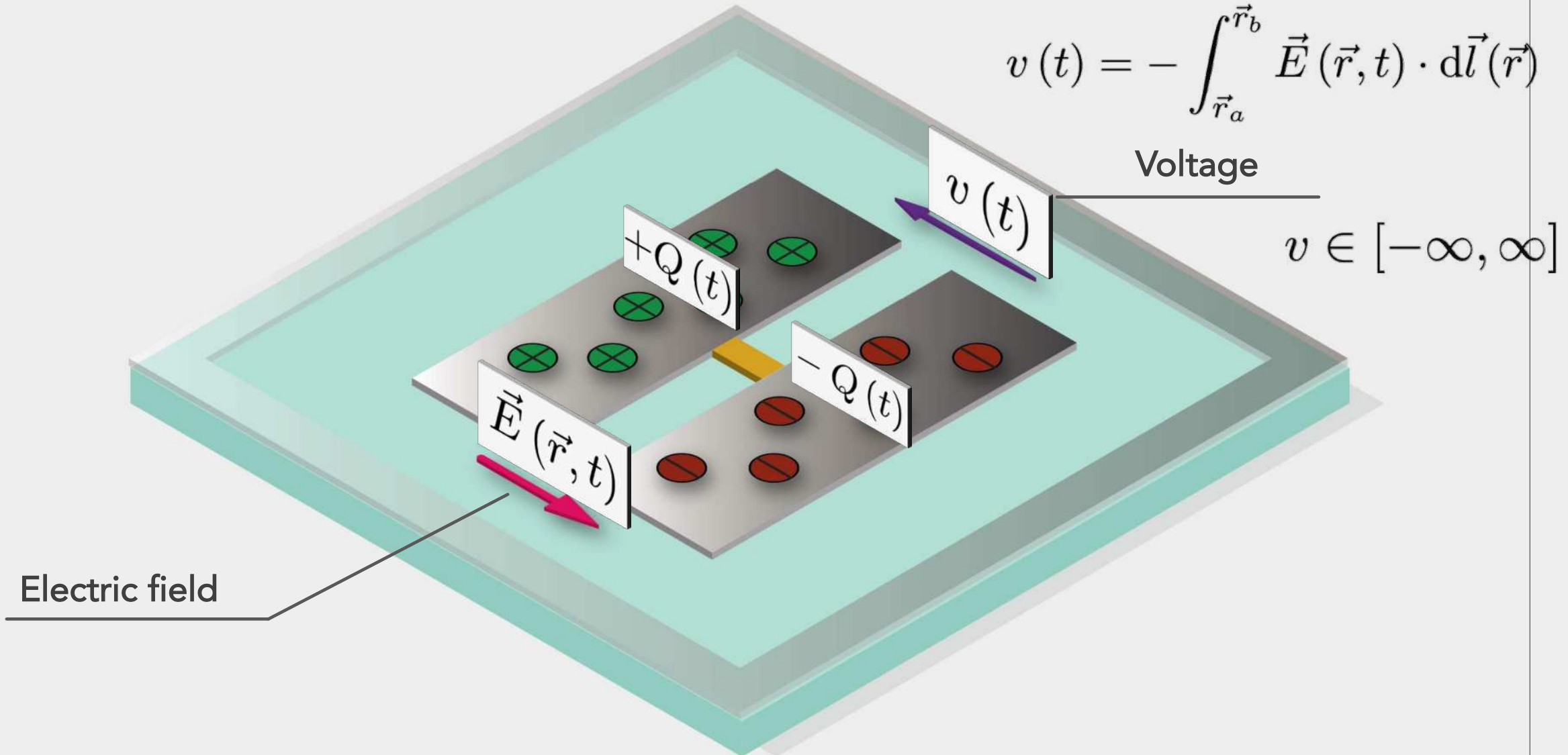
Transmon qubit



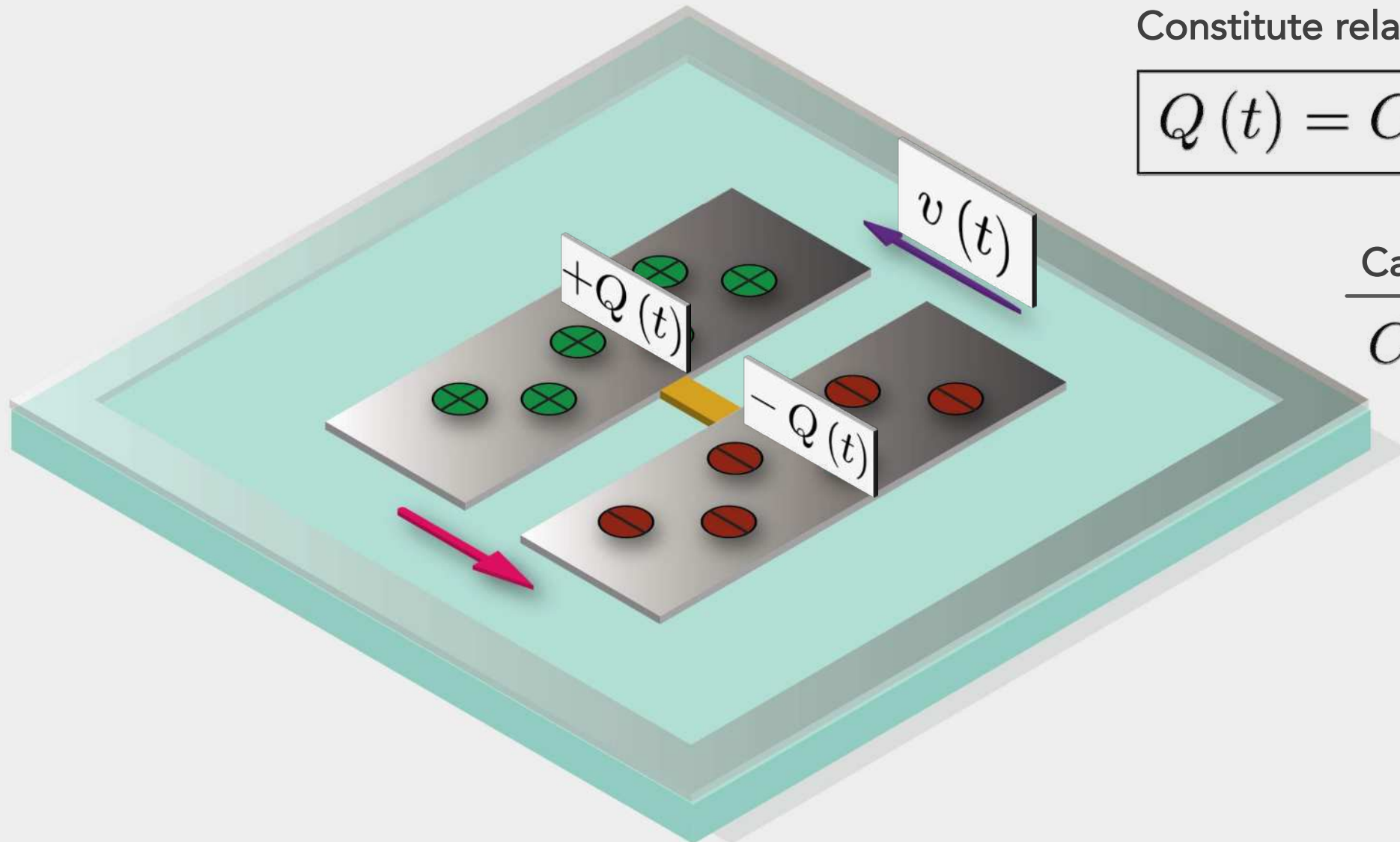
Transmon qubit: charge



Electric field and voltage



Charge and capacitance

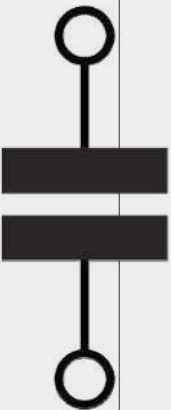


Constitute relationship

$$Q(t) = C v(t)$$

Capacitance

$$C \in (0, \infty)$$



For a good discussion, see "The Feynman Lectures on Physics Vol. II Ch. 22: AC Circuits." Caltech.

Charge and current

Conservation of charge

Universal relationship

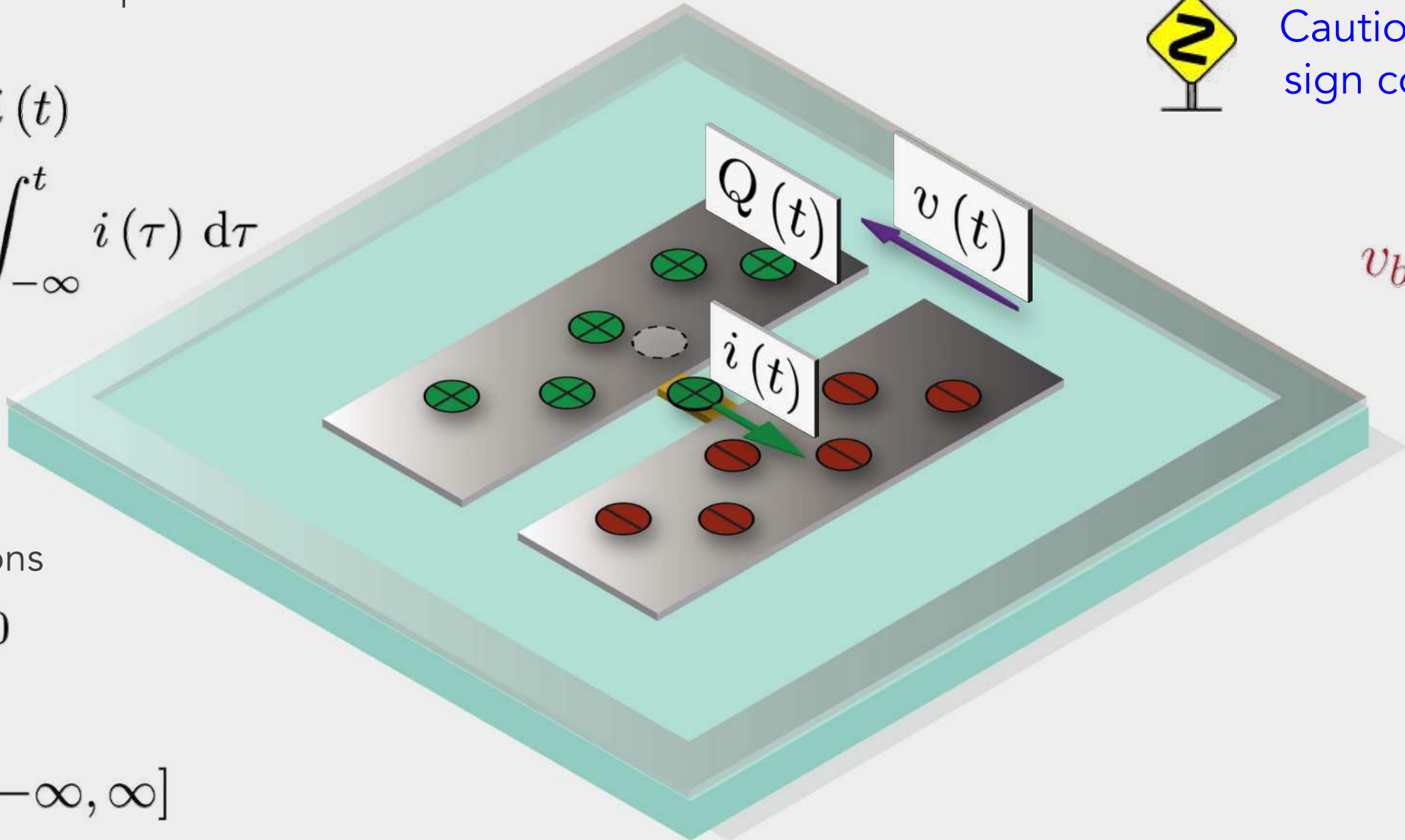
$$\frac{d}{dt}Q(t) = i(t)$$

$$Q(t) = \int_{-\infty}^t i(\tau) d\tau$$

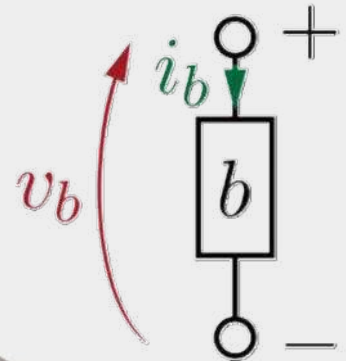
Initial conditions

$$Q(-\infty) = 0$$

$$i \in [-\infty, \infty]$$



Caution: Passive sign convention



Magnetic flux and inductance

Faraday's law of induction

Universal relationship

$$\Phi(t) = \int_{-\infty}^t v(\tau) d\tau$$

$$\frac{d}{dt}\Phi(t) = v(t)$$

Initial conditions

$$\Phi(-\infty) = 0$$

Magnetic flux

For kinetic inductors,
~98% of qubit inductive energy is not
in stored magnetic fields, but in
kinetic inductance

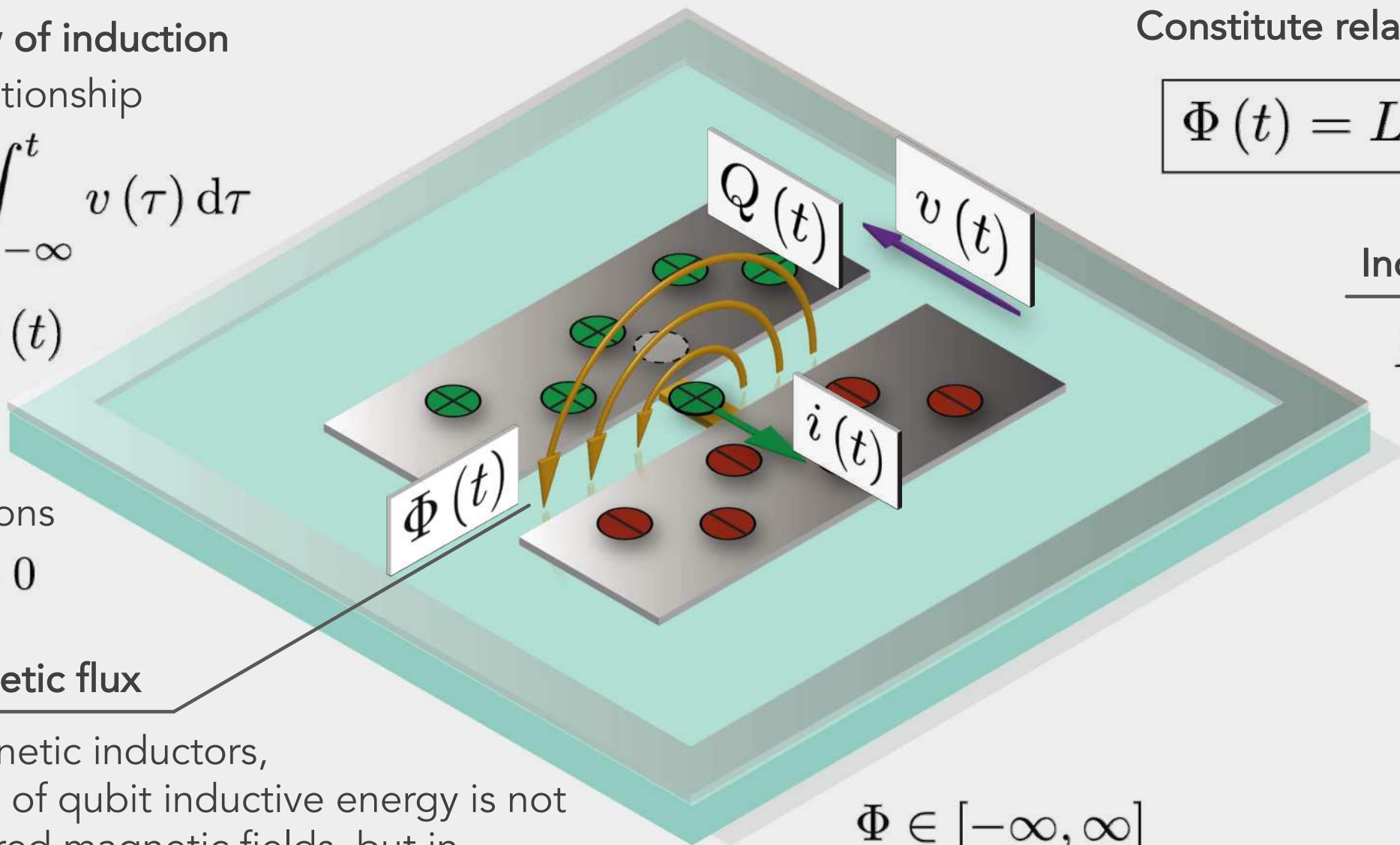
Constitute relationship

$$\Phi(t) = Li(t)$$

Inductance

$$L \in (0, \infty)$$

$$\Phi \in [-\infty, \infty]$$



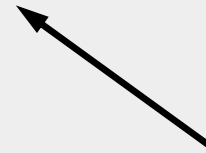
Power and energy

Universal

$$\frac{d}{dt}\mathcal{E}(t) = p(t) \equiv v(t)i(t)$$

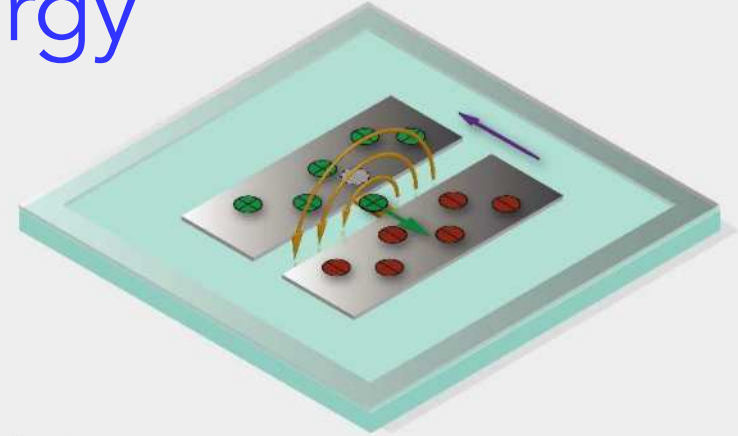


Energy stored *in*
(delivered *to*)
component



Instantaneous
power flowing
to component

$$\mathcal{E}(t) = \int_{-\infty}^t p(\tau) d\tau$$



$$\mathcal{E}_{\text{cap}}(\dot{\Phi}) = \frac{1}{2}C\dot{\Phi}^2$$

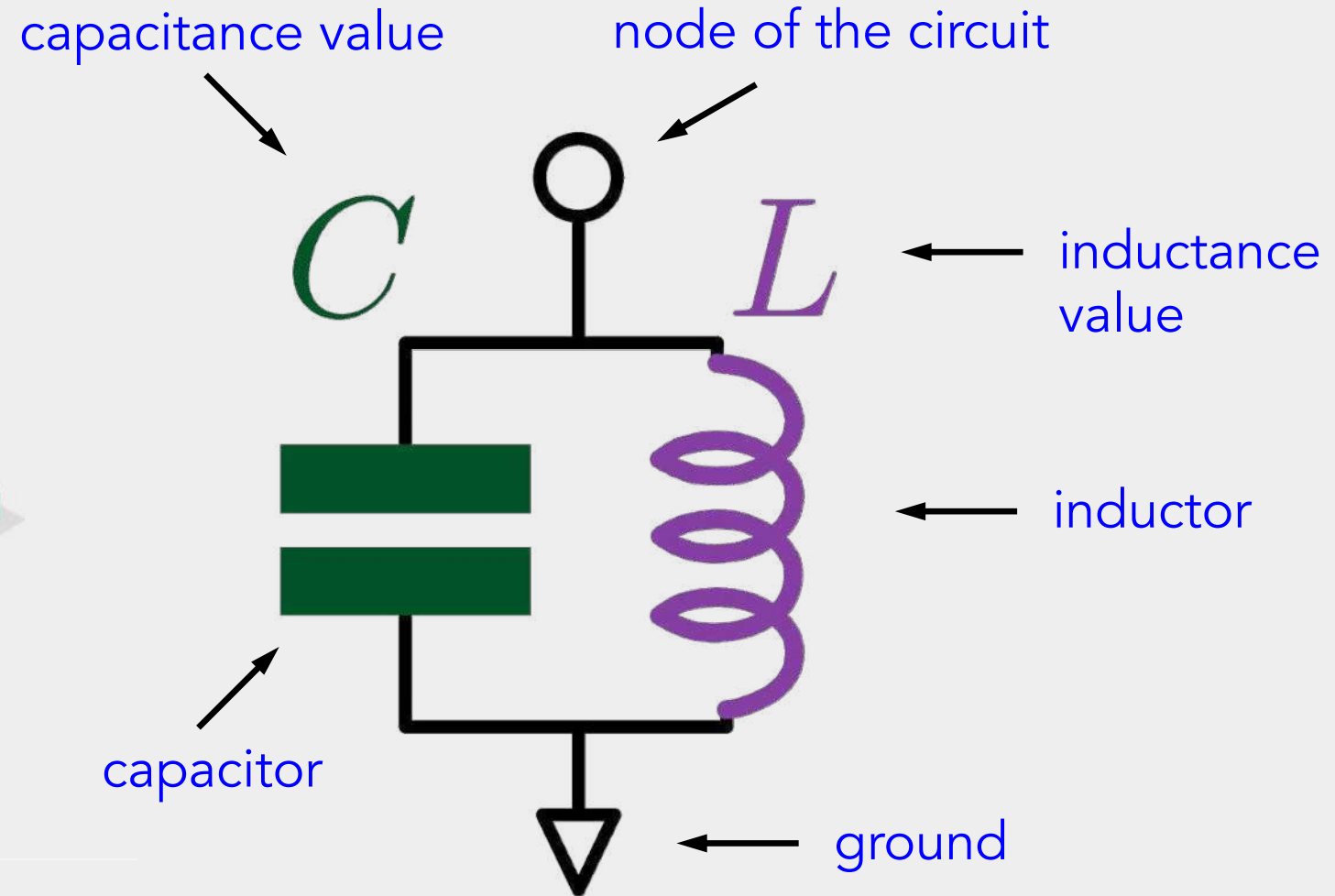
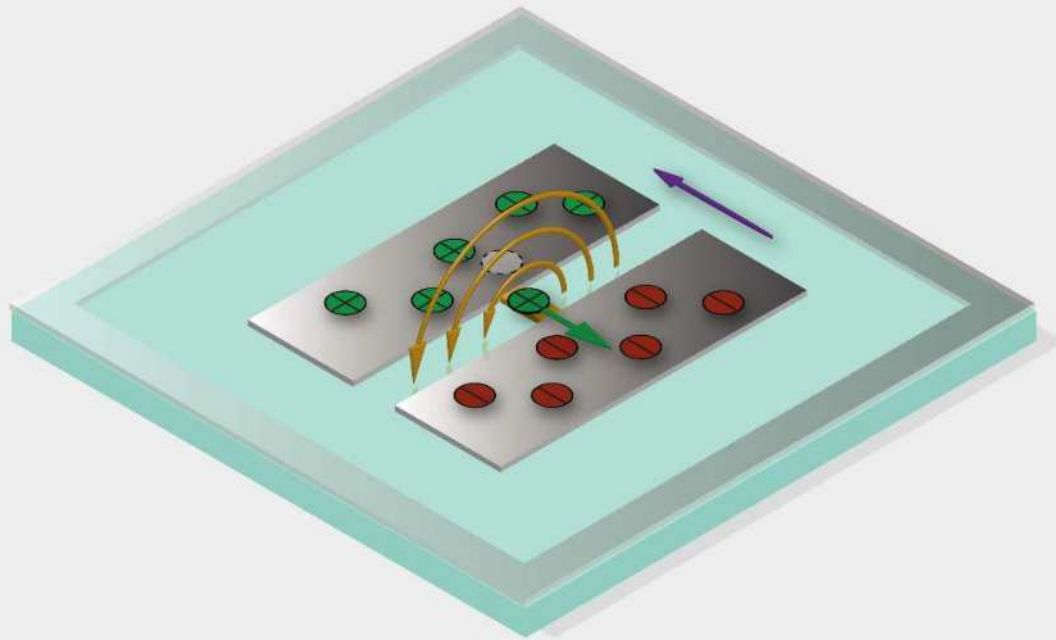
$$\mathcal{E}_{\text{cap}}(Q) = \frac{Q^2}{2C}$$



$$\mathcal{E}_{\text{ind}}(\Phi) = \frac{\Phi^2}{2L}$$

$$\mathcal{E}_{\text{ind}}(\dot{Q}) = \frac{1}{2}L\dot{Q}^2$$

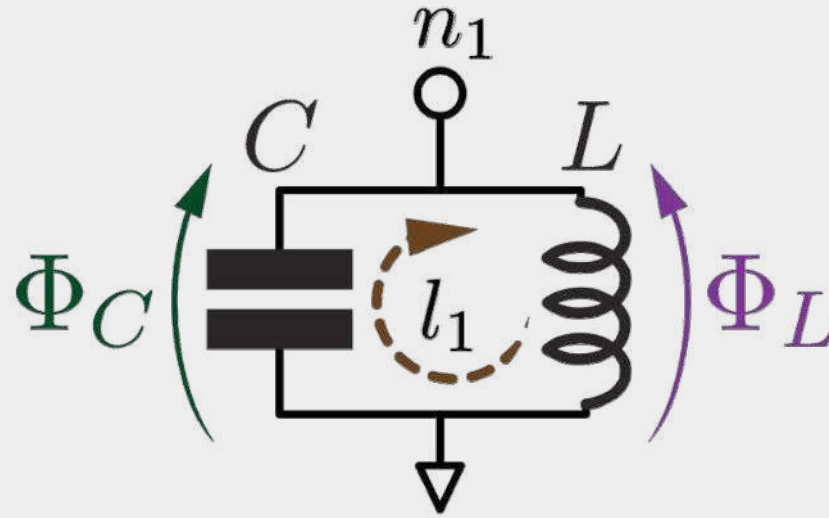
Electromagnetic oscillator



Kirchhoff's network laws*

Conservation of charge
Kirchhoff's current law

$$\sum_{b \in \text{node}} \pm \dot{Q}_b(t) = 0$$



Faraday's law of induction
Kirchhoff's voltage law

$$\sum_{b \in \text{loop}} \pm \dot{\Phi}_b(t) = 0$$

$$n_1 : \quad \dot{Q}_C + \dot{Q}_L = 0$$

$$C\ddot{\Phi} + L^{-1}\Phi = 0$$

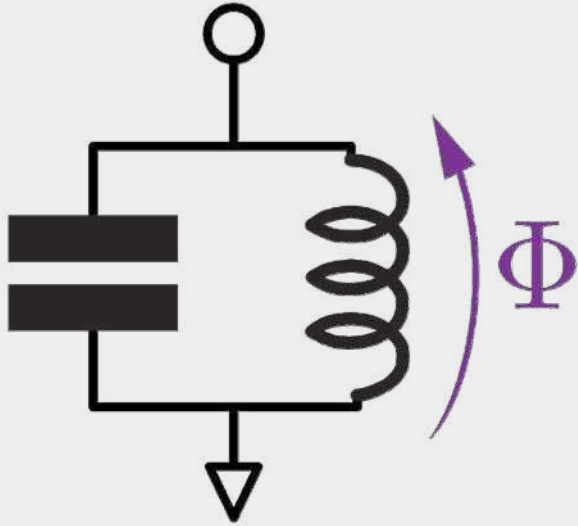
$$l_1 : \quad \dot{\Phi}_C - \dot{\Phi}_L = 0$$

$$\Phi_C = \Phi_L$$



As we will see later, for the Lagrangian description in flux basis, KVL acts as a set of holonomic constraints and KCL as the equations of motion

Oscillator analogy



$$C\ddot{\Phi} + L^{-1}\Phi = 0$$

$$\ddot{\Phi} = -\omega_0^2\Phi, \quad \omega_0 = \frac{1}{\sqrt{LC}}$$

Resonance
frequency

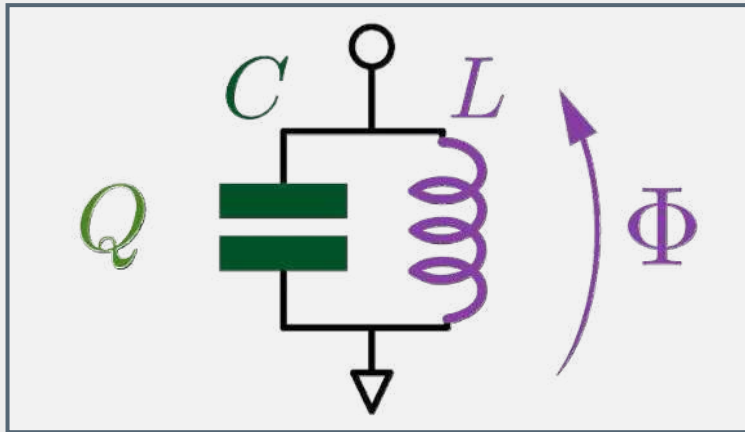


$\dot{\Phi}(t)$ Velocity

$\Phi(t)$ Deviation from equilibrium

Equilibrium position
 $x = 0, \quad \Phi = 0$

The LC classical harmonic oscillator

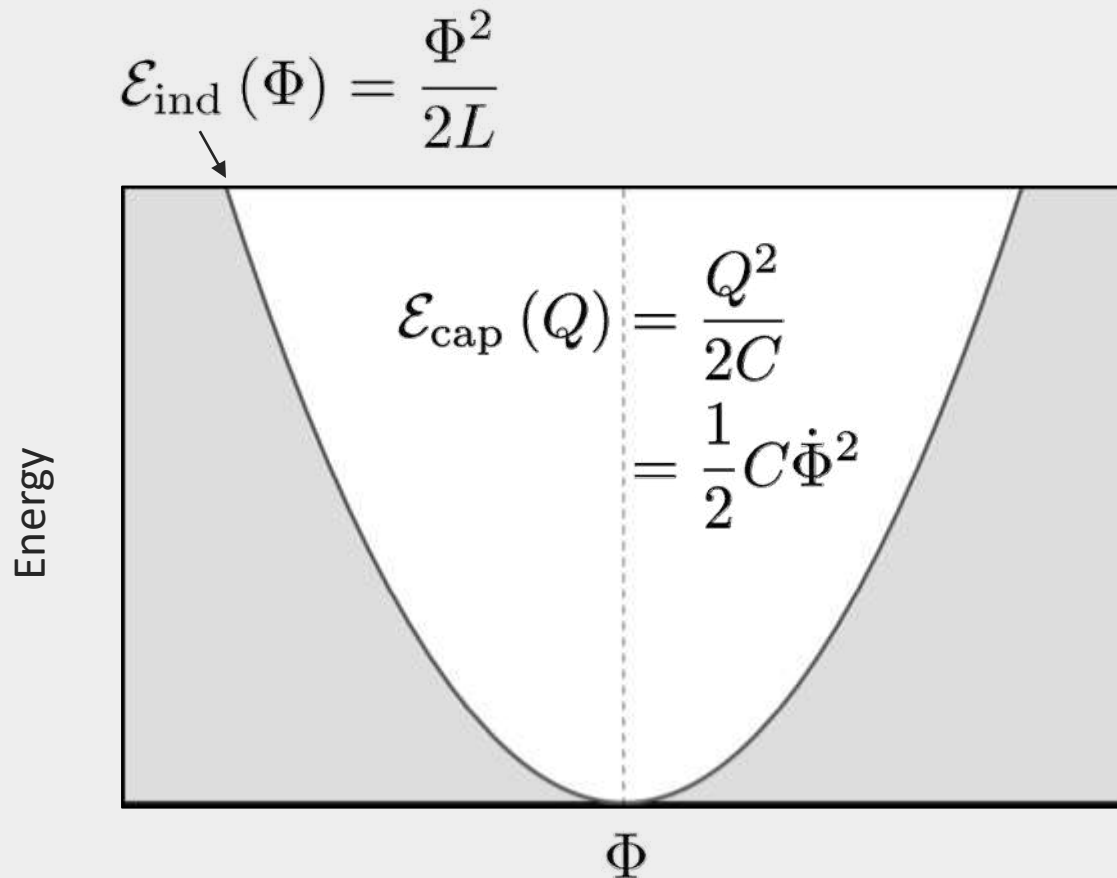


Position: $\Phi \mapsto x$
 Mass: $C \mapsto m$

Momentum: $Q \mapsto p$
 Spring constant: $L^{-1} \mapsto k$

$$\mathcal{H}(\Phi, Q) = \frac{Q^2}{2C} + \frac{\Phi^2}{2L}$$

$$\omega_0^2 = \frac{1}{LC} \quad Z_0 = \frac{L}{C}$$



Graph: Minev; Spring-mass: Svjo

“It is by *logic* that we prove,
but by *intuition* that we discover.”

Henri Poincaré

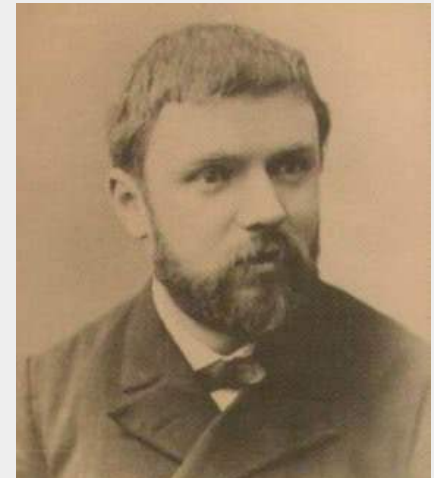


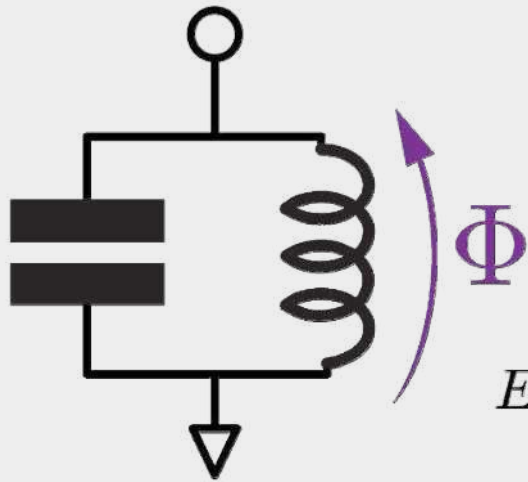
Photo by Eugène Pirou

Hamiltonian dynamics and phase space

$$\mathcal{H}(\Phi, Q) = \frac{Q^2}{2C} + \frac{\Phi^2}{2L}$$

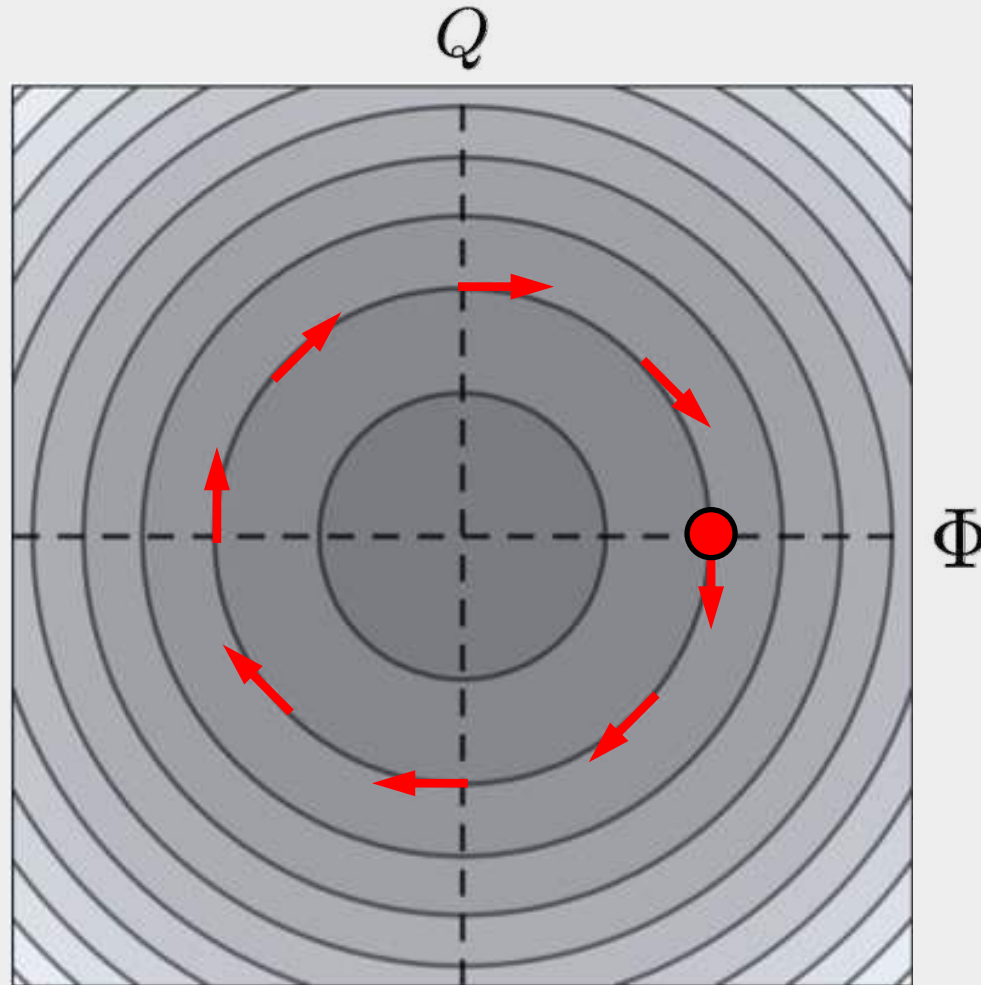
with $L = C = 1$

$$E = \hbar\omega_0 \left(n + \frac{1}{2} \right) = \frac{1}{2} (Q^2 + \Phi^2)$$



$$E = \frac{3\hbar\omega_0}{2}$$

$$E = \frac{\hbar\omega_0}{2}$$



$$\dot{\Phi} = +\frac{\partial \mathcal{H}}{\partial Q} = +Q$$

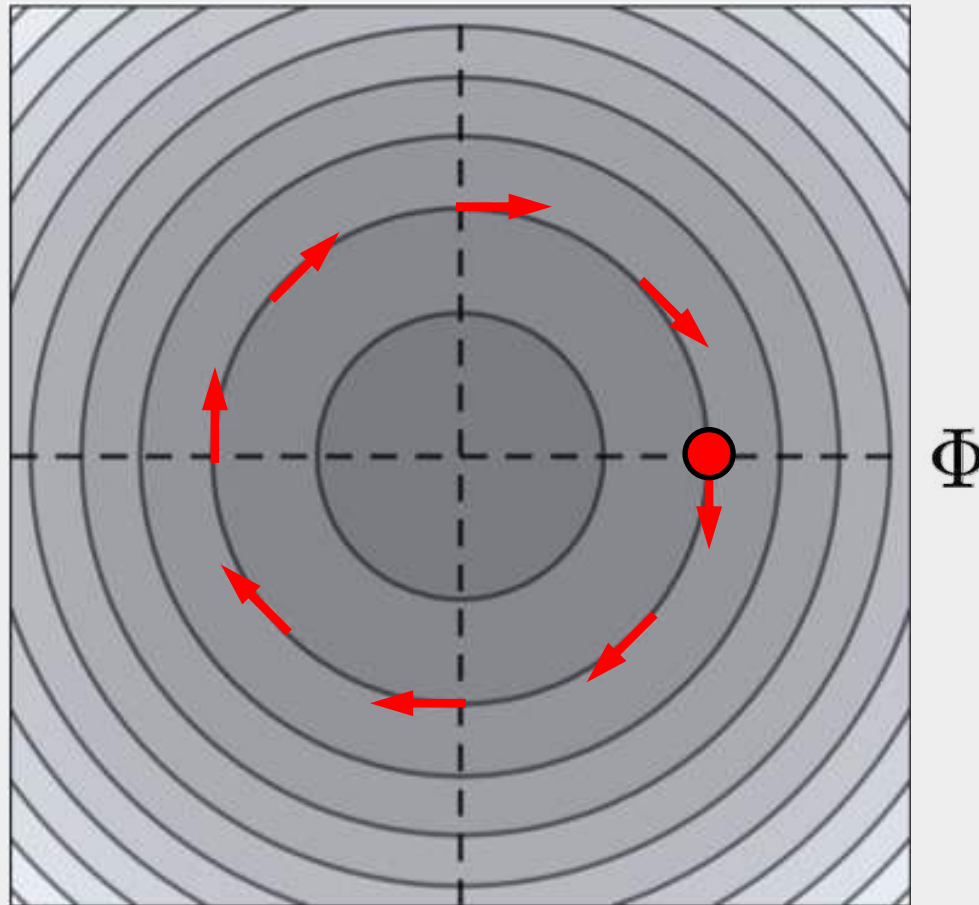
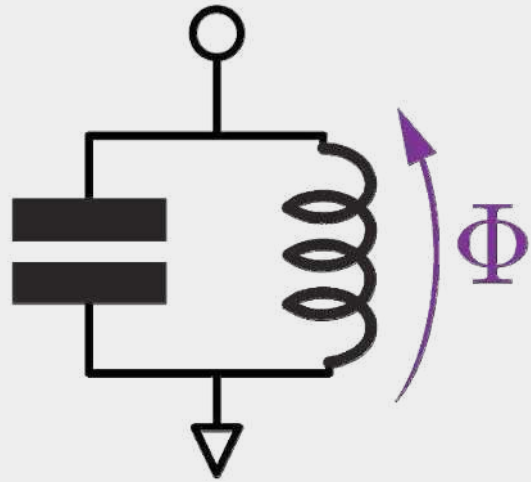
$$\dot{Q} = -\frac{\partial \mathcal{H}}{\partial \Phi} = -\Phi$$

Complex action-angle variable

$$\mathcal{H}(\Phi, Q) = \frac{Q^2}{2C} + \frac{\Phi^2}{2L} (\alpha^* \alpha + \alpha \alpha^*)$$

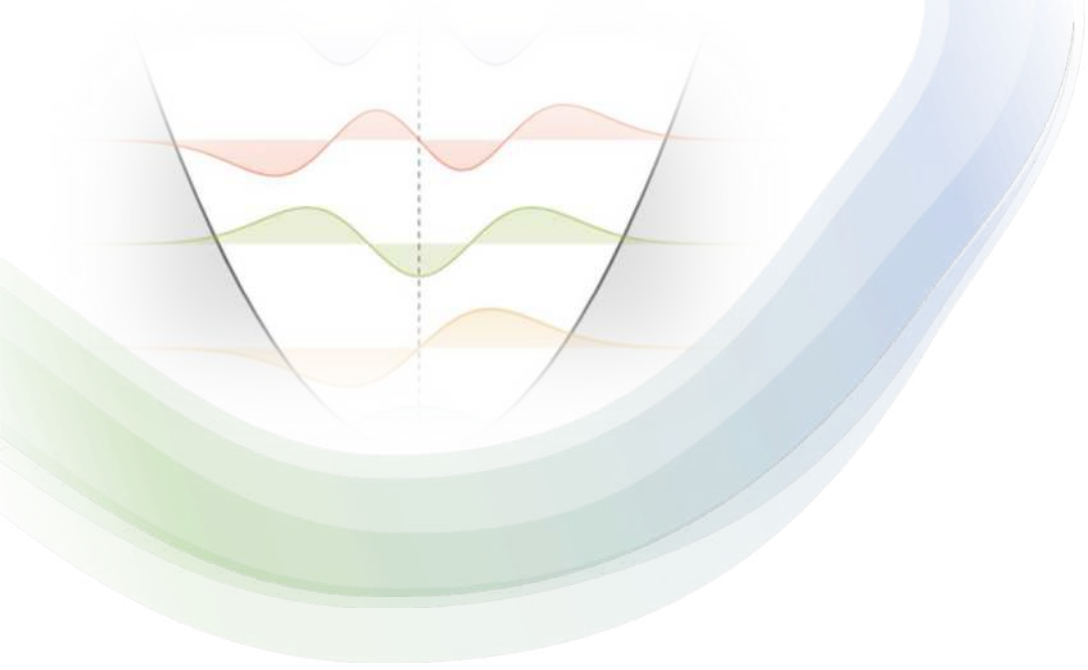
$$E = \hbar \omega_0 \left(n + \frac{1}{2} \right)$$

$$\alpha(t) = \sqrt{\frac{1}{2\hbar Z}} [\Phi(t) + iZQ(t)]$$



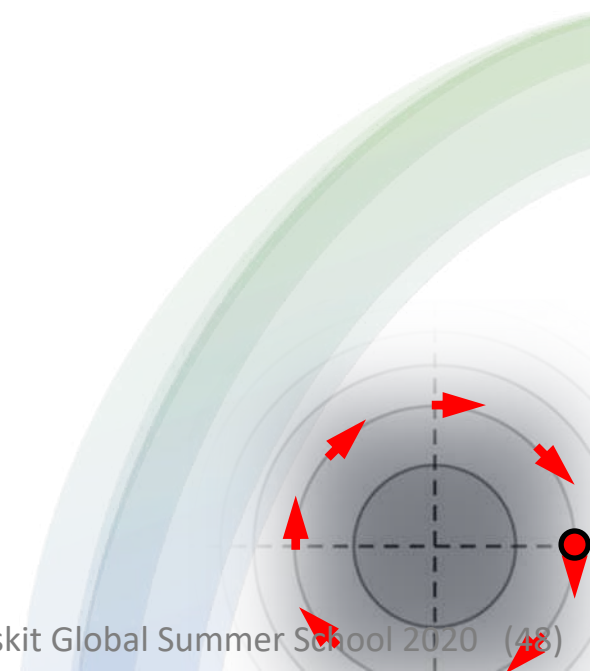
Classical analog of the bosonic ladder operator

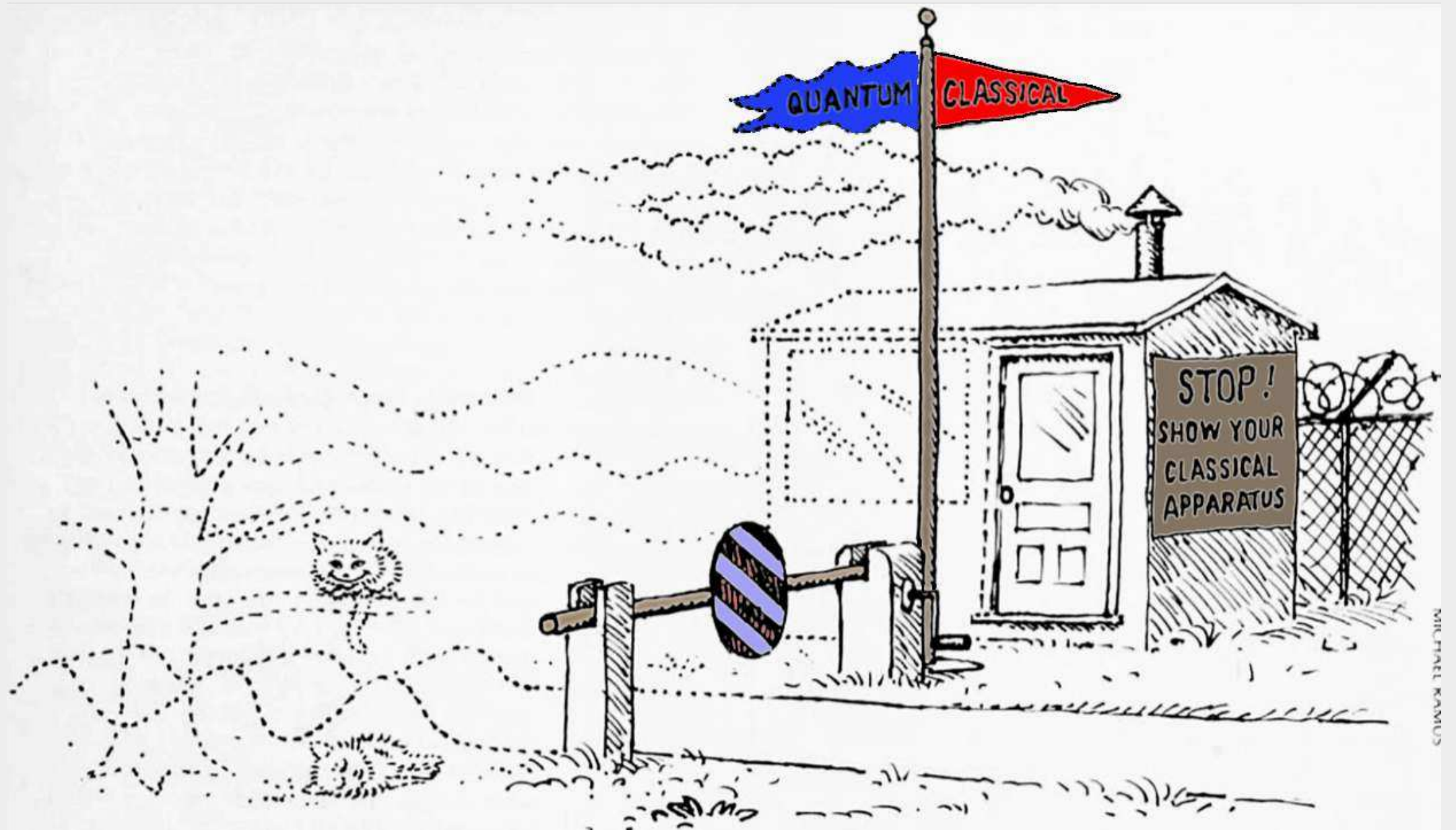
$$\alpha(t) = \alpha(0) e^{-i\omega_0 t}$$



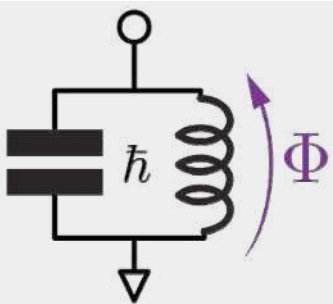
Unveiling the quantum

Quantum harmonic oscillator





Drawing: Zurek, Physics Today (1991)



The classical and quantum oscillator

Classical

Quantum

$$\Phi(t) \mapsto \hat{\Phi}$$

$$Q(t) \mapsto \hat{Q}$$

$$\mathcal{H}(\Phi, Q) \mapsto \hat{H}(\hat{\Phi}, \hat{Q})$$

$$\{\Phi, Q\} = 1 \mapsto [\hat{\Phi}, \hat{Q}] = i\hbar \hat{1}$$

$$\{\alpha, \alpha^*\} = 1/(i\hbar) \mapsto [\hat{a}, \hat{a}^\dagger] = \hat{1}$$

$$\{A, B\} = \frac{\partial A}{\partial \Phi} \frac{\partial B}{\partial Q} - \frac{\partial B}{\partial \Phi} \frac{\partial A}{\partial Q}$$

$$[\hat{A}, \hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A}$$

If I knew what I was doing, it
wouldn't be called research.

Albert Einstein
See Hawken *et al.* (2010)

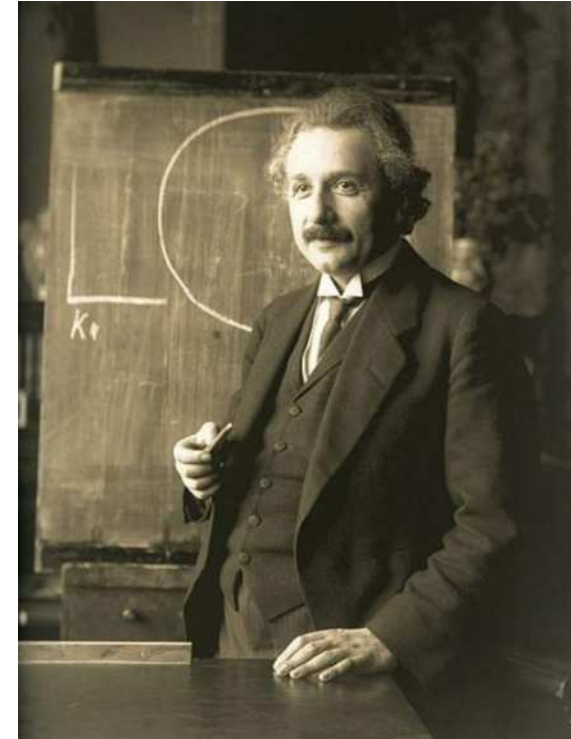
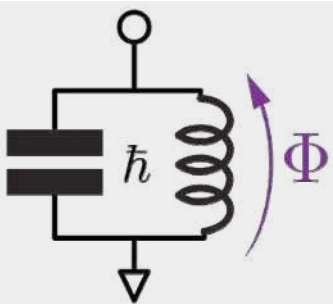
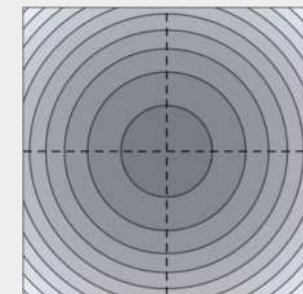


Photo: F. Schmutzer



The classical and quantum oscillator



Classical

Quantum

Hamiltonian

$$\begin{aligned}\mathcal{H} &= \frac{Q^2}{2C} + \frac{\Phi^2}{2L} \\ &= \frac{1}{2} \hbar \omega_0 (\alpha^* \alpha + \alpha \alpha^*)\end{aligned}$$

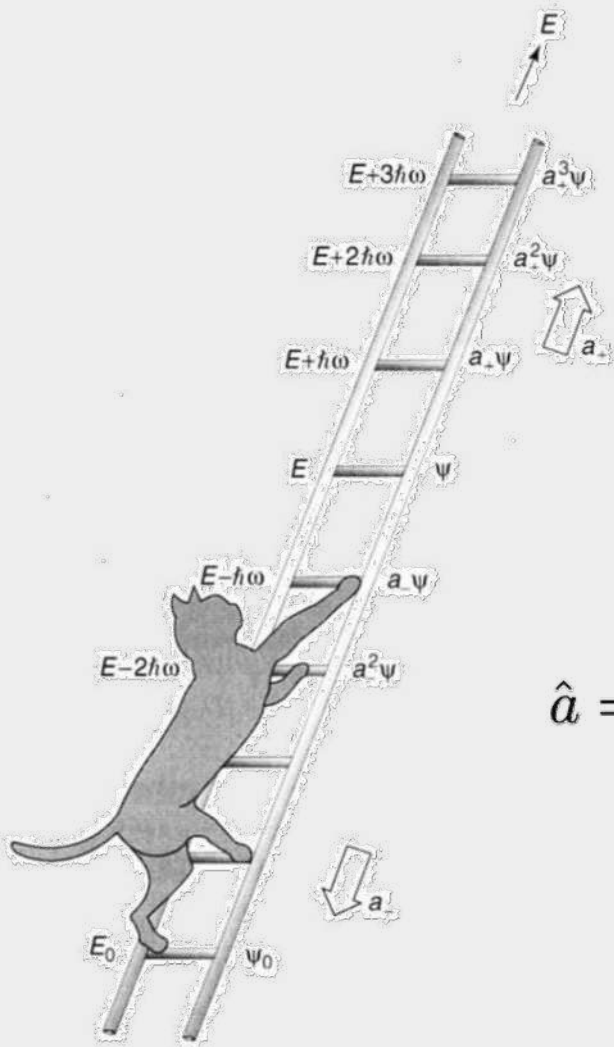
$$\begin{aligned}\hat{H} &= \frac{\hat{\Phi}^2}{2L} + \frac{\hat{Q}^2}{2C} \\ &= \hbar \omega_0 (\hat{a}^\dagger \hat{a} + \hat{a} \hat{a}^\dagger)\end{aligned}$$

Phase space

$$\begin{aligned}\alpha(t) &= \sqrt{\frac{1}{2\hbar Z}} [\Phi(t) + iZQ(t)] \\ \alpha(t) &= \alpha(0) e^{-i\omega_0 t} \\ \Phi(t) &= \sqrt{\frac{\hbar Z}{2}} (\alpha^*(t) + \alpha(t)) \\ Q(t) &= i\sqrt{\frac{\hbar}{2Z}} (\alpha^*(t) - \alpha(t))\end{aligned}$$

$$\begin{aligned}\hat{a} &= \sqrt{\frac{1}{2\hbar Z}} (\hat{\Phi} + iZ\hat{Q}) \\ \hat{a}(t) &= \hat{a}(0) e^{-i\omega_0 t} \quad (\text{Heisenberg picture}) \\ \hat{\Phi} &= \Phi_{\text{ZPF}} (\hat{a}^\dagger + \hat{a}) \\ \hat{Q} &= iQ_{\text{ZPF}} (\hat{a}^\dagger - \hat{a})\end{aligned}$$

Ladder operators and matrix representation



annihilation

$$\hat{a} |0\rangle = 0$$

$$\hat{a} |1\rangle = \sqrt{1} |0\rangle$$

creation

$$\hat{a}^\dagger |0\rangle = \sqrt{1} |1\rangle$$

$$\hat{a}^\dagger |1\rangle = \sqrt{2} |2\rangle$$

general hopping

$$\hat{a} |n\rangle = \sqrt{n} |n-1\rangle$$

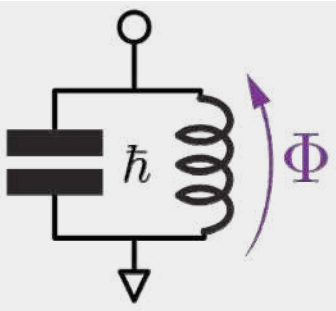
$$\hat{a}^\dagger |n\rangle = \sqrt{n+1} |n+1\rangle$$

$$\hat{a} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & \sqrt{2} & 0 \\ 0 & 0 & 0 & \sqrt{3} \\ 0 & 0 & 0 & \ddots \end{pmatrix}$$

$$\hat{a}^\dagger = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & \sqrt{2} & 0 & 0 \\ 0 & 0 & \sqrt{3} & \ddots \end{pmatrix}$$

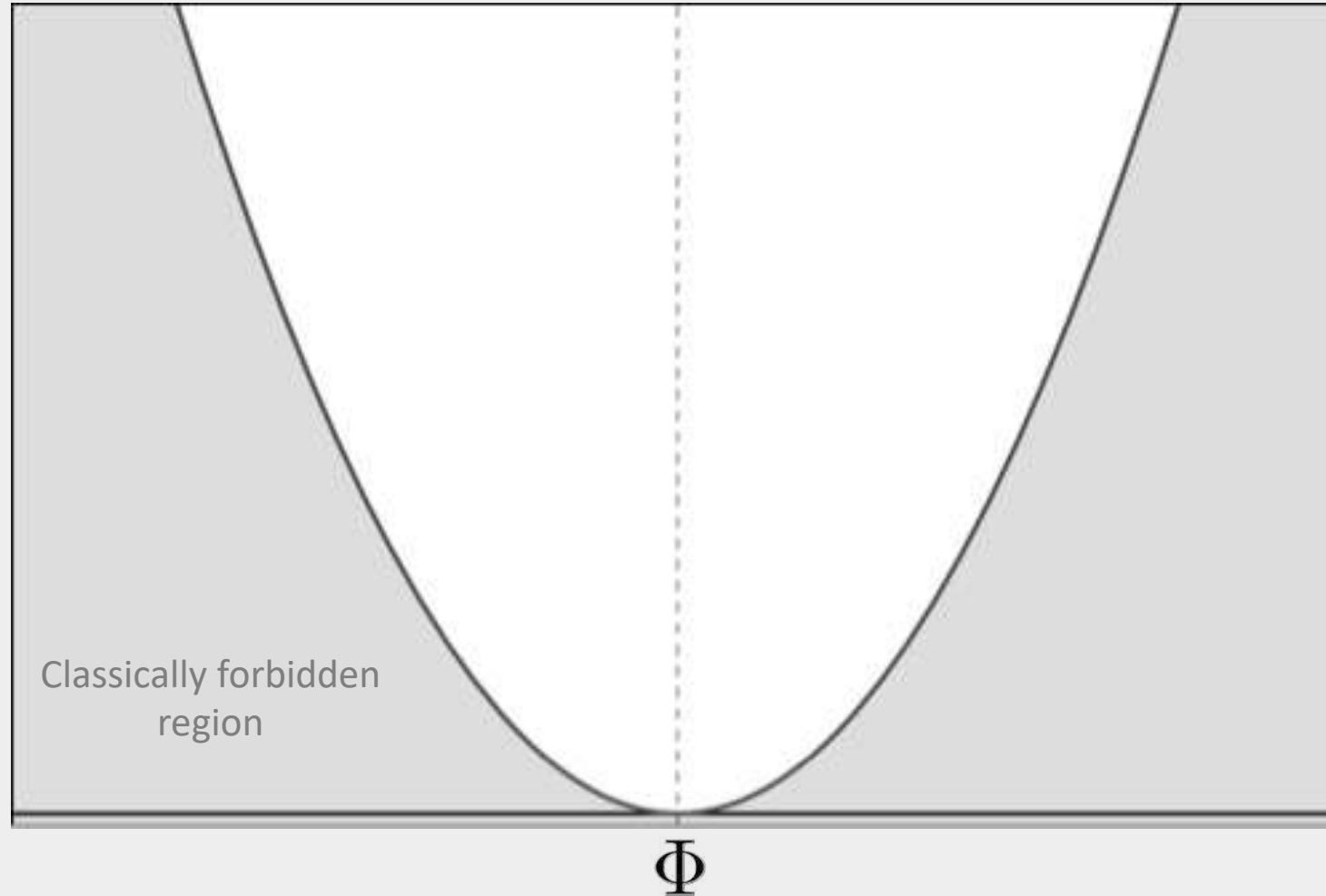
$$\hat{a}^\dagger \hat{a} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & \ddots \end{pmatrix}$$

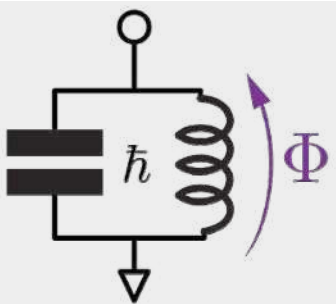
Wavefunctions of the quantum oscillator



$$\mathcal{E}_{\text{ind}}(\Phi) = \frac{\Phi^2}{2L}$$

Energy



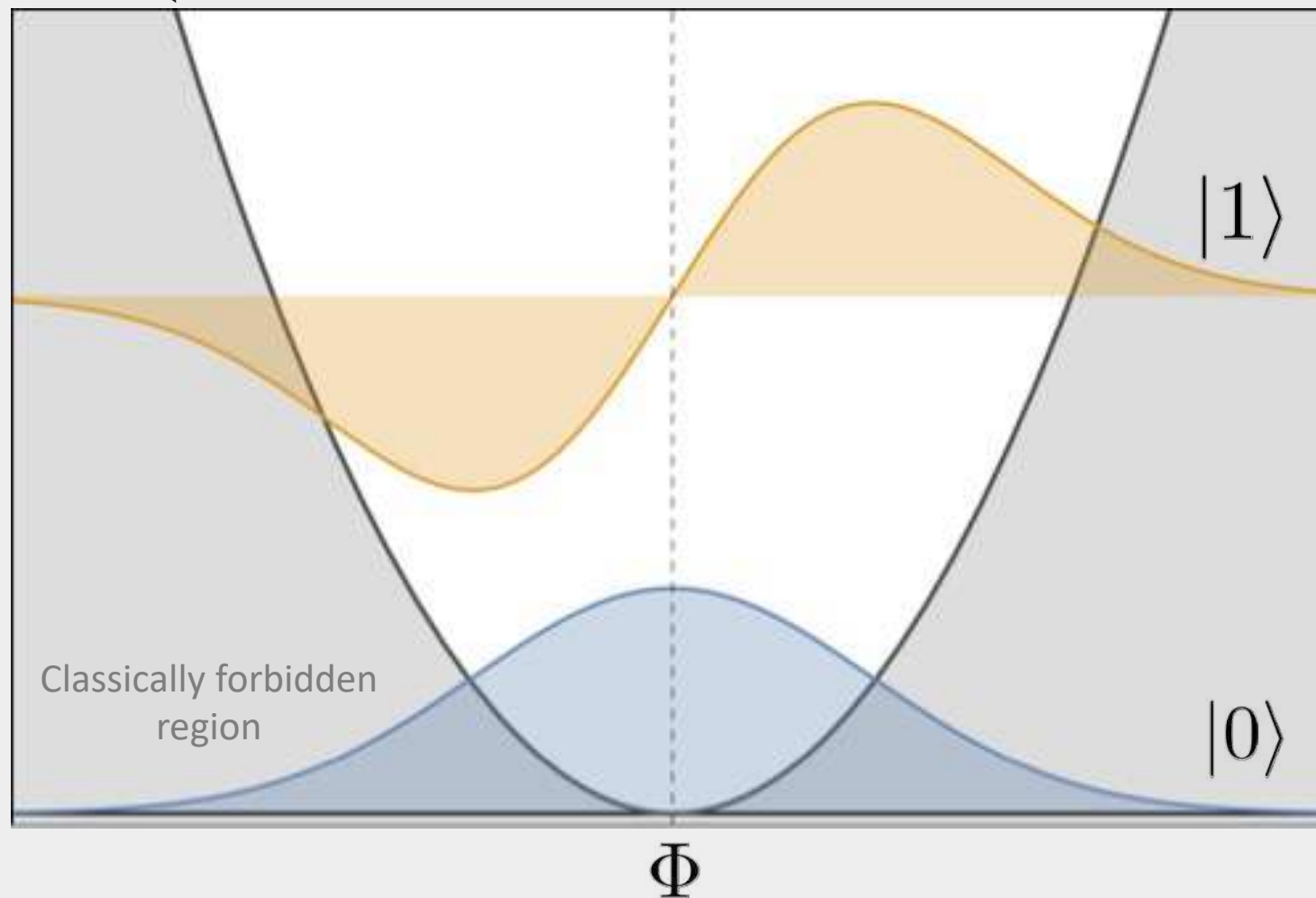


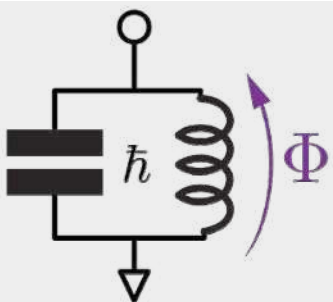
Wavefunctions of the quantum oscillator

$$\mathcal{E}_{\text{ind}}(\Phi) = \frac{\Phi^2}{2L}$$

$$\psi_n(\Phi) \equiv \langle \Phi | n \rangle$$

Energy /
Scaled
wavefunction
amplitude



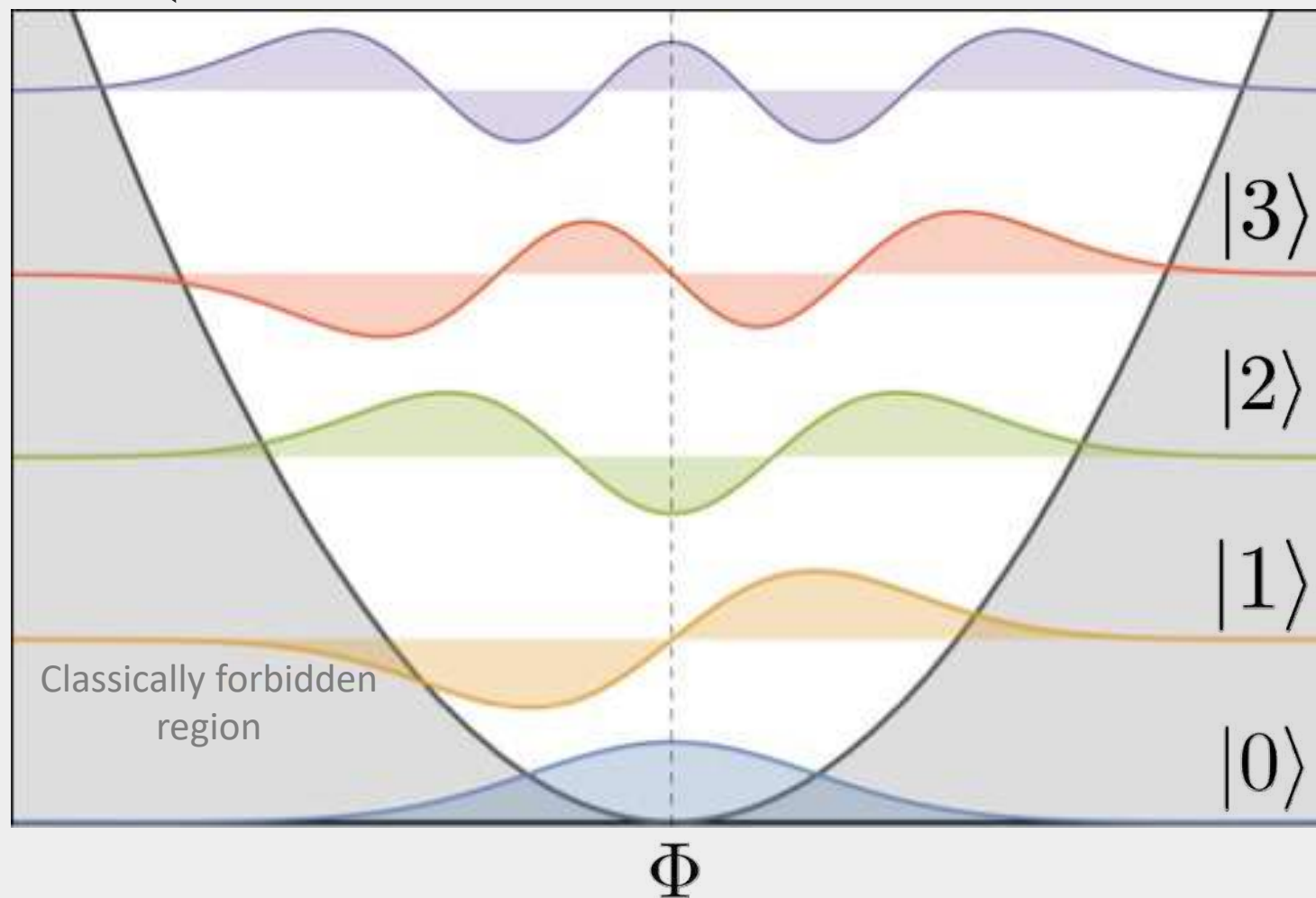


Wavefunctions of the quantum oscillator

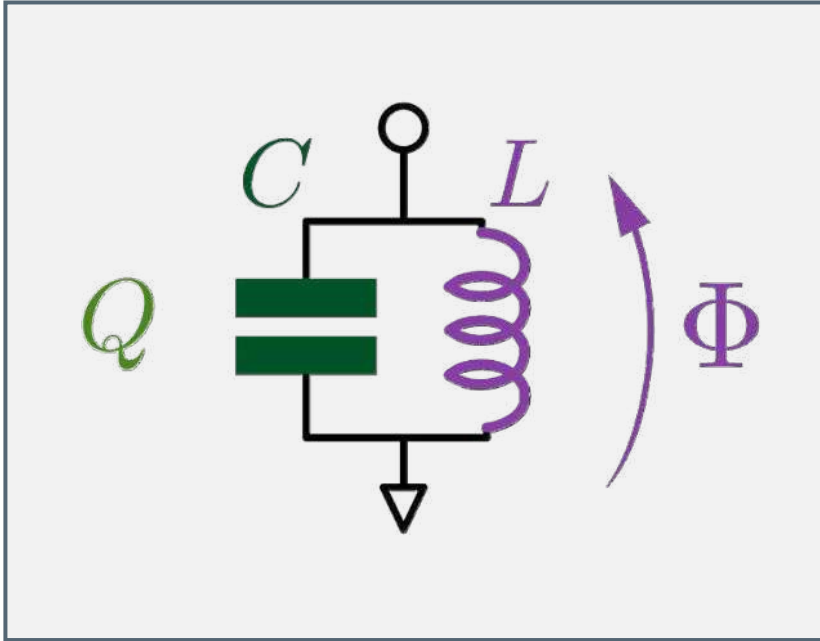
$$\mathcal{E}_{\text{ind}}(\Phi) = \frac{\Phi^2}{2L}$$

$$\psi_n(\Phi) \equiv \langle \Phi | n \rangle$$

Energy /
Scaled
wavefunction
amplitude



Pop-up question



The flux and charge operators are Hermitian observables.

How can some expectations, such as

$$\langle 0 | \hat{\Phi} \hat{Q} | 0 \rangle = \frac{1}{4} i ,$$

be imaginary?

Or, others be negative...?

Advanced questions



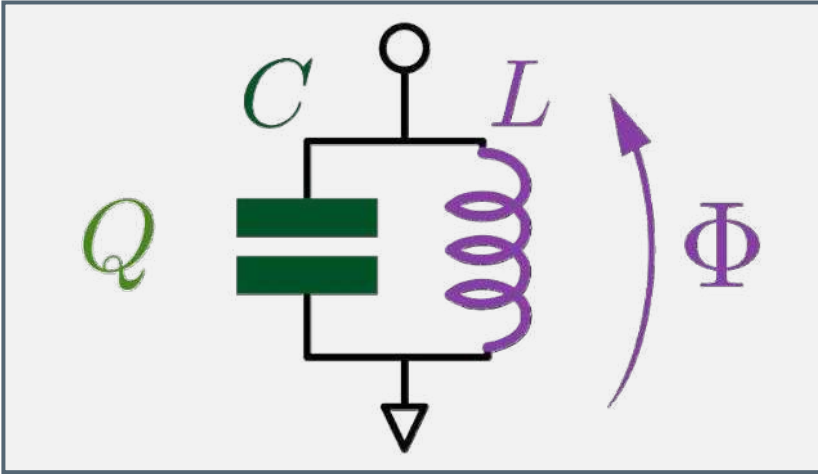
1. What is $\hat{\Phi}(t)$ in terms of $\hat{a}$ in the Heisenberg picture?
2. **Autocorrelation.** Find the autocorrelation operator $\hat{C}_{\Phi\Phi}(t) = \hat{\Phi}(t) \hat{\Phi}(0)$. What frequency components does it have?
 - (a) What is the flux autocorrelation expectation value $C_{\Phi\Phi}(t) = \langle n | \hat{\Phi}(t) \hat{\Phi}(0) | n \rangle$ for the Fock state $|n\rangle$? Is it real? Why not? What is the frequency spectrum?
 - (b) Repeat for a coherent state $|\alpha\rangle$.



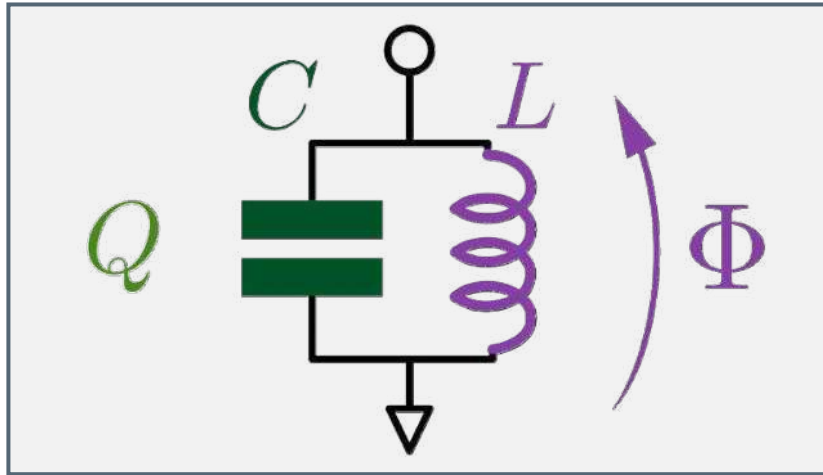
3. **Thermal state.** The thermal state is $\hat{\rho}_{\text{th}} = \exp[-\beta \hat{a}^\dagger \hat{a}] / \text{Tr}[\exp(-\beta \hat{a}^\dagger \hat{a})]$, where $\beta = \hbar\omega_0/k_B T$, and T is the temperature of the oscillator.
 - (a) What is the mean and variance of the flux $\hat{\Phi}$ and charge $\hat{Q}$ operators?
 - (b) How does the frequency spectrum of the autocorrelation $\langle \hat{C}_{\Phi\Phi}(t) \rangle = \text{Tr}[\hat{\rho}_{\text{th}} \hat{C}_{\Phi\Phi}(t)]$ change when with that for the state $|n\rangle$ and $|\alpha\rangle$?
 - (c) The spectrum is not symmetric in frequency. How can you interpret that positive and negative frequencies have different weights? How is this related to absorption and emission of the oscillator? (Consider Fermi's golden rule).
 - (d) What happens to the spectrum in the limit of high temperature, $k_B T \gg \hbar\omega_0$? How about low temperature, $k_B T \ll \hbar\omega_0$?
 - (e) How do the above conclusions change for charge; i.e., for $\langle \hat{C}_{QQ}(t) \rangle = \text{Tr}[\hat{\rho}_{\text{th}} \hat{C}_{QQ}(t)]$?

Will discuss
answers on my
blog sometime
soon

Linear harmonic oscillator summary



The LC quantum harmonic oscillator



Position: $\Phi \mapsto x$

Mass: $C \mapsto m$

Momentum: $Q \mapsto p$
Spring constant: $L^{-1} \mapsto k$

$$\hat{H}(\hat{\Phi}, \hat{Q}) = \frac{\hat{\Phi}^2}{2L} + \frac{\hat{Q}^2}{2C} = \hbar\omega_0 \left(\hat{a}^\dagger \hat{a} + \frac{1}{2} \right) \quad \omega_0^2 = \frac{1}{LC}$$

$$\hat{H}|n\rangle = \hbar\omega_0 \left(n + \frac{1}{2} \right) |n\rangle \quad Z_0 = \frac{L}{C}$$

$$\begin{aligned} \hat{\Phi} &= \Phi_{\text{ZPF}} (\hat{a}^\dagger + \hat{a}) & \Phi_{\text{ZPF}} &= \sqrt{\frac{\hbar}{2}} Z_0 = \Phi_0 \sqrt{\frac{z_0}{2\pi}}, \\ \hat{Q} &= iQ_{\text{ZPF}} (\hat{a}^\dagger - \hat{a}) & Q_{\text{ZPF}} &= \sqrt{\frac{\hbar}{2}} Z_0^{-1} = (2e) \sqrt{\frac{1}{2\pi z_0}}, \end{aligned}$$

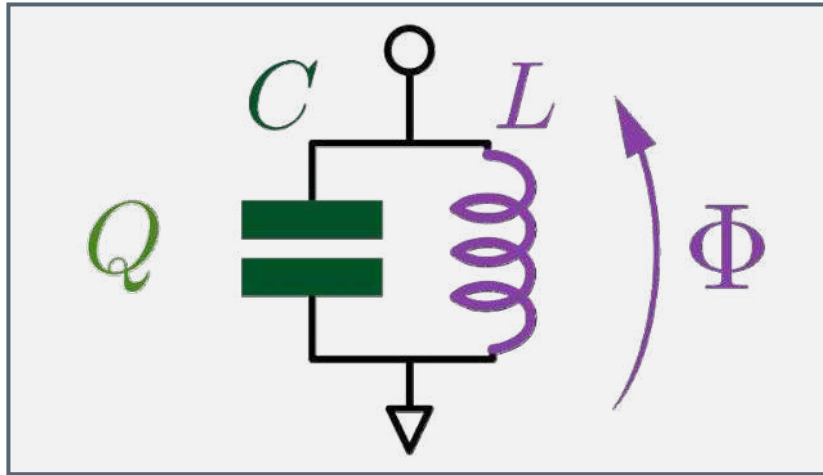
$$\mathcal{E}_{\text{ind}}(\Phi) = \frac{\Phi^2}{2L}$$

$$\begin{aligned} \mathcal{E}_{\text{cap}}(Q) &= \frac{Q^2}{2C} \\ &= \frac{1}{2} C \dot{\Phi}^2 \end{aligned}$$

Energy

Φ/ϕ_0

The LC quantum harmonic oscillator



Position: $\Phi \mapsto x$
Mass: $C \mapsto m$

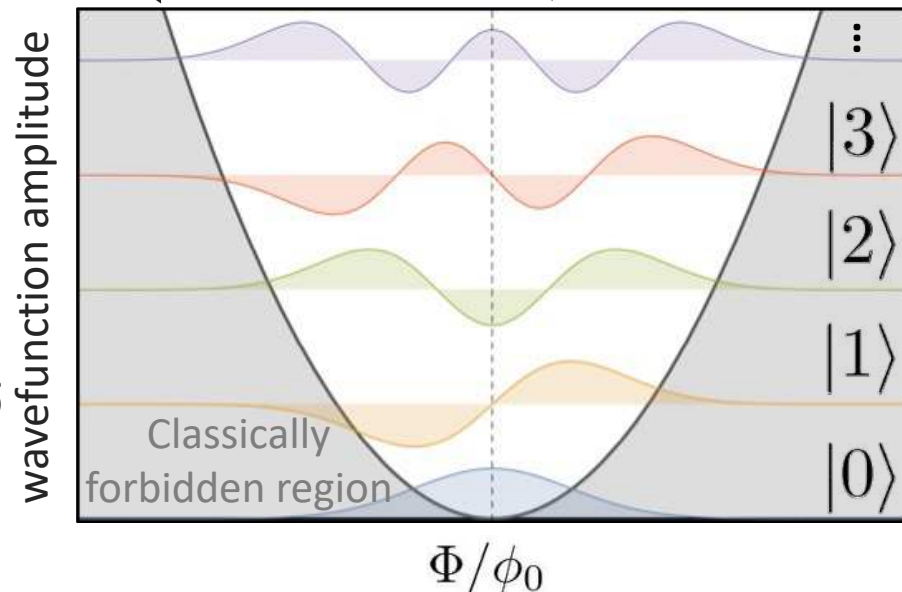
Momentum: $Q \mapsto p$
Spring constant: $L^{-1} \mapsto k$

$$\hat{H}(\hat{\Phi}, \hat{Q}) = \frac{\hat{\Phi}^2}{2L} + \frac{\hat{Q}^2}{2C} = \hbar\omega_0 \left(\hat{a}^\dagger \hat{a} + \frac{1}{2} \right) \quad \omega_0^2 = \frac{1}{LC}$$

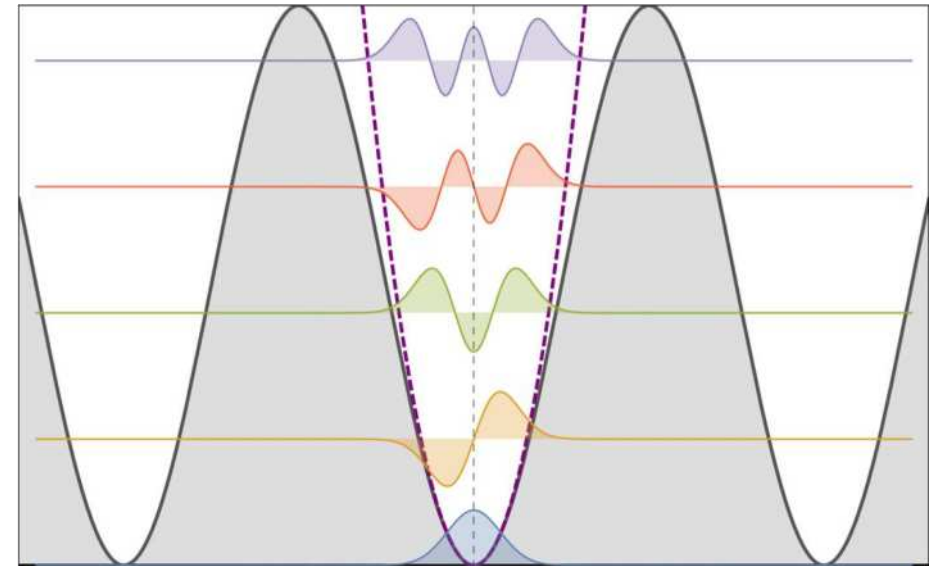
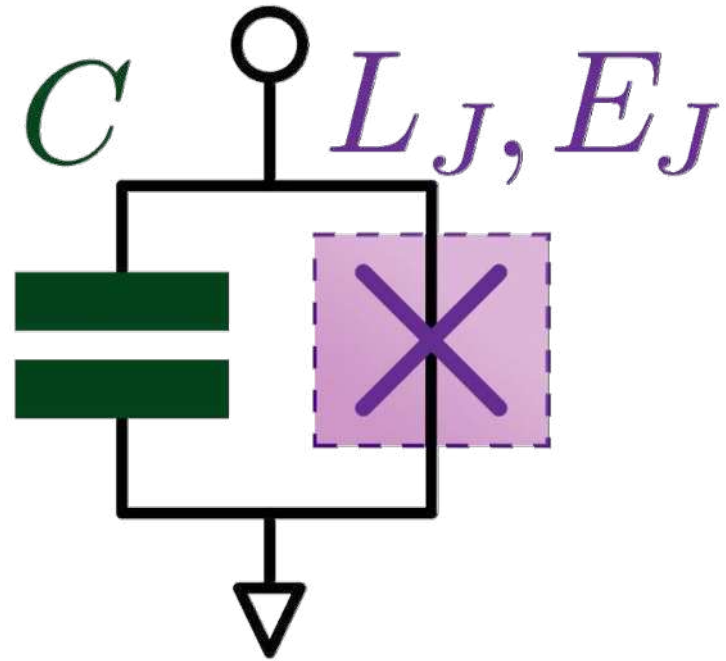
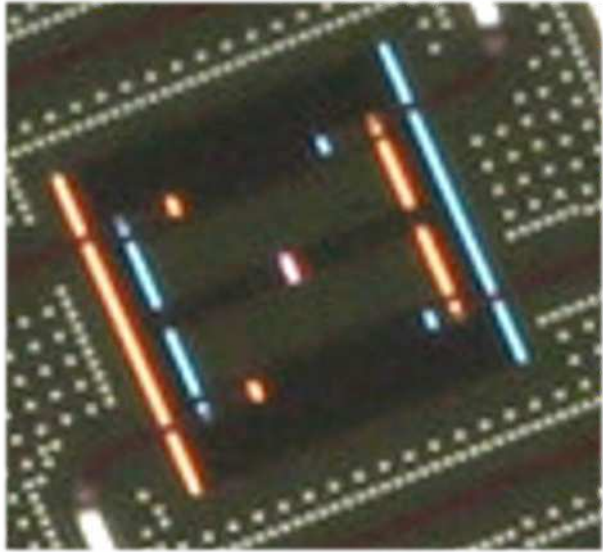
$$\hat{H}|n\rangle = \hbar\omega_0 \left(n + \frac{1}{2} \right) |n\rangle \quad \Phi_{\text{ZPF}} Q_{\text{ZPF}} = \frac{\hbar}{2} \quad Z_0 = \frac{L}{C}$$

$$\begin{aligned} \hat{\Phi} &= \Phi_{\text{ZPF}} (\hat{a}^\dagger + \hat{a}) & \Phi_{\text{ZPF}} &= \sqrt{\frac{\hbar}{2} Z_0} & \langle 0 | \hat{\Phi}^2 | 0 \rangle &= \Phi_{\text{ZPF}}^2 \\ \hat{Q} &= iQ_{\text{ZPF}} (\hat{a}^\dagger - \hat{a}) & Q_{\text{ZPF}} &= \sqrt{\frac{\hbar}{2} Z_0^{-1}} & \langle 0 | \hat{Q}^2 | 0 \rangle &= Q_{\text{ZPF}}^2 \end{aligned}$$

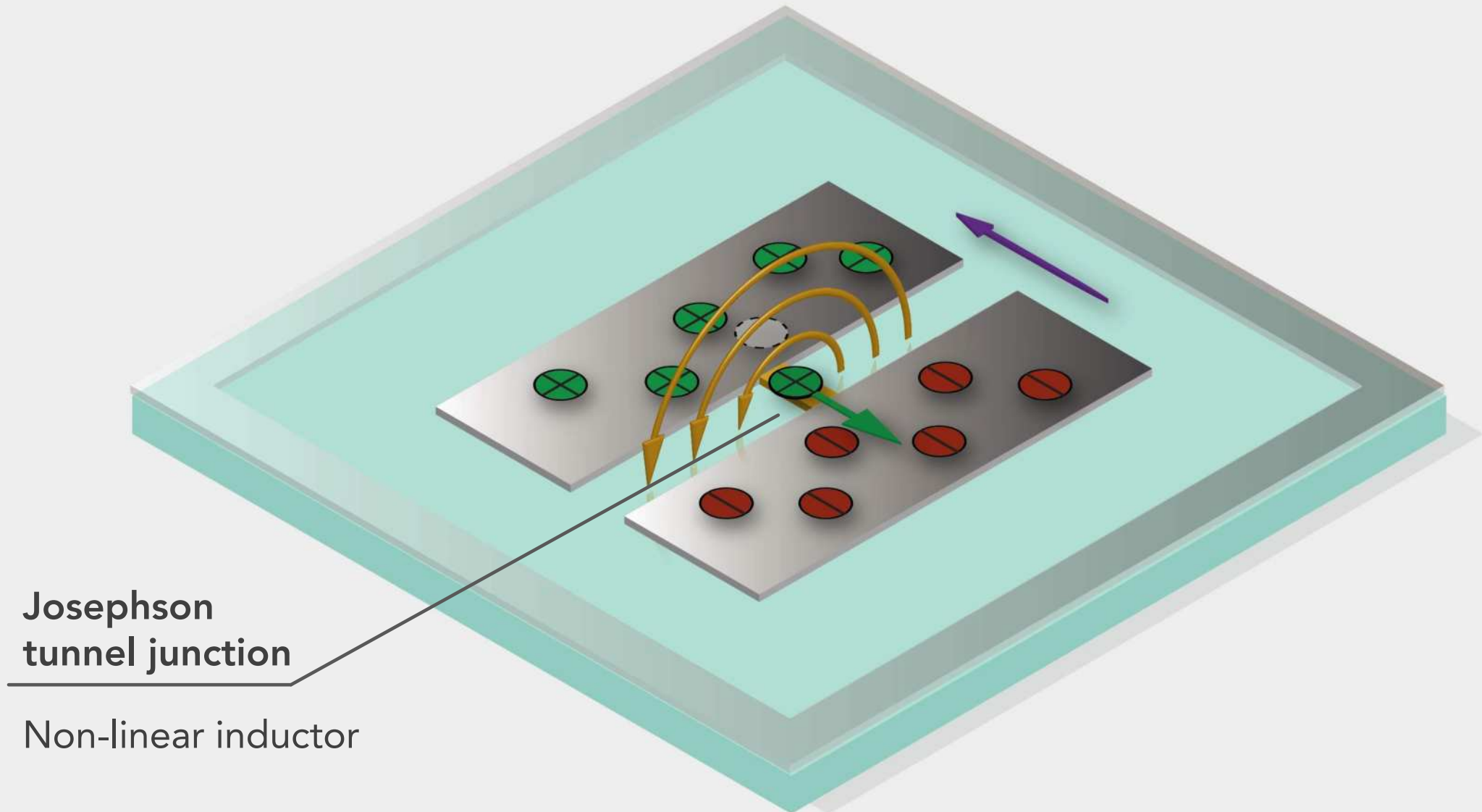
$$\mathcal{E}_{\text{ind}}(\Phi) = \frac{\Phi^2}{2L} \quad \psi_n(\Phi) \equiv \langle \Phi | n \rangle$$



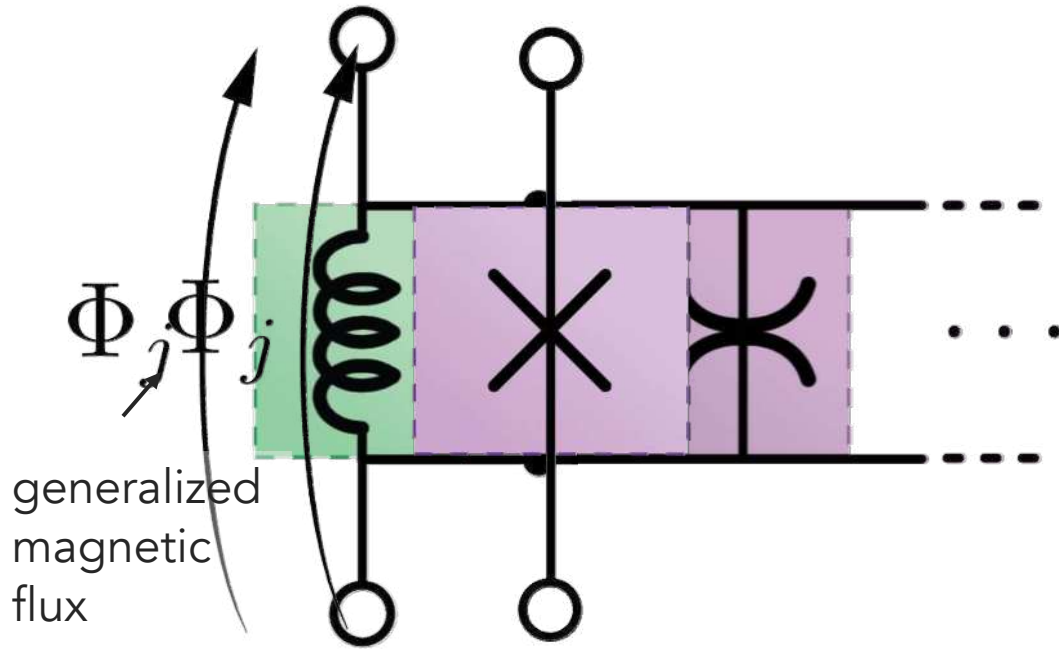
Transmon Qubit



The transmon as a non-linear oscillator



Josephson tunnel junction

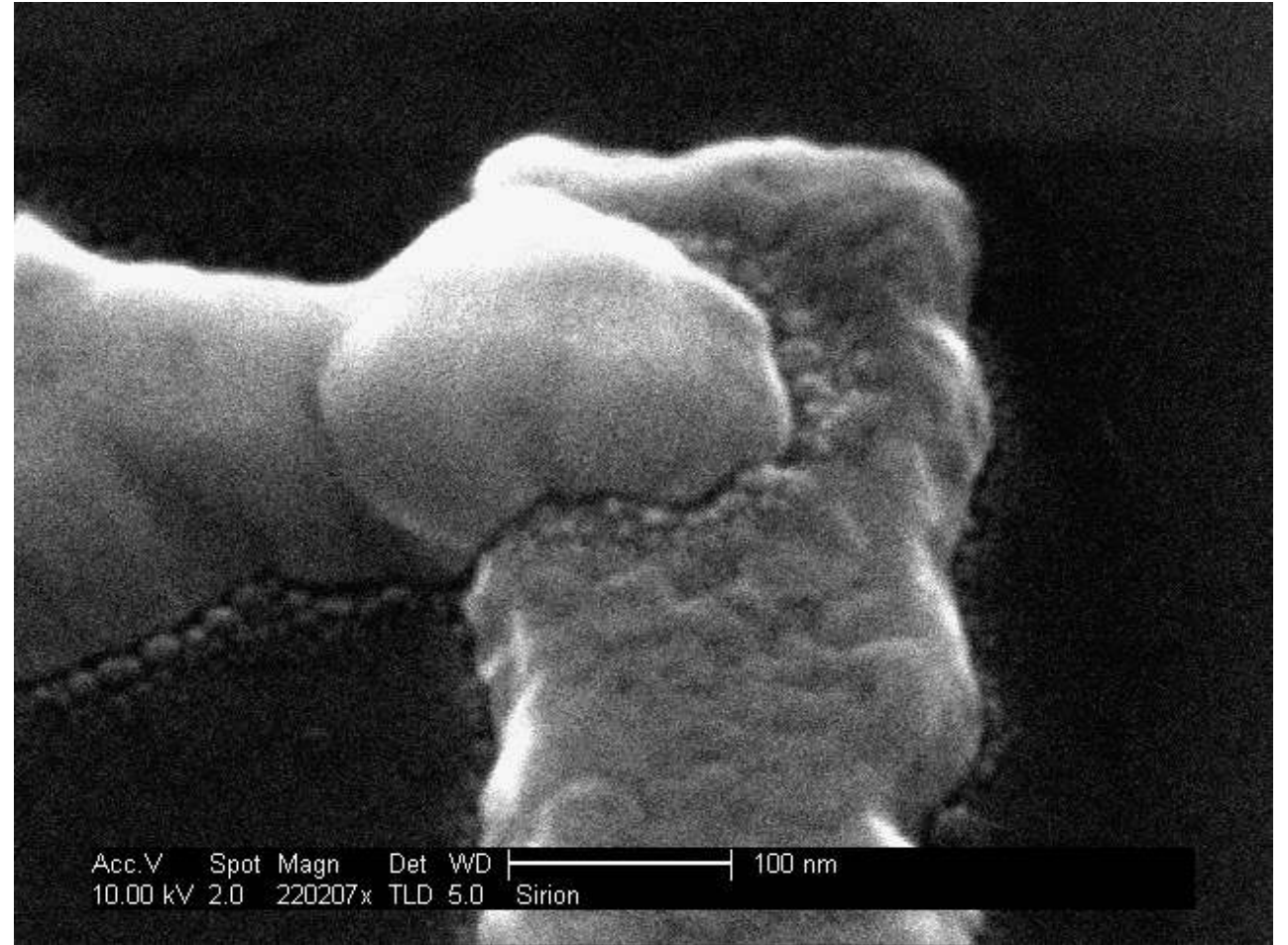


$$\mathcal{E}_j(\Phi_j) = E_j (1 - \cos(\Phi_j/\phi_0))$$

$$= \mathcal{E}_j^{\text{lin}}(\Phi_j) + \mathcal{E}_j^{\text{nl}}(\Phi_j)$$

$$\phi_0 \equiv \hbar/2e$$

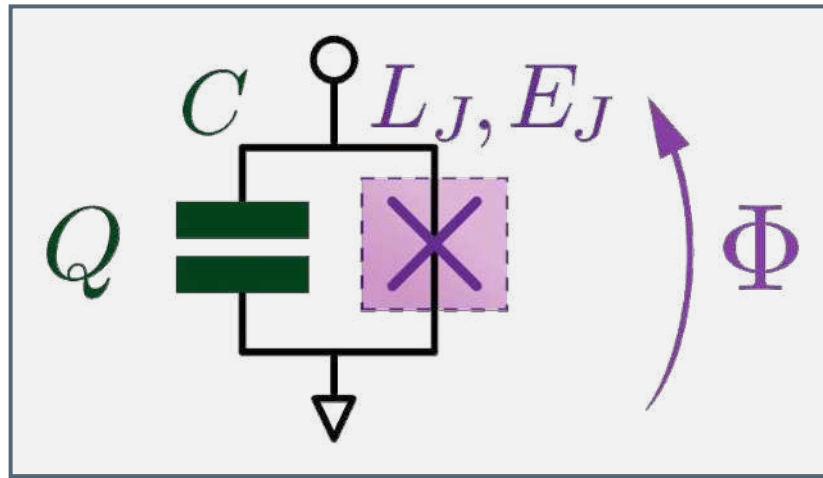
$$= \frac{E_j}{2} \left(\frac{\Phi_j}{\phi_0} \right)^2 - \frac{E_j}{4!} \left(\frac{\Phi_j}{\phi_0} \right)^4 + \mathcal{O}(\Phi_j^6)$$



SEM image: L. Frunzio

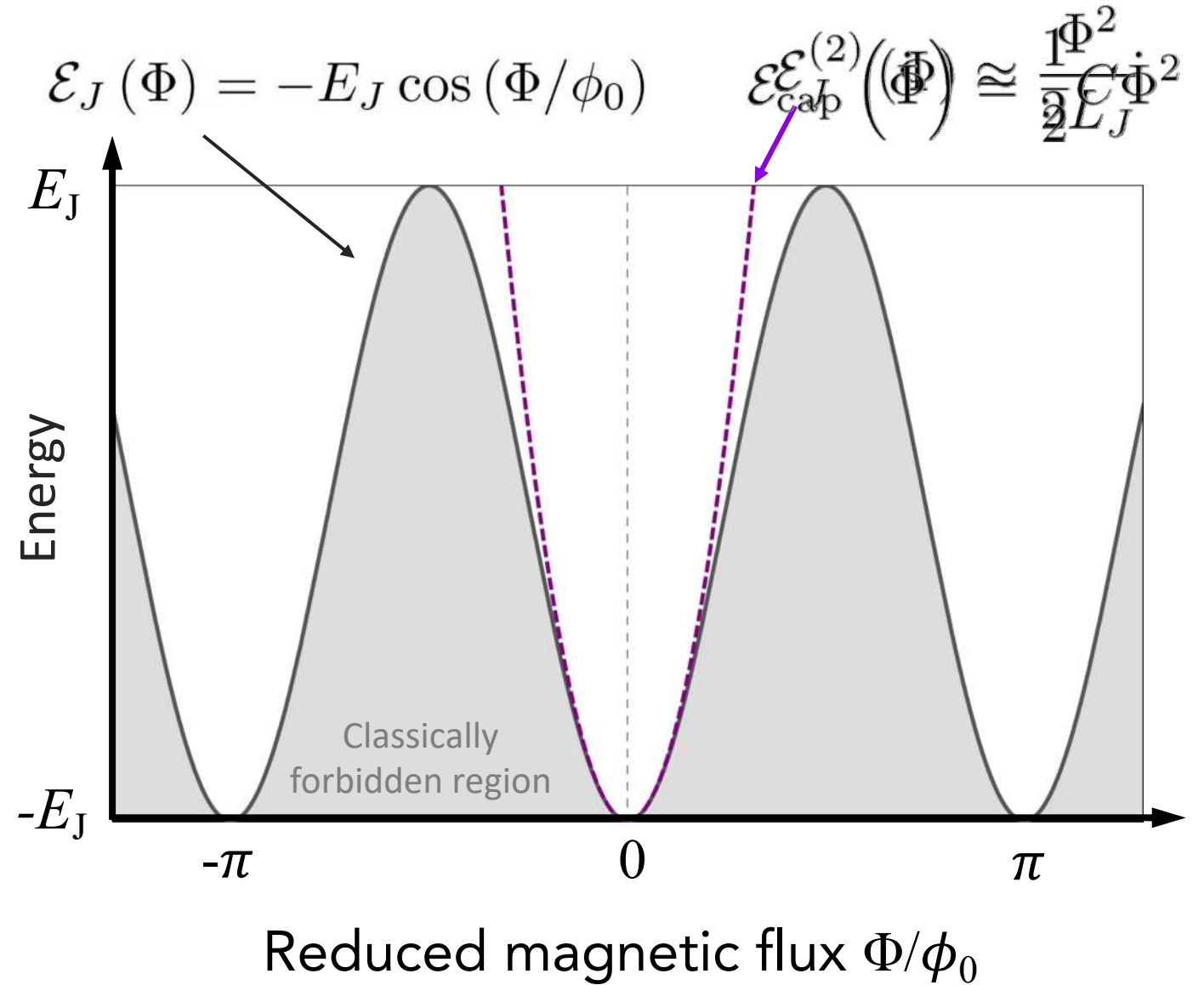
Circuit image: Mineev et al., EPR to appear (2020)

The Transmon qubit



$$E_J = \frac{\phi_0^2}{L_J}$$

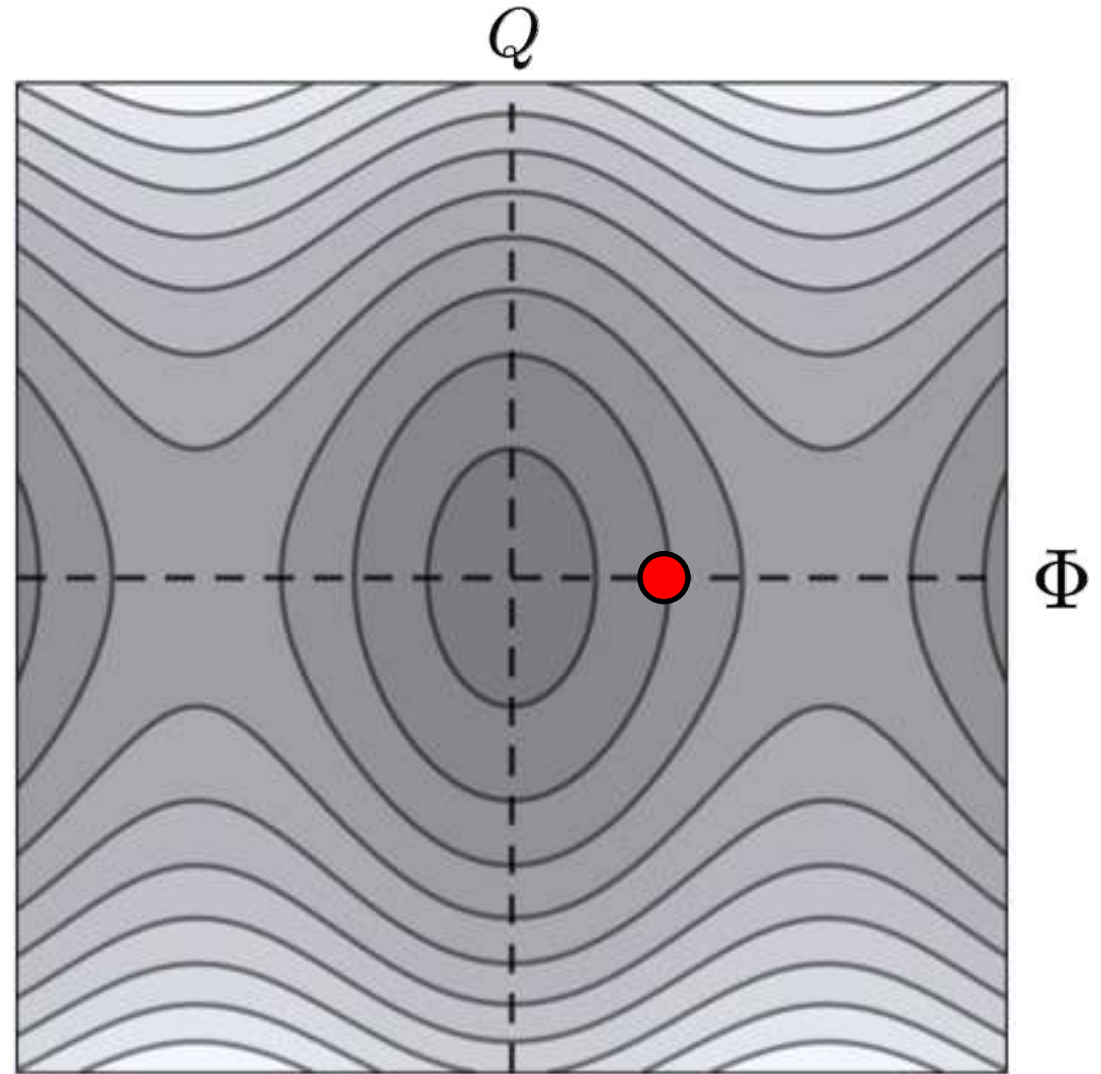
$$\phi_0 = \frac{\hbar}{2e} \approx 3.3 \times 10^{-16} \text{ Wb}$$



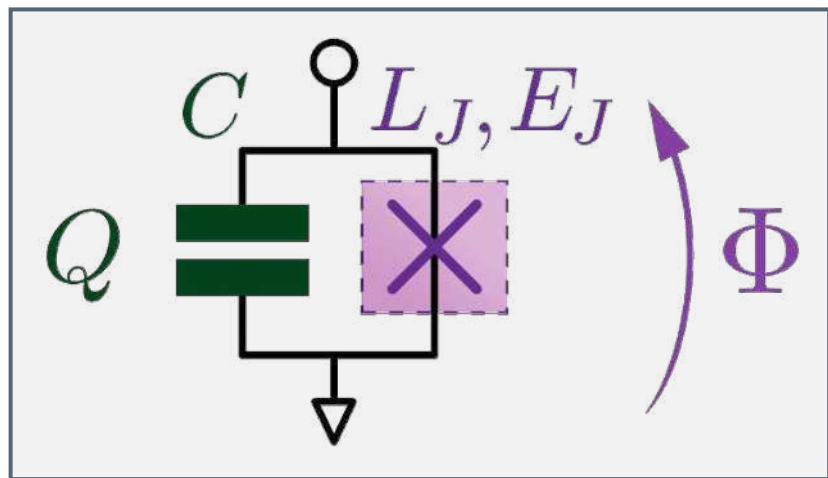
Semi-classical intuition: phase space picture

$$\mathcal{H}(\Phi, Q) = \frac{Q^2}{2C} - E_J \cos(\Phi/\phi_0)$$

$$\begin{aligned} E &= \hbar\omega_0 \left(n + \frac{1}{2} \right) \\ &= \frac{Q^2}{2C} - E_J \cos(\Phi/\phi_0) \end{aligned}$$



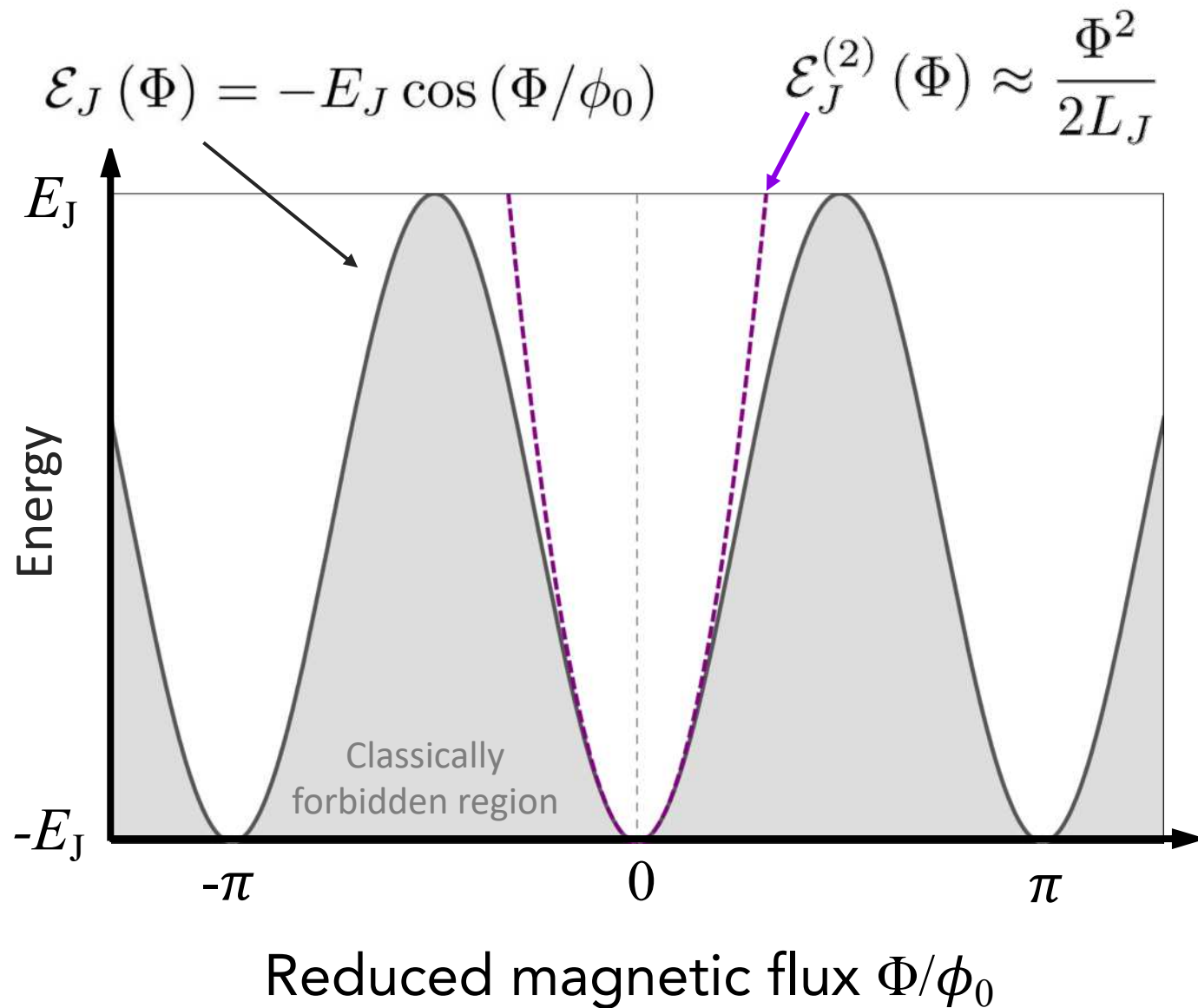
The Transmon qubit



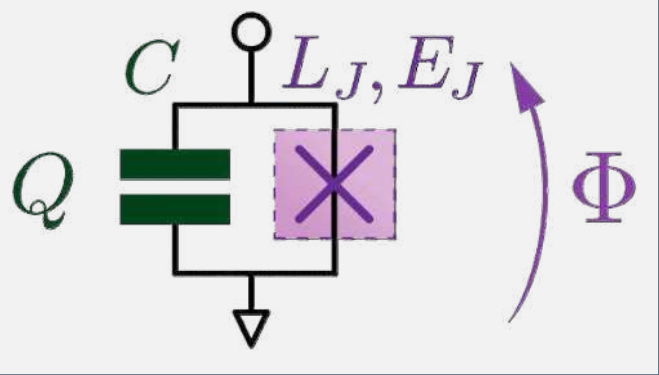
$$\hat{H} = \frac{\hat{Q}^2}{2C} - E_J \cos \left(\hat{\Phi} / \phi_0 \right)$$

$$\approx \frac{\hat{Q}^2}{2C} + \frac{\hat{\Phi}^2}{2L_J} - \frac{E_J}{4!} \left(\frac{\hat{\Phi}}{\phi_0} \right)^4$$

$$= \hbar \omega_0 \hat{a}^\dagger \hat{a} - \frac{E_J \phi_{\text{ZPF}}^4}{4!} (\hat{a} + \hat{a}^\dagger)^4$$



The Transmon qubit



$$= \hbar\omega_0 \hat{a}^\dagger \hat{a} - \frac{E_J \phi_{\text{ZPF}}^4}{4!} (3 + \cancel{6\hat{a}^2} + 12\hat{a}^\dagger \hat{a} + \cancel{6\hat{a}^{\dagger 2}} + \cancel{\hat{a}^4} + \cancel{4\hat{a}^\dagger \hat{a}^3} + \cancel{6\hat{a}^{\dagger 2} \hat{a}^2} + \cancel{4\hat{a}^{\dagger 3} \hat{a}} + \cancel{\hat{a}^{\dagger 4}})$$

RWA or 1st order PT

$$\hat{H}_4^{\text{RWA}} = \hbar\omega_0 \hat{a}^\dagger \hat{a} - \frac{E_J}{4!} \phi_a^4 (12\hat{a}^\dagger \hat{a} + 6\hat{a}^{\dagger 2} \hat{a}^2)$$

$$= \hbar(\omega_0 - \Delta_q) \hat{a}^\dagger \hat{a} - \frac{\hbar\alpha}{2} \hat{a}^{\dagger 2} \hat{a}^2$$

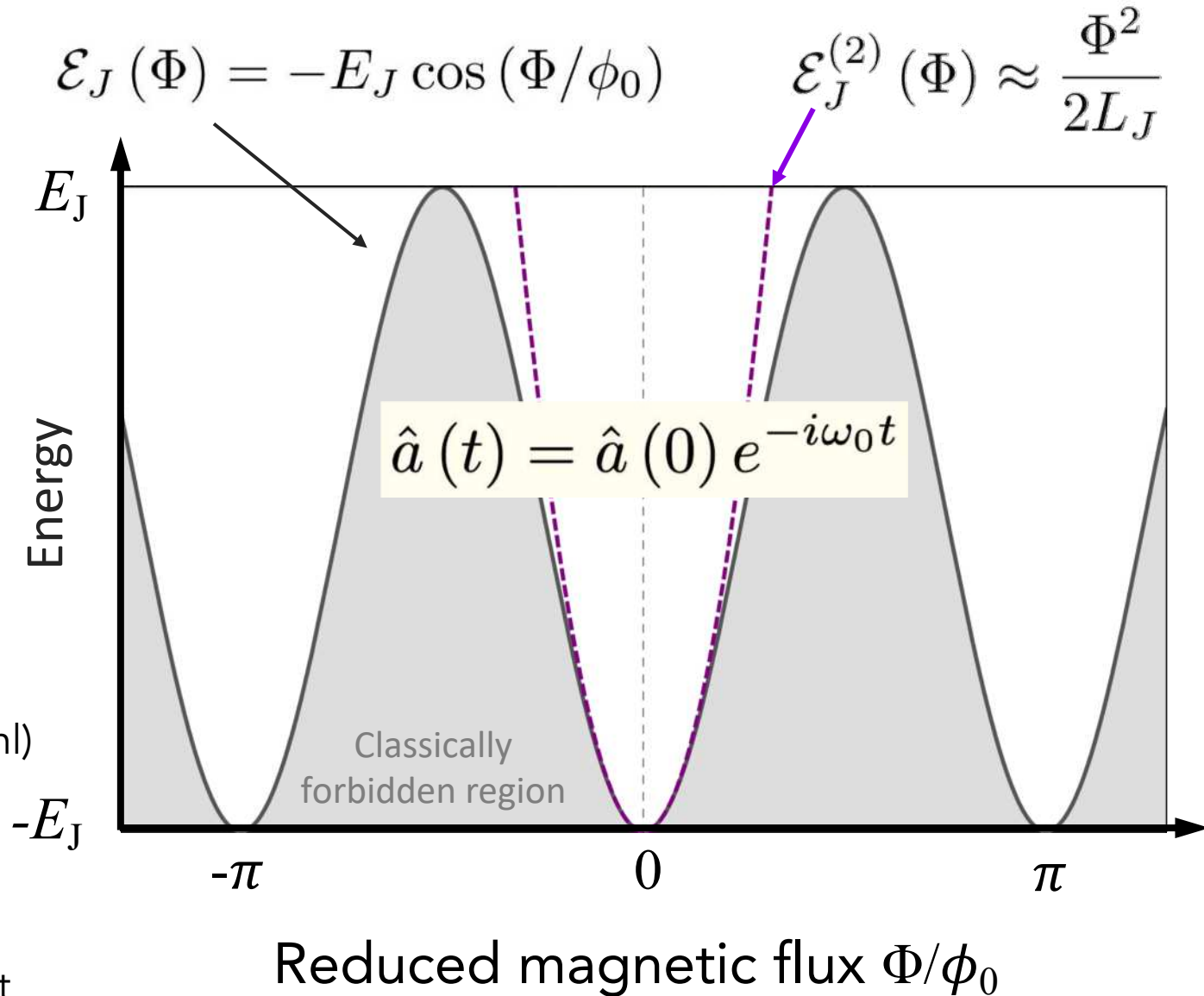
Due to nl & commutator Due to non-linearity (nl)

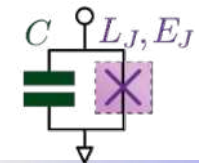
where $\hbar\Delta_q = \hbar\alpha \equiv \frac{1}{2} E_J \phi_{\text{ZPF}}^4$

"Lamb shift"
due to ZPF

Anharmonicity

Zero-point
fluctuations





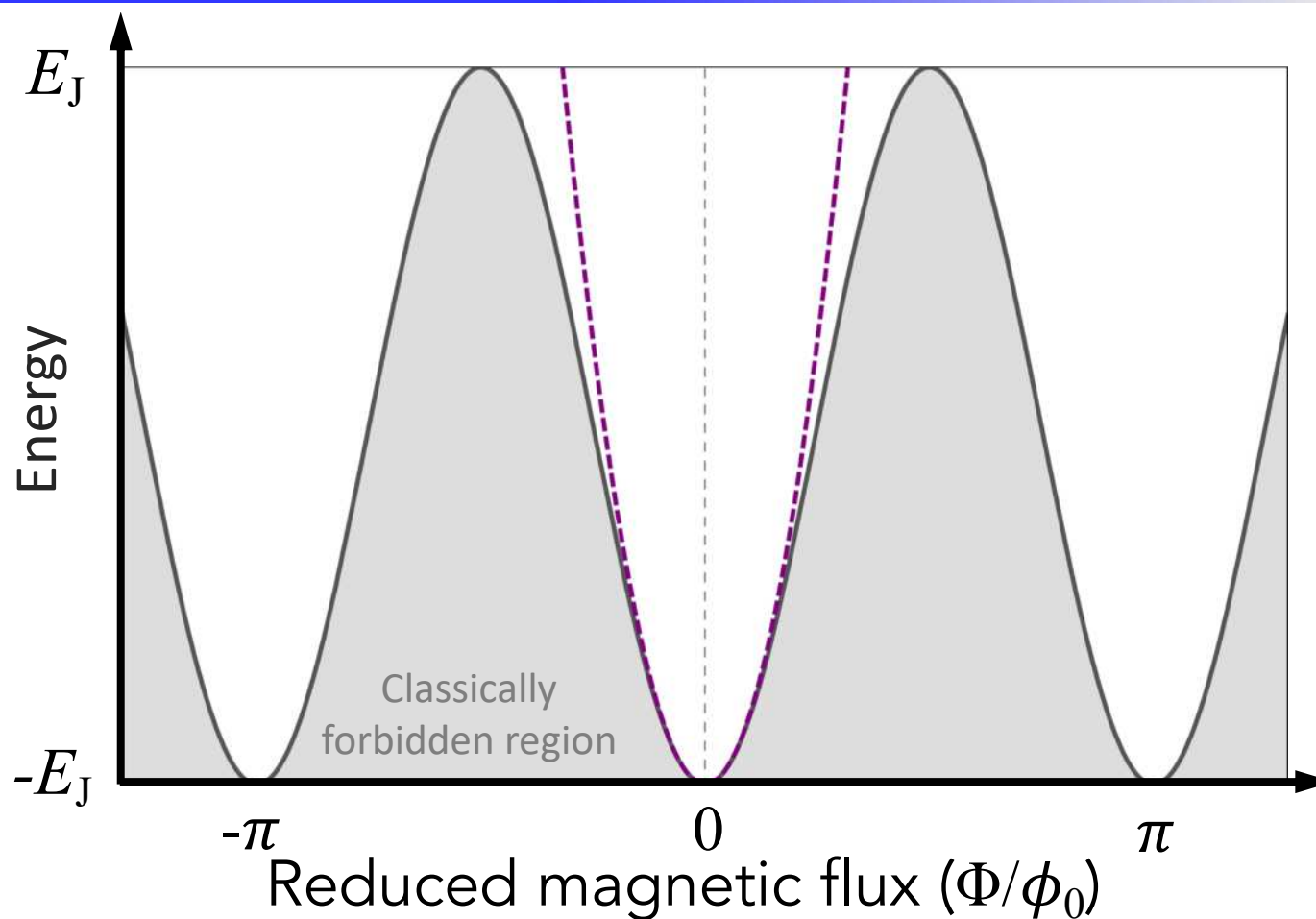
The Kerr Hamiltonian of the transmon

$$\hat{N} \equiv \hat{a}^\dagger \hat{a}$$

$$\hat{H}_4^{\text{RWA}} \approx \underbrace{\hbar\omega_q \hat{N}}_{\hat{H}_{\text{lin}}} - \underbrace{\frac{\hbar\alpha}{2} \hat{N} (\hat{N} - 1)}_{\hat{H}_{\text{nl}}}$$

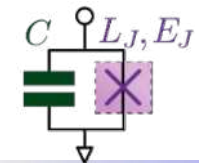
$\hat{H}_{\text{lin}}$ Solution known (SHO) $\hat{H}_{\text{nl}}$ New from nonlinearity

$$\hat{H}_4^{\text{RWA}} |n\rangle = \hbar\omega_q n \left(1 - \frac{\hbar\alpha_a}{2} (n - 1) \right) |n\rangle$$



1st order correction to energy: $E_n^{(1)} = \langle n^{(0)} | \hat{H}_{\text{nl}} | n^{(0)} \rangle$ and to eigenstates: $|n^{(1)}\rangle = \sum_{k \neq n} \frac{\langle k^{(0)} | \hat{H}_{\text{nl}} | n^{(0)} \rangle}{E_n^{(0)} - E_k^{(0)}} |k^{(0)}\rangle.$

To first order perturbation theory the eigenstates do not change! Only the energy changes. Dispersive.



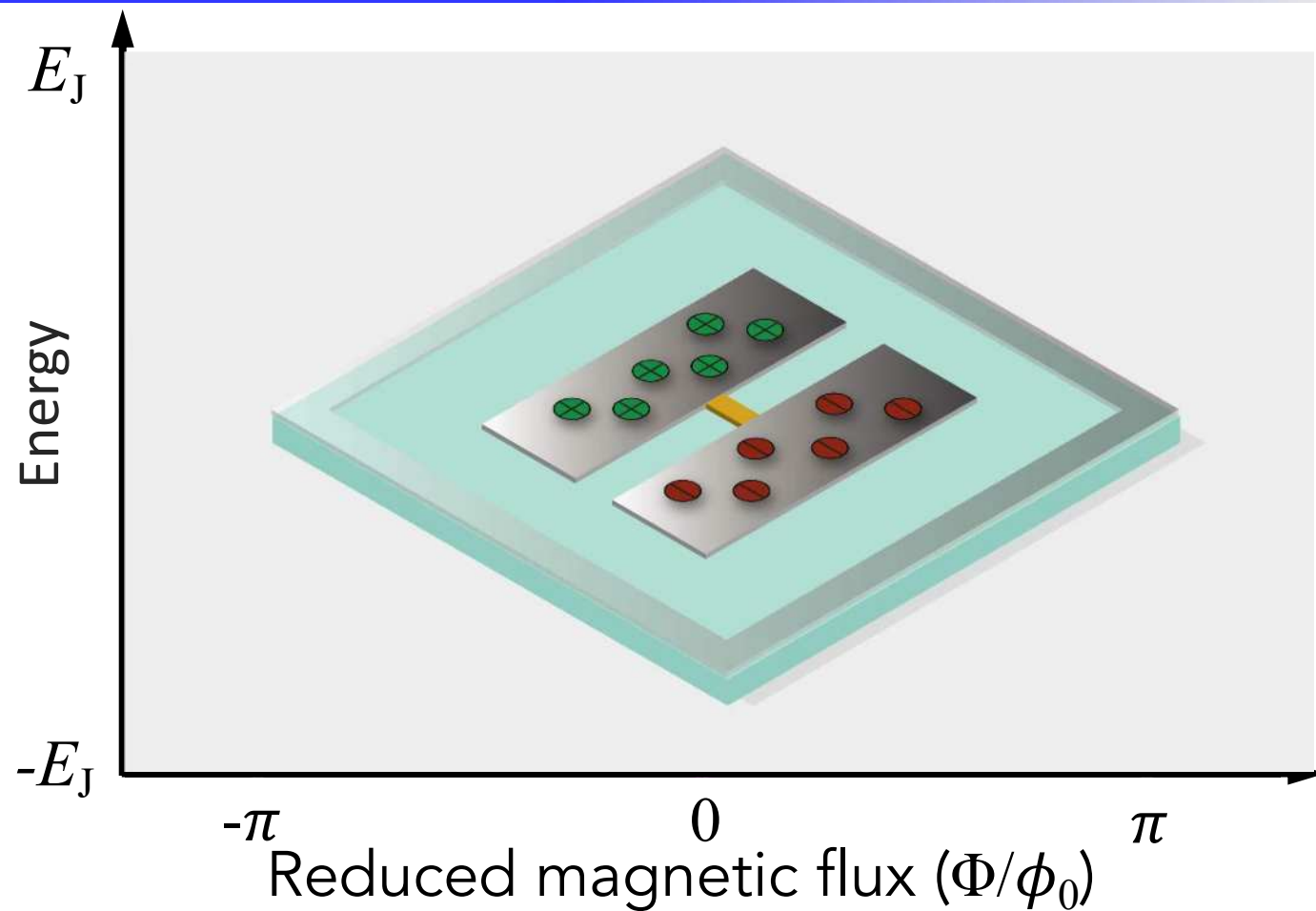
Exploring a real transmon qubit

$$\hat{N} \equiv \hat{a}^\dagger \hat{a}$$

$$\hat{H}_4^{\text{RWA}} \approx \underbrace{\hbar\omega_q \hat{N}}_{\hat{H}_{\text{lin}}} - \underbrace{\frac{\hbar\alpha}{2} \hat{N} (\hat{N} - 1)}_{\hat{H}_{\text{nl}}}$$

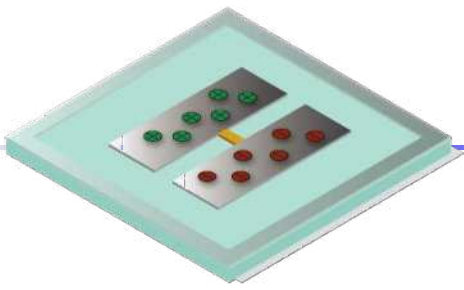
$\nearrow \hat{H}_{\text{lin}}$
 Solution known (SHO)

$\nwarrow \hat{H}_{\text{nl}}$
 New from nonlinearity



Experimental parameters

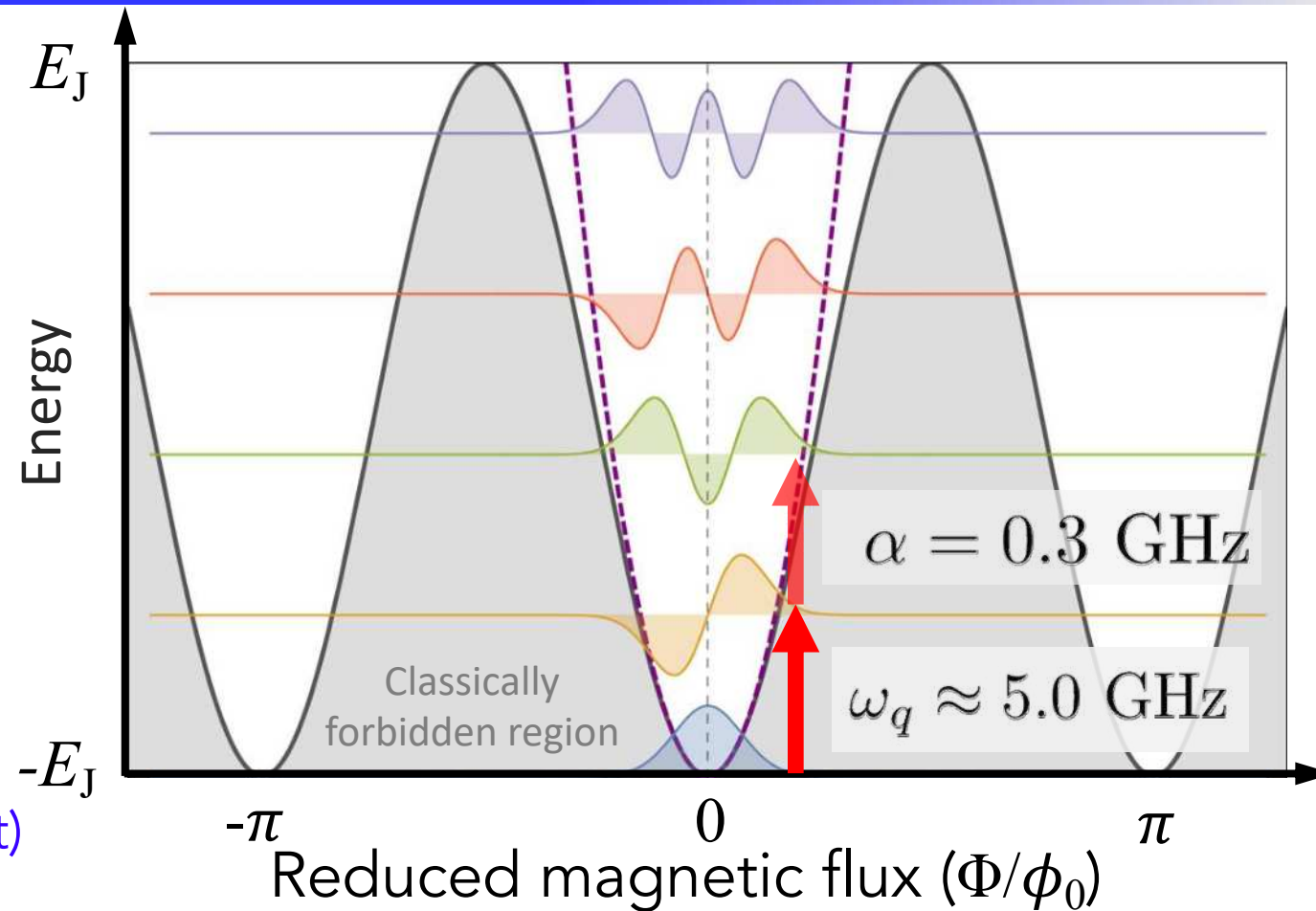
$$\begin{aligned}
 L_J &= 14 \text{ nH} & E_J &= \frac{\phi_0^2}{L_J} = 12 \text{ GHz} \\
 C_J &= 65 \text{ fF} & E_C &= \frac{e^2}{2C} = 0.3 \text{ GHz}
 \end{aligned}$$



Exploring a real transmon qubit

$$\hat{H}_4^{\text{RWA}} \approx \hbar\omega_q \hat{N} - \frac{\hbar\alpha}{2} \hat{N} (\hat{N} - 1)$$

Dispersive, states didn't change to 1st order
"Lamb shift" due to ZPF



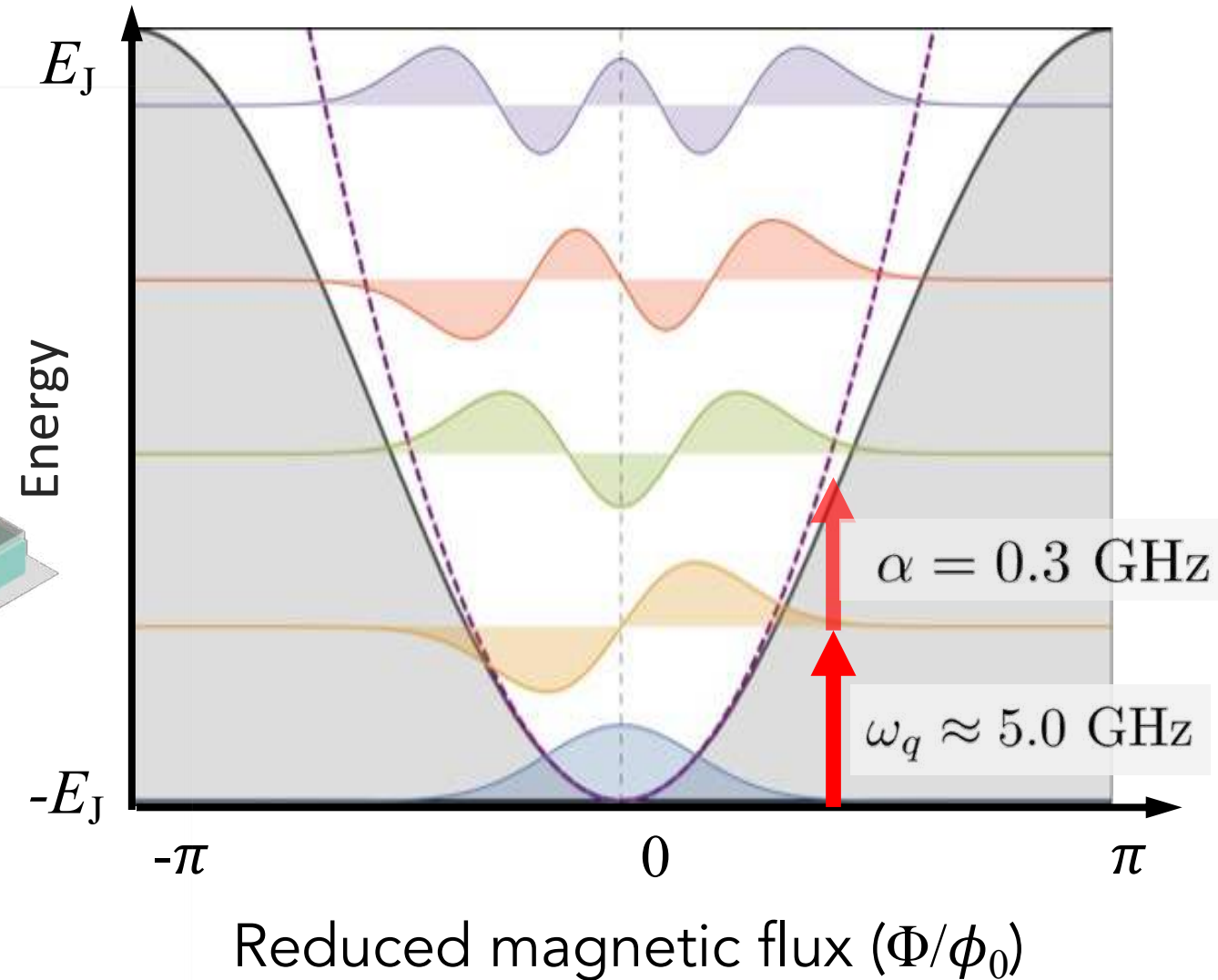
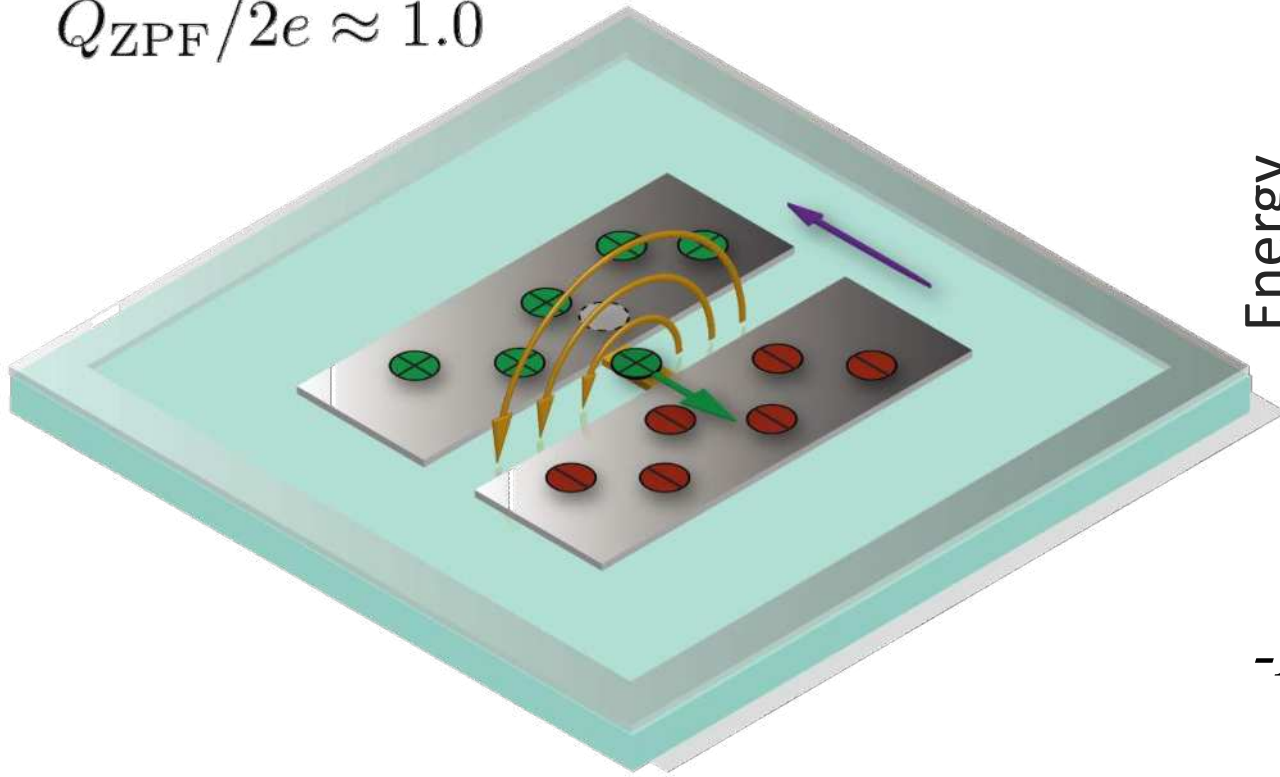
Parameters used in figure (of a measured qubit)

$$\begin{aligned} L_J &= 14 \text{ nH} & E_J &= \frac{\phi_0^2}{L_J} = 12 \text{ GHz} & \omega_0 &= \sqrt{\frac{1}{LC}} = 5.3 \text{ GHz} \\ C_J &= 65 \text{ fF} & E_C &= \frac{e^2}{2C} = 0.3 \text{ GHz} & Z &= \sqrt{\frac{L}{C}} \approx 450 \text{ } \Omega \end{aligned}$$

Quantum fluctuations of the transmon qubit

$$\Phi_{\text{ZPF}}/\phi_0 \approx 0.5$$

$$Q_{\text{ZPF}}/2e \approx 1.0$$

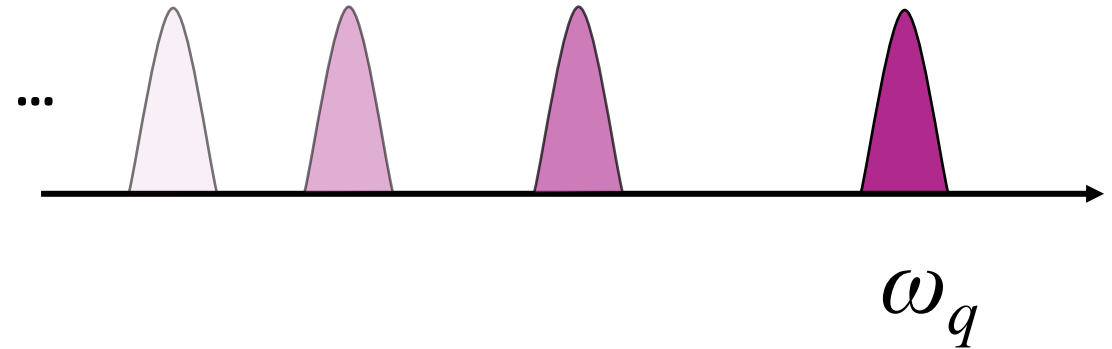


Energy diagram and transition spectrum

Energy levels

$$\hat{H} \approx \omega_0 \hat{a}^\dagger \hat{a} - \frac{\alpha}{2} \hat{a}^{\dagger 2} \hat{a}^2$$

Transition spectrum



The Transmon qubit: restricting Hilbert space

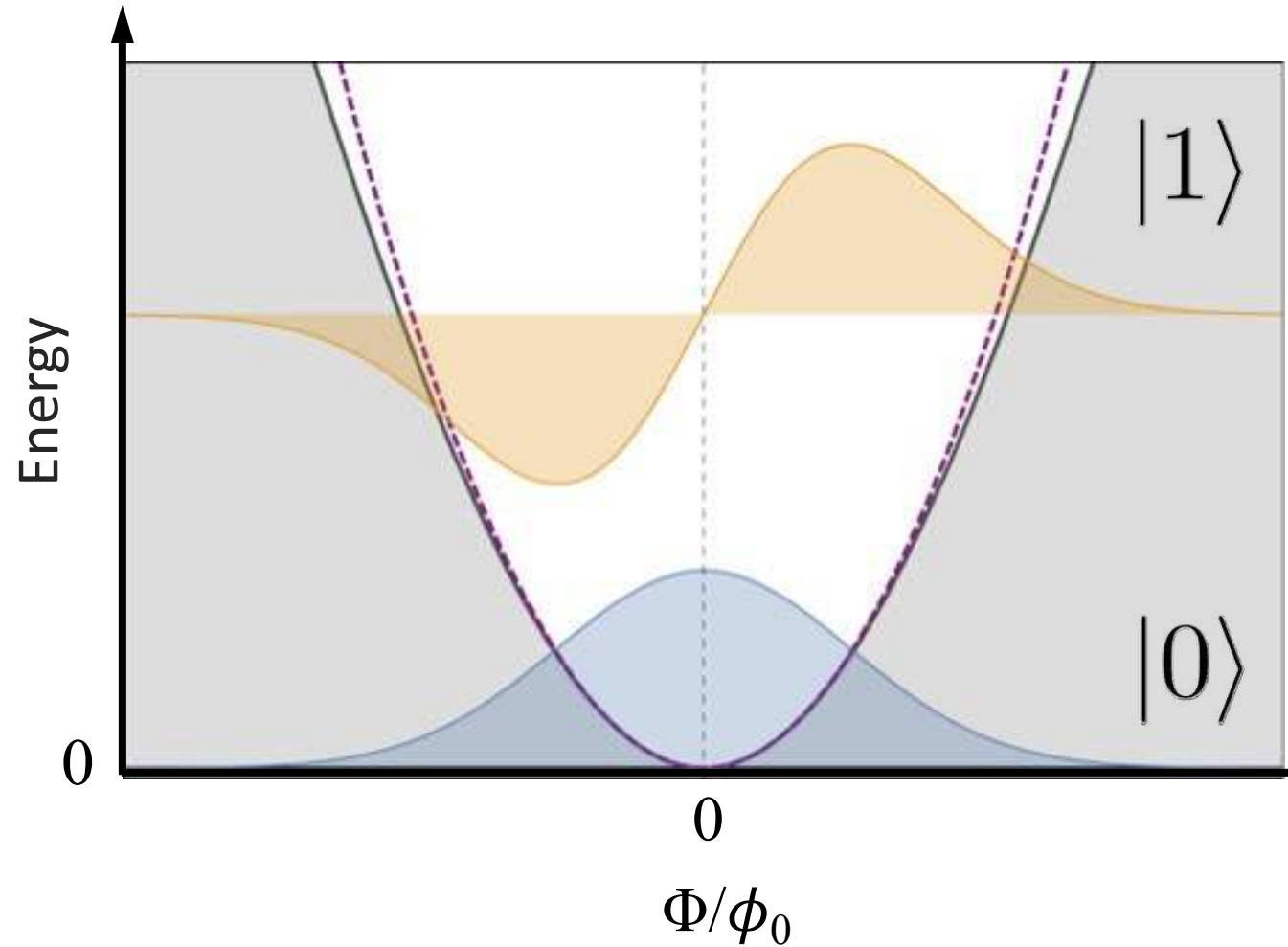
$$\hat{H}_4^{\text{RWA}} \approx \hbar\omega_q \hat{N} - \frac{\hbar\alpha}{2} \hat{N} (\hat{N} - 1)$$

$$\hat{N} \equiv \hat{a}^\dagger \hat{a}$$

Restrict to qubit subspace of $|0\rangle$ and $|1\rangle$

$$\hat{a}^\dagger \hat{a} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & \ddots \end{pmatrix}$$

$$\hat{a} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & \sqrt{2} & 0 \\ 0 & 0 & 0 & \sqrt{3} \\ 0 & 0 & 0 & \ddots \end{pmatrix}$$



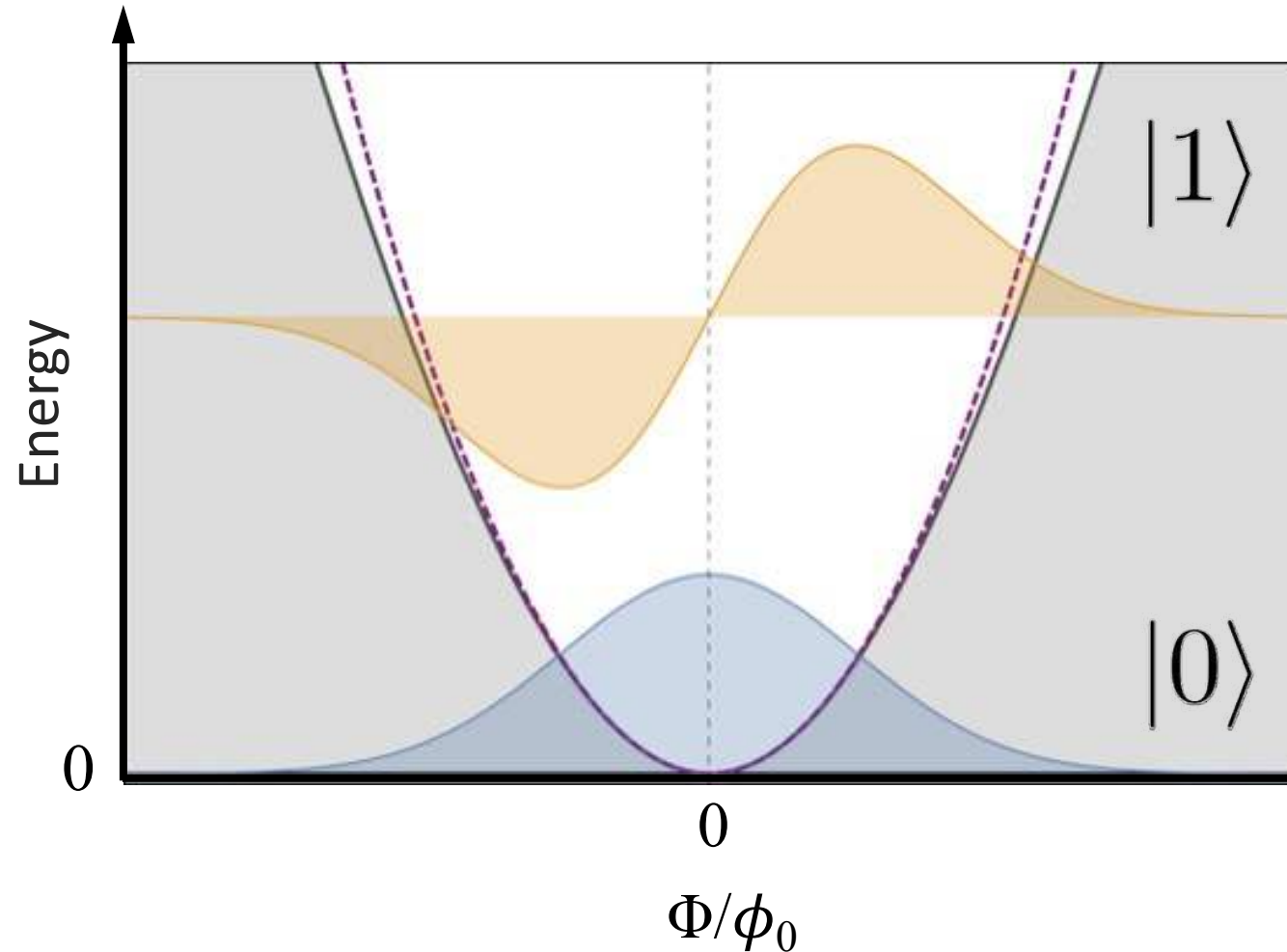
The Transmon qubit: restricting Hilbert space

$$\hat{H}_4^{\text{RWA}} \approx \hbar\omega_q \hat{N} - \frac{\hbar\alpha}{2} \hat{N} (\hat{N} - 1)$$

$$\hat{N} \equiv \hat{a}^\dagger \hat{a}$$

Restrict to qubit subspace of $|0\rangle$ and $|1\rangle$

$$\begin{array}{ccc} \left(\hat{N} - \frac{1}{2} \hat{I} \right) & \mapsto & -\frac{1}{2} \hat{Z} \quad \hat{a} \mapsto \hat{\sigma}_- = \frac{1}{2} (\hat{X} - i\hat{Y}) \\ \uparrow & & \uparrow \qquad \qquad \uparrow \qquad \uparrow \\ \text{Fock number} & & \text{Qubit Pauli Z} & & \text{Qubit Pauli X and Y} \\ \text{operator} & & \text{operator} & & \text{operators} \\ \begin{pmatrix} 1 & 0 & \cdots \\ 0 & 2 & \cdots \\ \vdots & \vdots & \ddots \end{pmatrix} & & \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} & & \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \end{array}$$



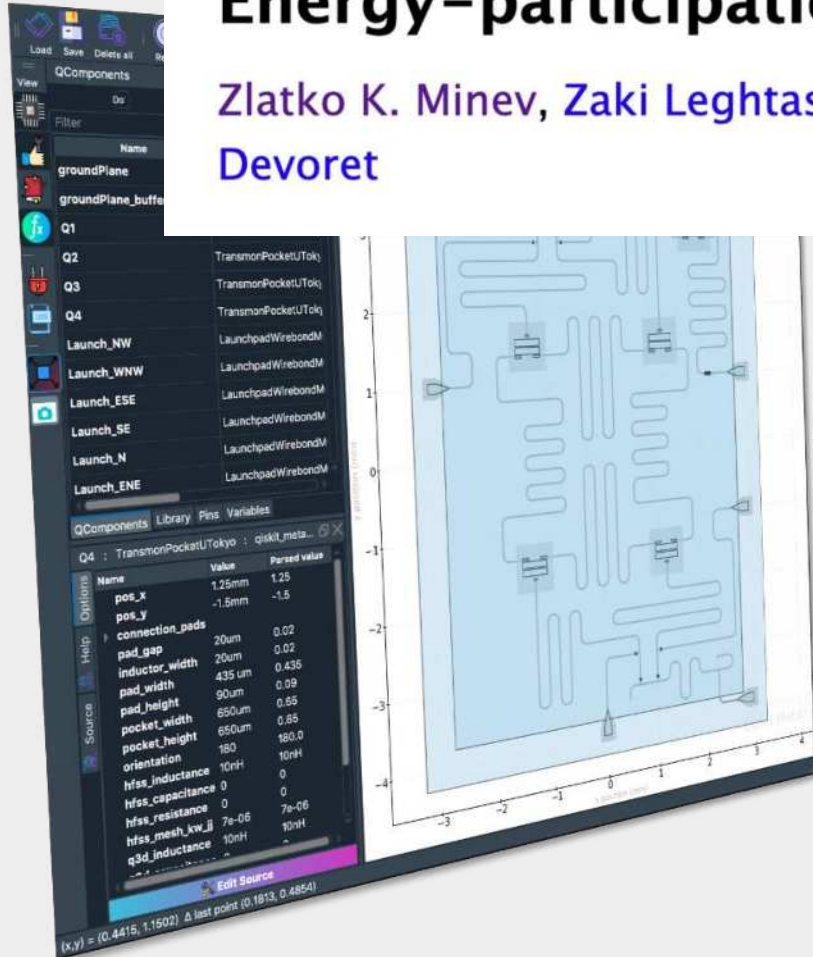
Quantum Physics

[Submitted on 1 Oct 2020]

Energy-participation quantization of Josephson circuits

Zlatko K. Minev, Zaki Leghtas, Shantanu O. Mundhada, Lysander Christakis, Ioan M. Pop, Michel H. Devoret

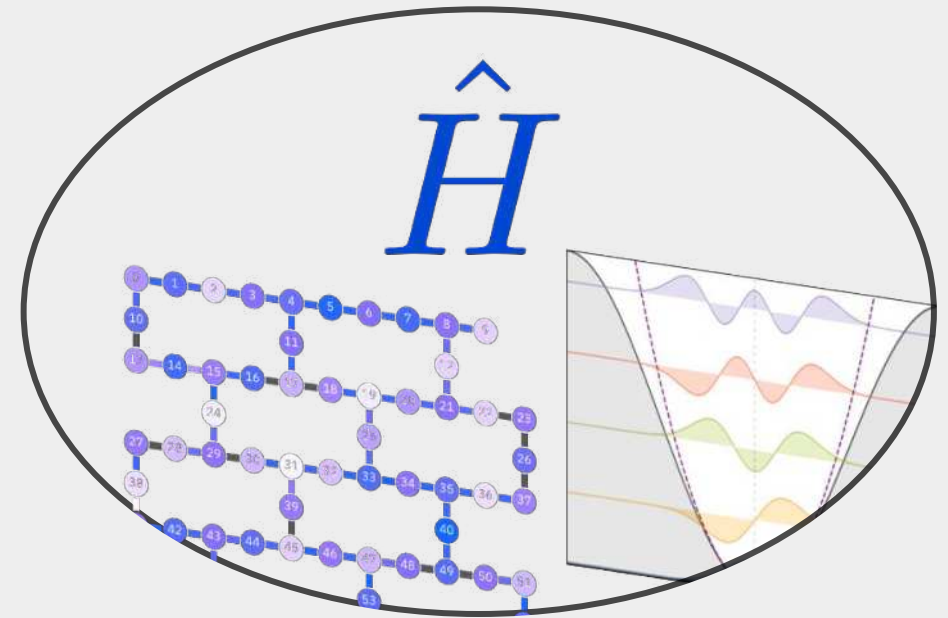
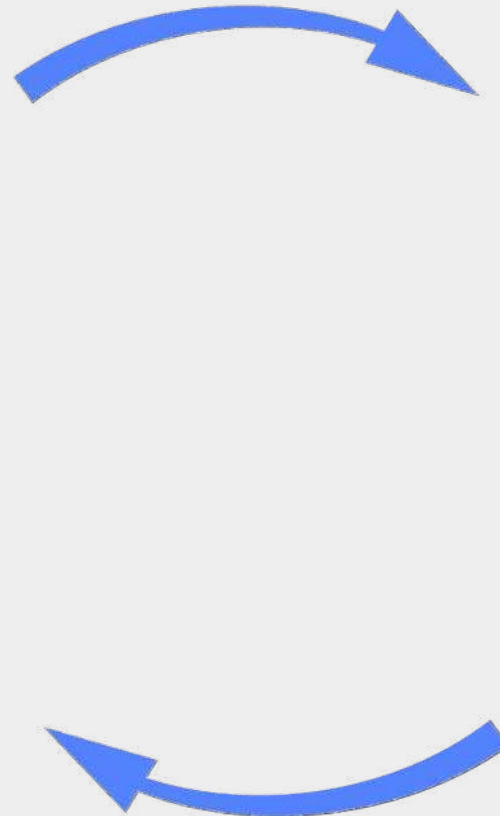
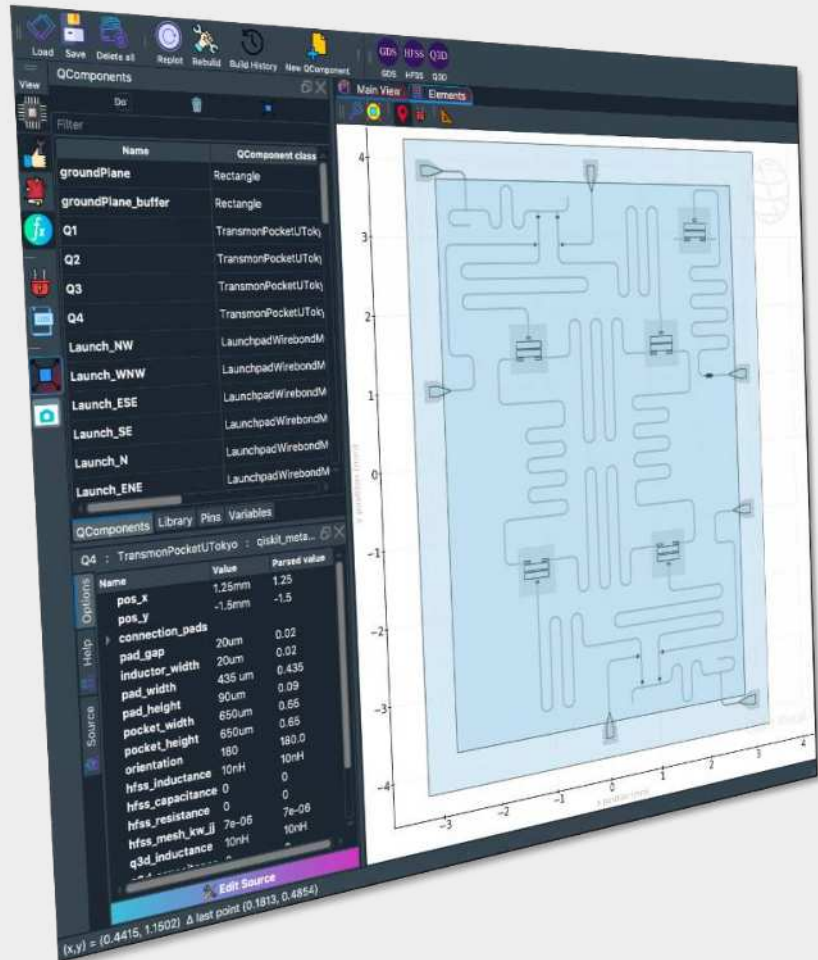
Accepted: *Nature npj qi*



$$\hat{H}_{\text{tot}} = \hat{H}_{\text{lin}} + \hat{H}_{\text{nl}}$$

See also Minev
Yale dissertation Sec. 4.1
(arXiv: 1902.10355)

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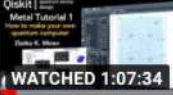
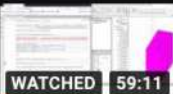
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[10.5281/zenodo.4618153](#)

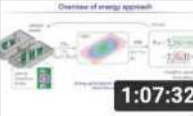
Quantum hardware design and analysis



Tutorials so far

- 1  **Qiskit Metal E01 - Overview**
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- 2  **Qiskit Metal E02 - End to end example of Quantum Chip Design - Part 1 of 2**
Qiskit
WATCHED 1:02:00
- 3  **Qiskit Metal E03 End to end example of Quantum Chip Design - Part 2 of 2**
Qiskit
WATCHED 59:11
- 4  **Qiskit Metal E04 - QComponents for parametric design**
Qiskit
52:46

New analysis (pre-recorded)

- 5  **Qiskit Metal E05.1 - Analysis - Capacitance and Frequency Control**
Qiskit
1:09:45
- 6  **Qiskit Metal E05.2 - Analysis - Eigenmode and Energy Participation**
Qiskit
1:07:02
- 7  **Qiskit Metal E05.3 - Analysis - EPR Theory**
Qiskit
1:18:26
- 8  **Qiskit Metal E05.4 - Analysis - Summary EPR Quantization with Code Example**
Qiskit
1:07:32
- 9  **Qiskit Metal E05.5 - Analysis - Finish Eigenmode Start Impedance Analysis**
Qiskit
54:27
- 10  **Qiskit Metal E05.6 - Analysis - Extracting S Parameters for a Hanging Resonator**
Qiskit
40:18

Transmon Qubit CPB Hamiltonian Charge Basis Demo

localhost:8888/notebooks/tutorials/Appendix/Quick Topic Tutorials Notebooks/Transmon Qubit CPB Hamiltonian Charge Basis Demo.ipynb#Experimental

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jupyter Transmon Qubit CPB Hamiltonian Charge Basis Demo Last Checkpoint: 05/19/2021 (autosaved)

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Run

Modeling transmon qubit Cooper-pair box Hamiltonian in the charge basis

(Zlatko Mineev, Christopher Warren, Nick Lanzillo 2021)

This module models the transmon qubit in the cooper-pair charge basis, assuming wrapped junction phase variable. The Hamiltonian is given by:

$$\hat{H} = 4E_C (\hat{n}_I - n_g) - E_J \cos(\hat{\phi}),$$

where E_C is the charging energy, E_J is the Josephson energy, $\hat{n}$ is the number of Cooper pairs transferred between charge islands, $\hat{\phi}$ is the gauge-phase difference between charge islands, and n_g is effective offset charge of the device. Expressions for the charging energy, Josephson energy and charge can be written as:

$$E_C = \frac{e^2}{2C_\Sigma}, \quad n_g = -\frac{C_d \Phi_s(t)}{2e}, \quad E_J = \frac{\phi_0^2}{L_J},$$

where $C_\Sigma = C_J + C_B + C_g$ (the sum of the Josephson capacitance, shunting capacitance and gate capacitance), L_J is the inductance of the Jose junction, and ϕ is the magnetic flux.

The variables are

$$\hat{\phi} \equiv \frac{\hat{\Phi}}{\phi_0}, \quad \hat{n} \equiv \frac{\hat{Q}}{2e},$$

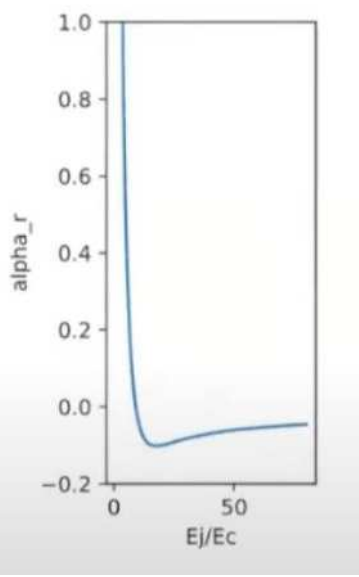
Observe that $\hat{\phi}$ and $\hat{n}$ are both dimensiunless, and they obey the commutation relationship:

$$[\hat{\phi}, \hat{n}] = i$$

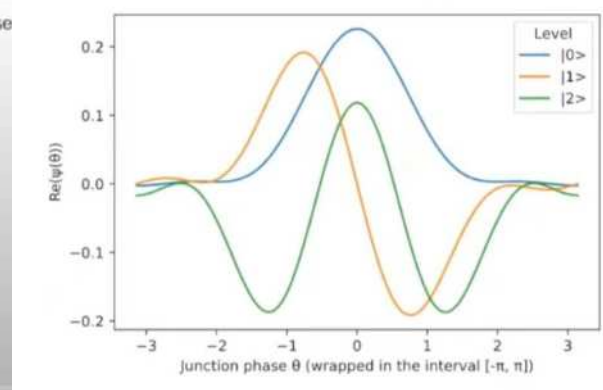
The Hamiltonian can be written in the charge ($\hat{n}$) basis as:

$$H = 4E_C (\hat{n} - n_g)^2 - \frac{1}{2} E_J \sum_n (|n\rangle \langle n+1| + \text{h.c.}),$$

Where $\hat{n} = \sum_n |n\rangle \langle n|$



```
import matplotlib.pyplot as plt
for k in range(3):
    psi, theta = H.psi_k(k, 101)
    plt.plot(theta, psi.real+psi.imag, label=f'|{k}>')
plt.xlabel("Junction phase theta (wrapped in the interval [-pi, pi])")
plt.ylabel("Re(psi(theta))")
plt.legend(title="Level")
```





Docs & tutorials

qiskit.org/documentation/metal

Tutorial videos

YouTube – see docs

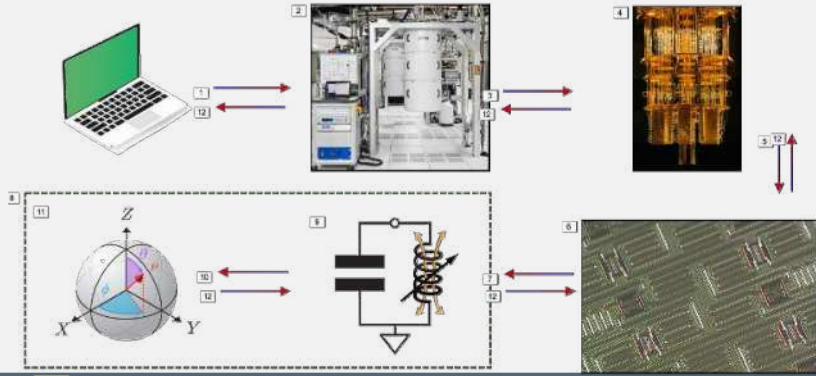
Slack

[#metal](#) (qiskit workspace)

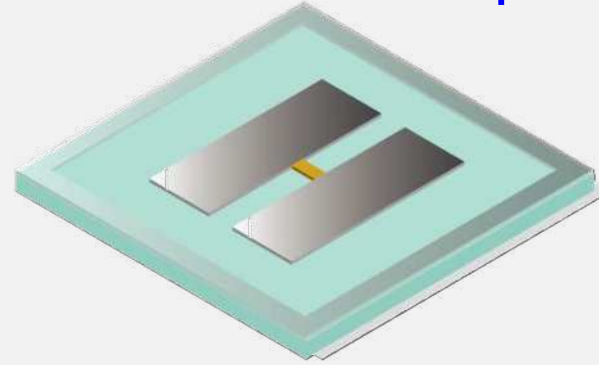
Live weekly tutorials

The road behind and ahead

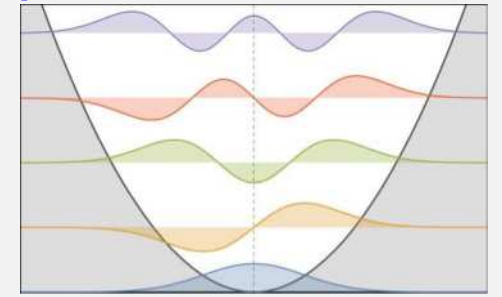
Qubit in the cloud



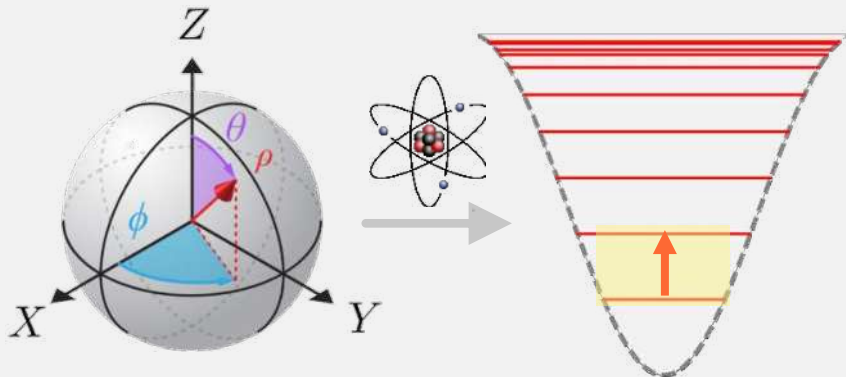
cQED: Transmon qubit



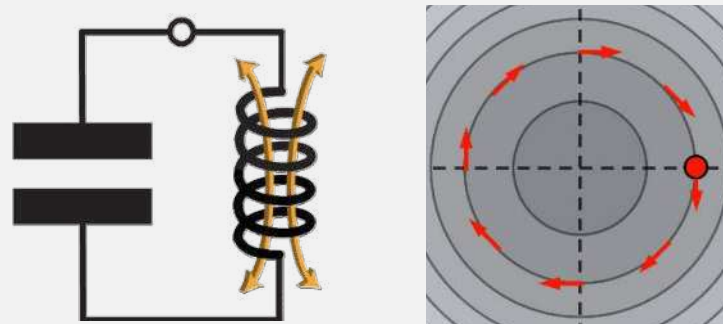
Unveiling the quantum oscillator



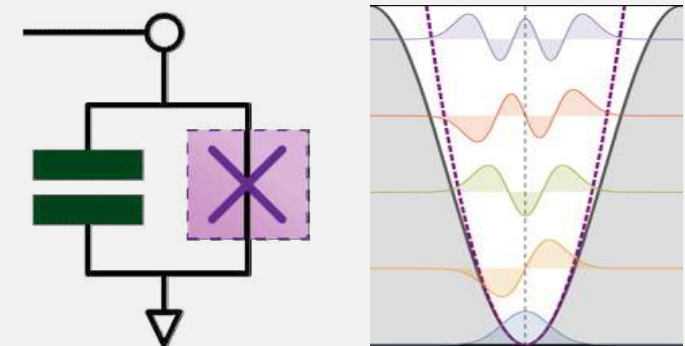
Qubit from atom / oscillator



Classical circuits & the LC



Transmon qubit



Next steps

Tightly integrated lab work with Dr. Thomas McConkey!
More depth on qubit control
Run experiments on real devices

Check out references, problems given in the lecture,
dangerous bends

Break away from the rules of today



Thank you!

Zlatko K. Minev



@zlatko_minev



zlatko-minev.com

IBM Quantum

The important thing is not to stop questioning.
Curiosity has its own reason for existence.

One cannot help but be in awe when he
contemplates the mysteries of eternity, of life, of the
marvelous structure of reality.

It is enough if one tries merely to comprehend a
little of this mystery each day.

Albert Einstein

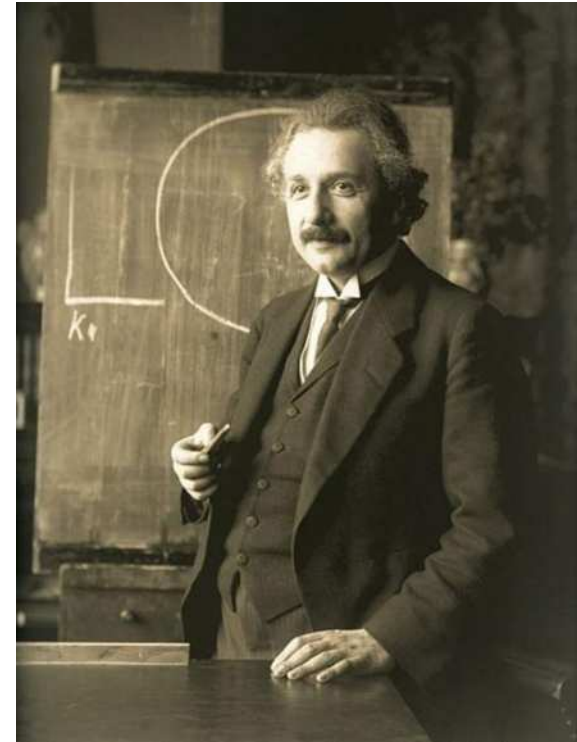


Photo: F. Schmutzer



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